

EIDGENÖSSISCHE TECHNISCHE HOCHSCHULE ZÜRICH

Analysis II: Several Variables

Lecture Notes — Spring Semester 2024

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Preface

This notes are the continuation of 'Analysis I: One variable' and follow the same format and general spirit.

Originally crafted in German for the academic year 2016/2017 by Manfred Einsiedler and Andreas Wieser, they were designed for the Analysis I and II courses in the Interdisciplinary Natural Sciences, Physics, and Mathematics Bachelor programs. In the academic year 2019/2020, a substantial revision was undertaken by Peter Jossen.

For the academic year 2023/2024, Joaquim Serra has developed this English version. It differs from the German original in several aspects: reorganization and alternative proofs of some materials, rewriting and expansion in certain areas, and a more concise presentation in others. This version strictly aligns with the material presented in class, offering a streamlined educational experience.

The courses Analysis I/II and Linear Algebra I/II are fundamental to the mathematics curriculum at ETH and other universities worldwide. They lay the groundwork upon which most future studies in mathematics and physics are built.

Throughout Analysis I/II, we will delve into various aspects of differential and integral calculus. Although some topics might be familiar from high school, our approach requires minimal prior knowledge beyond an intuitive understanding of variables and basic algebraic skills. Contrary to high-school methods, our lectures emphasize the development of mathematical theory over algorithmic practice. Understanding and exploring topics such as differential equations and multidimensional integral theorems is our primary goal. However, students are encouraged to engage with numerous exercises from these notes and other resources to deepen their understanding and proficiency in these new mathematical concepts.

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Chapter 9

Metric spaces

In Analysis I, we focused primarily on functions that operate between real numbers, $\mathbb{R}$ to $\mathbb{R}$. Consequently, we deeply explored the real number line, $\mathbb{R}$, and its properties.

Now, in Analysis II, we expand our scope to functions that map from $\mathbb{R}^n$ to $\mathbb{R}^m$, where n and m are positive integers. In this context, we start by delving into the properties of $\mathbb{R}^n$ (or $\mathbb{R}^m$, as we can use them interchangeably since n and m are arbitrary).

We'll discover that $\mathbb{R}^n$ follows the axioms of a metric space when equipped with the standard Euclidean distance, which we'll define later. This means that $\mathbb{R}^n$ is a metric space. Understanding this concept is vital because it lays the foundation for many fundamental definitions and results we want to establish for $\mathbb{R}^n$ and can be extended to a broader range of metric spaces.

Additionally, this will allow us to revisit (and review) the crucial concept of convergence, which we have seen in Analysis I for $\mathbb{R}$, within the broader context of metric spaces. We'll emphasize the essential properties of metric spaces and their topology, particularly focusing on compactness.

Furthermore, we'll introduce and explore standard results related to normed vector spaces. As we will see, these spaces are more general than Euclidean space $\mathbb{R}^n$, but they possess a higher degree of structure (i.e., more axioms) than metric spaces.

9.1 Basics of Metric Spaces

In this class, our primary focus will be on $\mathbb{R}^n$, but we'll find that certain definitions and proofs become clearer when viewed within the broader context of metric spaces, which have fewer axioms.

However, when we're dealing with a general metric space X , it often helps to initially visualize it as our familiar 2-dimensional or 3-dimensional spaces, $\mathbb{R}^2$ or $\mathbb{R}^3$. This approach can simplify the intuitive understanding of various arguments. If an argument relies on only a limited set of fundamental properties of $\mathbb{R}^3$ (excluding aspects like angles and vector space structure), it may be applicable to general metric spaces as well.

9.1.1 The Euclidean space $\mathbb{R}^n$

For some integer $n \geq 1$, we denote by $\mathbb{R}^n$ the set of all ordered n -tuples of real numbers. A general element $x \in \mathbb{R}^n$ is thus of the form $x = (x_1, \dots, x_n)$, where the x_i 's are real numbers. (Even more rigorously $\mathbb{R}^n := \{x: \{1, \dots, n\} \rightarrow \mathbb{R}\}$.)

$\mathbb{R}^n$ is a vector space over the field of real numbers with the coordinate-wise addition and multiplication by a scalar. Rigorously, given $x, y \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$ we have

$$x + y := (x_1 + y_1, \dots, x_n + y_n), \quad \lambda x := (\lambda x_1, \dots, \lambda x_n).$$

DEFINITION 9.1: EUCLIDEAN STRUCTURE OF $\mathbb{R}^n$

Given $x, y \in \mathbb{R}^n$ we define the standard **scalar product** of x and y as

$$x \cdot y = \langle x, y \rangle := \sum_{1 \leq i \leq n} x_i y_i,$$

the **Euclidean norm** of x as

$$\|x\| := \sqrt{\sum_{1 \leq i \leq n} x_i^2},$$

and the **Euclidean distance** of x and y as

$$d(x, y) := \|x - y\| = \sqrt{\sum_{1 \leq i \leq n} (x_i - y_i)^2}.$$

We key property of the Euclidean distance that we want to abstract is

PROPOSITION 9.2: TRIANGLE INEQUALITY IN $\mathbb{R}^n$

For all $x, y, z \in \mathbb{R}^n$

$$\|x - z\| \leq \|x - y\| + \|y - z\|.$$

Equivalently, for all $x, y \in \mathbb{R}^n$, $\|x + y\| \leq \|x\| + \|y\|$.

Proof. To prove the equivalence of the two statements consider $a = x - y$ and $b = y - z$ so that $a + b = x - z$.

We will see later a more general proof that applies to all inner product spaces (see Corollary 9.100), based on the Cauchy-Schwarz inequality (see Proposition 9.98). Let us give here a hands-on argument.

Pick two points in $\mathbb{R}^n$, $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ we would like to show

$$\left(\sum_{i=1}^n (x_i + y_i)^2 \right)^{1/2} \leq \left(\sum_{i=1}^n x_i^2 \right)^{1/2} + \left(\sum_{i=1}^n y_i^2 \right)^{1/2},$$

Taking squares, this is equivalent to

$$\sum_{i=1}^n x_i^2 + 2x_i y_i + y_i^2 \leq \sum_{i=1}^n x_i^2 + 2 \left(\sum_{i=1}^n x_i^2 \right)^{1/2} \left(\sum_{i=1}^n y_i^2 \right)^{1/2} + \sum_{i=1}^n y_i^2,$$

and, canceling terms, to

$$\sum_{i=1}^n x_i y_i \leq \left(\sum_{i=1}^n x_i^2 \right)^{1/2} \left(\sum_{i=1}^n y_i^2 \right)^{1/2}. \quad (9.1)$$

Therefore, the Proposition will follow if we can establish the validity of (9.1), or (squaring it) of:

$$\left(\sum_{i=1}^n x_i y_i \right)^2 \leq \left(\sum_{i=1}^n x_i^2 \right) \left(\sum_{i=1}^n y_i^2 \right).$$

But this last inequality is easily established summing over all pairs $i, j \in \{1, \dots, n\}$ the inequalities

$$2x_i x_j y_i y_j \leq x_i^2 y_j^2 + x_j^2 y_i^2 \quad \Leftrightarrow \quad (x_i y_j - x_j y_i)^2 \geq 0;$$

and observing that $\sum_{i,j=1}^n x_i^2 y_j^2 = \sum_{i,j=1}^n x_j^2 y_i^2$. □

9.1.2 Definition of metric Space

DEFINITION 9.3: METRIC SPACE

A **metric space** (X, d) is a nonempty set X together with a nonnegative function $d : X \times X \rightarrow [0, \infty)$, called the **distance** (or **metric**) on X , which satisfies:

- (1) For all $x, y \in X$, $d(x, y) = 0$ if and only if $x = y$ (Definiteness).
- (2) For all $x, y \in X$, $d(x, y) = d(y, x)$ (Symmetry).
- (3) For all $x, y, z \in X$, $d(x, z) \leq d(x, y) + d(y, z)$ (Triangle inequality).

9.4. — A metric d on a set X assigns to each pair of points their **distance**. In this interpretation, the definiteness condition states that the only point at zero distance from a given point $x \in X$ is x itself. The symmetry condition states that the distance from $x \in X$ to $y \in X$ is the same as from y to x . Interpreting the distance between two points as the length of a shortest path from one point to the other, the triangle inequality states that the length of a shortest path from x to y is at most the length of a path one takes by first going to y and then from there to zx .

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9.5. — When there is no possible confusion, we will often say “Let X be a metric space...”, leaving the distance function unspecified. This is a shorter version of the more precise sentence “Let (X, d) be a metric space...”.

Furthermore, we may refer to the set X as a **space** and the elements of X as **points**. This is because we have in mind that X is some sort of geometric space, like a subset of the plane or the surface of a sphere. In this setting, “spaces” and “points” will be synonymous with “sets” and “elements”.

9.6. — Notice that the Euclidean space $(\mathbb{R}^n, d)$, with $d(x, y) := \|x - y\|$, is a metric space. In particular $\mathbb{R}$, equipped with the absolute value distance $|x - y|$ is a metric space.

EXERCISE 9.7. — Let (X, d) be a metric space and let $\phi : [0, \infty) \rightarrow [0, \infty)$ a function which is concave, increasing, $\phi(0) = 0$, and not identically zero. Show that $(X, \phi \circ d)$ is again a metric space. For example one can take $\phi(t) = \sqrt{t}$, $\phi(t) = \arctan t$ or $\phi(t) = \frac{t}{1+t}$. Notice that the last two choices always give bounded distances.

EXAMPLE 9.8. — When $X \subset \mathbb{R}$, we can take the **standard metric** d defined by

$$d(x, y) = |x - y| \text{ for all } x, y \in X.$$

EXAMPLE 9.9. — Let X be a set, and $d : X \times X \rightarrow \mathbb{R}$ defined by

$$d(x, y) = \begin{cases} 1 & \text{if } x \neq y \\ 0 & \text{if } x = y \end{cases}$$

for $x, y \in X$. Then, (X, d) is a metric space. Indeed, d is definite and symmetric by definition. Furthermore, d satisfies the triangle inequality: Let x, y, z be points in X . If $d(x, z) = 0$, then $d(x, z) \leq d(x, y) + d(y, z)$ is trivially satisfied. If $d(x, z) = 1$, then $x \neq z$, and y is at least different from one point in $\{x, z\}$, so the triangle inequality also holds. This metric d is called the **discrete metric** on the set X .

EXAMPLE 9.10. — Let $X = \mathbb{R}^2$, and define the **Manhattan metric** on X by

$$d_{NY}(x, y) = |x_1 - y_1| + |y_1 - y_2|$$

where we put $x = (x_1, x_2)$ and $y = (y_1, y_2)$. It can be verified (exercise) that d_{NY} satisfies all axioms of a metric. The reason why d_{NY} is called the Manhattan metric is that in grid-like places such as Manhattan, one can reach from (x_1, x_2) to (y_1, y_2) in the following way: first move ‘horizontally’ (i.e., with constant second coordinate from $x = (x_1, x_2)$ to (y_1, x_2) and then ‘vertically’ (with constant first coordinate) from (y_1, x_2) to $y = (y_1, y_2)$, or vice versa: $x = (x_1, x_2)$ to (x_1, y_2) , and then to $y = (y_2, y_2)$. Since all streets in Manhattan run either from west to east or from north to south, d_{NY} measures the relevant distance between two points.

EXERCISE 9.11. — Let X be the set of all continuous real-valued functions defined on $[0, 1] \subset \mathbb{R}$. For $f, g \in X$ set

$$d_1(f, g) := \max\{|f(x) - g(x)| \mid x \in [0, 1]\} \quad \text{and} \quad d_2(f, g) := \int_0^1 |f(x) - g(x)| dx.$$

Show that (X, d_1) and (X, d_2) are metric spaces.

EXAMPLE 9.12. — If (X, d) is a metric space and $X_0 \subset X$ is some subset then X_0 inherits an structure of metric space from X . Indeed, one can easily verify that (X_0, d_0) , where d_0 the restriction of d to $X_0 \times X_0 \subset X \times X$ is a metric space.

For a more concrete instance of this take $X = \mathbb{R}^3$ with the Euclidean distance d and let X_0 be the sphere $\{x \in \mathbb{R}^3 \mid x_1^2 + x_2^2 + x_3^2 = R^2\}$, for some $R > 0$. Then for any pair of points x, y in the sphere we have

$$d_0(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + (x_3 - y_3)^2} = \sqrt{2R^2 - 2x \cdot y}$$

An arguably more natural metric d_1 on the sphere X_0 can be define measuring the length of the geodesic arc joining x and y . One can see that this metric is given by:

$$d_1(x, y) = R \arccos \left(\frac{x \cdot y}{R^2} \right) \in [0, \pi R].$$

9.1.3 Sequences, limits, and completeness

The definition of sequence in a set was given in Analysis I. We recall it next:

DEFINITION 9.13: SEQUENCES IN A SET

Let X be a set, a **sequence** in X is a function $x : \mathbb{N} \rightarrow X$. The image $x(n)$ of $n \in \mathbb{N}$ is also denoted as x_n and referred to as the n -th **term** of x .

Instead of $x : \mathbb{N} \rightarrow X$, one often writes $(x_n)_{n \in \mathbb{N}}$, $(x_n)_{n \geq 0}$, or $(x_n)_{n=0}^\infty$.

We introduce the following vocabulary, which is useful if used precisely.

DEFINITION 9.14: “EVENTUALLY”

Let $(x_n)_{n \geq 0} \subset X$ be a sequence and let $\mathcal{P} : X \rightarrow \{\text{true}, \text{false}\}$ be a property that an element in X might have or not. Then one says that “ x_n satisfies $\mathcal{P}$ eventually” there exist $N \in \mathbb{N}$ such that $\mathcal{P}(x_n)$ is true for all $n \geq N$. In other words, if $\mathcal{P}(x_n)$ holds true along the sequence with only finitely many exceptions.

DEFINITION 9.15: CONVERGENT SEQUENCE

Let (X, d) be a metric space, $x \in X$ and $(x_n)_{n \in \mathbb{N}}$ be a sequence in X . We say that $(x_n)_{n \in \mathbb{N}}$ **converges** to x , or that x is the **limit** of the sequence $(x_n)_{n \in \mathbb{N}}$, if

$$\lim_n d(x_n, x) = 0.$$

In other words, for any $\varepsilon > 0$ eventually $d(x_n, x) < \varepsilon$.

9.16. — When the metric space (X, d) is clear from the context, we may write $\lim_n x_n = x$ or even $x_n \rightarrow x$ to express that $(x_n)_{n \in \mathbb{N}}$ converges to x .

EXERCISE 9.17. — In the setting of Exercise 9.7, show that a sequence converges in (X, d) if and only if it converges in $(X, \phi \circ d)$.

LEMMA 9.18: UNIQUENESS OF THE LIMIT

In a metric space, a convergent sequence has only one limit.

Proof. Let (X, d) be a metric space and let $A, B \in X$ be limits of some sequence $(x_n)_{n=0}^\infty$, we mean to show that $A = B$. Take $\varepsilon > 0$, then, we can find $N_A, N_B \in \mathbb{N}$ such that $d(x_n, A) < \frac{\varepsilon}{2}$

for all $n \geq N_A$, and $d(x_n, B) < \frac{\varepsilon}{2}$ for all $n \geq N_B$. Then, for $N := \max\{N_A, N_B\}$, we have that

$$d(A, B) \leq d(A, x_N) + d(x_N, B) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

where we used the triangular inequality. Since $\varepsilon > 0$ was arbitrary, it follows that $d(A, B) = 0$, and thus $A = B$ because of the definiteness of d . $\square$

We recall the notions of subsequences and accumulation points

DEFINITION 9.19: SUBSEQUENCE

Let $(x_n)_{n=0}^\infty$ be a sequence in a set X .

A **subsequence** of $(x_n)_{n=0}^\infty$ is a sequence of the form $(x_{f(k)})_{k=0}^\infty$, where $f : \mathbb{N} \rightarrow \mathbb{N}$ is a strictly increasing function. It is standard to denote subsequences by $(x_{n_k})_{k \in \mathbb{N}}$, $(x_{n_k})_{k \geq 0}$, or $(x_{n_k})_{k=0}^\infty$ (i.e., $n_k := f(k)$).

DEFINITION 9.20: ACCUMULATION POINTS

Let (X, d) be a metric space.

- Given $Y \subset X$, we say that $x \in X$ is an **accumulation point** of Y if there exists a sequence $(y_n)_{n \geq 0} \subset Y$ converging to x .
- Given a sequence $(x_n)_{n \geq 0}$ in X , we say that x is an **accumulation point** of the sequence if there is some subsequence converging to x .

LEMMA 9.21: ACCUMULATIONS POINTS OF A CONVERGING SEQUENCE

Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in a metric space X , and let $x \in X$. $(x_n)_{n \in \mathbb{N}}$ converges to x if and only if every subsequence of $(x_n)_{n \in \mathbb{N}}$ converges to x .

In other words: a sequence in a metric space is convergent if and only if it has a unique accumulation point.

Proof. We first prove the “only if” part. Let $(x_{f(n)})_{n \in \mathbb{N}}$ be a subsequence, i.e., $f : \mathbb{N} \rightarrow \mathbb{N}$ is some strictly increasing map. Given $\varepsilon > 0$ there is N such that $d(x_n, x) < \varepsilon$ for all $n \geq N$. Hence, $d(x_{f(n)}, x) < \varepsilon$ for all $n \geq N$, as $f(n) \geq n$.

We now prove the “if” part we can simply use that $(x_n)_{n \geq 0}$ is a subsequence (i.e., $f(n) = n$) and hence it converges to x . $\square$

9.22. — A stronger version of the previous Lemma that is useful in some contexts asserts the following: a sequence $(x_n)_{n \in \mathbb{N}}$ in a metric space converges to x if and only if every partial sequence $(x_{f(n)})_{n \in \mathbb{N}}$ (f increasing) has a sub-subsequence $(x_{g(f(n))})_{n \in \mathbb{N}}$ (g increasing) converging to x .

While the proof of the “only if” part is similar ($g(f(n)) \geq n$) the “if” part is less trivial than in the previous lemma. One can argue by contraposition: If x_n does **not** converge to x then

we want to find a subsequence such that we **cannot** extract a sub-subsequence converging to x .

To do so we start by the negation of “ x_n converges to x ”. Recall:

$$x_n \rightarrow x \quad \Leftrightarrow \quad \forall \varepsilon > 0 \quad \exists N \in \mathbb{N} \quad \forall n \geq N \quad d(x_n, x) < \varepsilon$$

The negation of this is:

$$\exists \varepsilon > 0 \quad \forall N \in \mathbb{N} \quad \exists n \geq N \quad d(x_n, x) \geq \varepsilon$$

In other words, the set of $\{n \in \mathbb{N} \mid d(x_n, x) \geq \varepsilon\}$ and hence there is an increasing sequence $(n_k)_{k \geq 0}$ such that $d(x_{n_k}, x) \geq \varepsilon$. Notice that any sub-subsequence will still remain at distance $\geq \varepsilon$ from x and hence will not converge to x .

This stronger version can be used, for example, to prove that a continuous function $f : [0, 1] \rightarrow \mathbb{R}$ has a unique minimum point if and only if all sequences $(x_n)_{n \geq 0} \subset [0, 1]$ such that $f(x_n) \rightarrow \min_{[0,1]}$ converge to the same limit point.

LEMMA 9.23: CONVERGENCE IN $\mathbb{R}^n$

A sequence in $\mathbb{R}^n$ converges (in the Euclidean distance) if and only if it converges coordinate-wise.

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Proof. Let $\{x_k\}_{k \in \mathbb{N}} \subset \mathbb{R}^n$ be a sequence. For $j = 1, \dots, n$, we denote with $x_{k,j}$ the j -th component of the vector x_k .

Assume that $x_k \rightarrow x \in \mathbb{R}^n$. By definition, given $\varepsilon > 0$ and any j , it holds

$$|x_{k,j} - x_j| \leq \left(\sum_{i=1}^n (x_{k,i} - x_i)^2 \right)^{1/2} = \|x_k - x\| \leq \varepsilon \quad \text{eventually in } k.$$

This proves that, for each $j = 1, \dots, n$, $x_{k,j} \rightarrow x_j$ when $k \rightarrow \infty$ (as sequences of real numbers, with the standard absolute value distance).

Assume now that for each $j = 1, \dots, d$ it holds

$$x_{k,j} \rightarrow x_j \quad \text{as } k \rightarrow \infty,$$

for some numbers $x_j \in \mathbb{R}$. We prove that $x_k \rightarrow x$ in $\mathbb{R}^d$ where $z := (z_1, \dots, z_d)$. Given $\varepsilon > 0$, for each $j \in \{1, \dots, d\}$, there exists an $N_j \in \mathbb{N}$ such that

$$|x_{k,j} - x_j| < \frac{\varepsilon}{n} \quad \text{for all } k \geq N_j.$$

This means that

$$\left(\sum_{i=1}^n (x_{k,i} - x_i)^2 \right)^{1/2} \leq \sqrt{n} \frac{\varepsilon}{n} < \varepsilon \quad \text{for all } k \geq \max\{N_1, \dots, N_j\},$$

which proves that $x_k \rightarrow x$ with respect to the Euclidean distance. $\square$

We introduce the concept of **completeness** for metric spaces. This concept does not conflict with the notion of completeness that we gave for $\mathbb{R}$. We will soon show that $\mathbb{R}$, as well as $\mathbb{C}$, is complete as a metric space. In contrast, the metric space $\mathbb{Q}$ is not complete.

DEFINITION 9.24: CAUCHY SEQUENCE

A sequence $(x_n)_{n=0}^{\infty}$ in a metric space (X, d) is a **Cauchy sequence** if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $d(x_m, x_n) < \varepsilon$ for all pair of integers m, n with $n \geq N$ and $m \geq N$.

EXERCISE 9.25. — Prove the following elementary facts about Cauchy sequences in a metric space (X, d) :

- A Cauchy sequence is bounded (meaning that $\{d(x_n, x_0)\} \subset \mathbb{R}$ is a bounded set)
- Every convergent sequence is a Cauchy sequence.
- A Cauchy sequence converges if and only if it has a converging subsequence.

DEFINITION 9.26: COMPLETE METRIC SPACE

A metric space (X, d) is called **complete** if every Cauchy sequence in (X, d) converges.

EXAMPLE 9.27. — The interval $(0, 1) \subset \mathbb{R}$, endowed with the standard distance $d(x, y) = |x - y|$, is *not* complete. However, $\mathbb{N} \subset \mathbb{R}$ is complete, as well as $[0, \infty)$.

EXERCISE 9.28. — Show that $\mathbb{Q}$, with the distance inherited from the standard distance on $\mathbb{R}$, is **not** a complete metric space.

EXERCISE 9.29. — Show that the space X of all bounded sequences $(x_n)_{n \geq 0}$ of real numbers, equipped with the distance

$$d((x_n)_{n \geq 0}, (y_n)_{n \geq 0}) = \sup_{n \geq 0} |x_n - y_n|$$

is complete. Show also that the subspace X_0 of sequences with limit 0 is **not** complete.

THEOREM 9.30: COMPLETENESS OF $\mathbb{R}^n$

For all $n \geq 1$, $\mathbb{R}^n$ (with the Euclidean distance) is complete. In particular, $\mathbb{R}$ and $\mathbb{C}$ are complete.

Proof. Similar to the proof Lemma 9.23, a sequence (x_k) in $\mathbb{R}^n$ is a Cauchy if and only if $x_{k,j}$, $j = 1, \dots, n$ are Cauchy (in $\mathbb{R}$). It then follows from Theorem 2.124 (Cauchy sequences in $\mathbb{R}$ converge) that x_k converges coordinate-wise. Thus, by Lemma 9.23 x_k converges in $\mathbb{R}^n$. $\square$

9.31. — Completion of metric space (extra material) Let (X, d) be a metric space. We write $\mathcal{C}_X$ for the set of all Cauchy sequences in X and define an equivalence relation on $\mathcal{C}_X$ by

$$(x_n)_{n=0}^\infty \sim (y_n)_{n=0}^\infty \iff \lim_{n \rightarrow \infty} d(x_n, y_n) = 0.$$

The quotient set $\overline{X} = \mathcal{C}_X / \sim$, equipped with the metric $\overline{d}$ given by

$$\overline{d}([(x_n)_{n=0}^\infty], [(y_n)_{n=0}^\infty]) = \lim_{n \rightarrow \infty} d(x_n, y_n),$$

is called the **completion** of (X, d) . The injection $\iota : X \rightarrow \overline{X}$, mapping $x \in X$ to the class of the constant sequence with value x , is called the **canonical embedding**. For all $x, y \in X$, we have

$$d(x, y) = \overline{d}(\iota(x), \iota(y)),$$

which implies that ι is injective.

EXERCISE 9.32. — Show that the objects introduced in 9.31 are well-defined. In particular, verify that $\overline{d}$ is indeed a metric on $\overline{X}$.

EXERCISE 9.33. — As the name suggests, the completion $(\overline{X}, \overline{d})$ of a metric space is complete, meaning that every Cauchy sequence in $\overline{X}$ converges. A sequence in $\overline{X}$ is essentially a sequence of sequences, i.e.,

$$[(x_{m,n})_{n=0}^\infty]_{m=0}^\infty.$$

Show that $[(x_{n,n})_{n=0}^\infty]$ is a limit of this sequence.

EXERCISE 9.34. — Let (X, d) be a metric space with completion $(\overline{X}, \overline{d})$. Let (Y, d_Y) be a complete metric space, and let $f : X \rightarrow Y$ be a function such that

$$d(x, y) = d_Y(f(x), f(y))$$

for all $x, y \in X$. Show that there exists a unique function $\overline{f} : \overline{X} \rightarrow Y$ such that $f = \overline{f} \circ \iota_X$ and

$$\overline{d}(\overline{x}, \overline{y}) = d_Y(\overline{f}(\overline{x}), \overline{f}(\overline{y}))$$

for all $\overline{x}, \overline{y} \in \overline{X}$. This can be interpreted as: “ $\overline{X}$ is the *smallest* complete metric space containing X .”

9.1.4 *The Reals as the Completion of Rationals (extra material; cf. Grundstrukturen)

In the first semester, we defined $\mathbb{R}$ as any complete ordered field, postulating its existence (Definitions 2.18 and 2.19). The idea of completion of metric spaces allows one to easily construct a model of $\mathbb{R}$. This construction shows, in particular, the existence of a complete ordered field. One can also prove with a bit of patience (although it is not hard to do so) that actually there is only one model of $\mathbb{R}$, in the sense that any two complete ordered fields must be isomorphic.

The completion of $\mathbb{Q}$ serves as a model for a field of real numbers. First, note that the construction of the completion of $\mathbb{Q}$ does not necessarily require a field of real numbers (as the target space for the standard metric on $\mathbb{Q}$). The set of all Cauchy sequences $\mathcal{C}$ in $\mathbb{Q}$ is the set of all sequences of rational numbers $(q_n)_{n=0}^{\infty}$ such that

$$\forall k \in \mathbb{N} \exists N \in \mathbb{N} : m, n \geq N \implies |q_n - q_m| < 2^{-k}.$$

The set $\mathcal{C}_{\mathbb{Q}}$ is a vector space over $\mathbb{Q}$ with component-wise operations, and

$$\mathcal{N} = \{(q_n)_{n=0}^{\infty} \in \mathcal{C} \mid \lim_{n \rightarrow \infty} q_n = 0\}$$

is a linear subspace. The equivalence relation $(p_n - q_n)_{n=0}^{\infty} \in \mathcal{N}$ in 9.31 translates to $(p_n - q_n)_{n=0}^{\infty} \in \mathcal{N}$. We define the set $\mathbb{R}$ as the quotient

$$\mathbb{R} = \mathcal{C} / \sim = \mathcal{C} / \mathcal{N}$$

in the sense of linear algebra. Thus, $\mathbb{R}$ is a vector space over $\mathbb{Q}$. We denote the injective linear map $\iota : \mathbb{Q} \rightarrow \mathbb{R}$ by the canonical embedding, which assigns to $q \in \mathbb{Q}$ the class of the constant sequence with value q . From now on, elements of $\mathbb{R}$ are called *real numbers*, and we consider $\mathbb{Q}$ as a subset of $\mathbb{R}$ via the canonical embedding ι .

We define a product on $\mathbb{R}$ by component-wise multiplication. That is, for elements $x = [(p_n)_{n=0}^{\infty}]$ and $y = [(q_n)_{n=0}^{\infty}]$, we define

$$x \cdot y = [(p_n q_n)_{n=0}^{\infty}].$$

It can be verified that this gives a well-defined commutative operation on $\mathbb{R}$, satisfying the distributive law with respect to addition, and compatible with the multiplication of rational numbers via the canonical embedding. In particular, $1_{\mathbb{R}} = \iota(1) = [(1)_{n=0}^{\infty}]$ is the multiplicative identity in $\mathbb{R}$. If $x = [(q_n)_{n=0}^{\infty}]$ is non-zero, then $(q_n)_{n=0}^{\infty}$ is a Cauchy sequence in $\mathbb{Q}$ that does not converge to zero. Therefore, $q_n \neq 0$ for all but finitely many $n \in \mathbb{N}$. The class of the sequence $(p_n)_{n=0}^{\infty}$ given by

$$p_n = \begin{cases} 1 & \text{if } q_n = 0 \\ q_n^{-1} & \text{otherwise} \end{cases}$$

serves as a multiplicative inverse for x . This shows that $\mathbb{R}$ is a field with the given operations.

We use the usual order relation on $\mathbb{Q}$ to construct an order relation on $\mathbb{R}$. For elements $x = [(p_n)_{n=0}^\infty]$ and $y = [(q_n)_{n=0}^\infty]$ in $\mathbb{R}$, we declare

$$x \leq y$$

if there exists a sequence $(r_n)_{n=0}^\infty \in \mathcal{N}$ such that $p_n - r_n \leq q_n$ for all $n \in \mathbb{N}$. It is left to the diligent reader to verify that this indeed defines a well-defined order relation on $\mathbb{R}$ that is compatible with the field structure on $\mathbb{R}$. Thus, $\mathbb{R}$ is equipped with the structure of an ordered field.

It remains to show that the ordered field $\mathbb{R}$ is complete in the sense of Definition ???. It is easy to see that $\mathbb{R}$ satisfies the Archimedean Principle: Let $x = [(q_n)_{n=0}^\infty] \in \mathbb{R}$ be positive. Then, $(q_n)_{n=0}^\infty$ is not a null sequence. Thus, there exists a $k \in \mathbb{N}$ such that $|q_n| > 2^{-k}$ for infinitely many $n \in \mathbb{N}$. However, $(q_n)_{n=0}^\infty$ is also a Cauchy sequence, so there exists $N \in \mathbb{N}$ such that $m, n \geq N \implies |q_n - q_m| < 2^{-k-1}$. This shows that $|q_n| > 2^{-k-1}$ and even $q_n > 2^{-k-1}$ for all but finitely many $n \in \mathbb{N}$, since $x > 0$. This demonstrates $\iota(2^{-k-1}) \leq x$, or simply $2^{-k-1} \leq x$ as we consider $\mathbb{Q}$ as a subset of $\mathbb{R}$. Thus, the Archimedean Principle holds, as stated in Corollary ??. Now, let $X, Y \subset \mathbb{R}$ be non-empty subsets such that $x \leq y$ for all $x \in X$ and $y \in Y$. We want to find a real number $z = [(r_n)_{n=0}^\infty] \in \mathbb{R}$ between X and Y . To do this, we first choose arbitrary $a_0, b_0 \in \mathbb{Q}$ such that $[a_0, b_0] \cap X \neq \emptyset$ and $[a_0, b_0] \cap Y \neq \emptyset$, and set $r_0 = \frac{1}{2}(a_0 + b_0)$. If $x \leq r_0 \leq y$ for all $x \in X$ and $y \in Y$, we set $z = r_0$ and we are done. Otherwise, we define a_1 and b_1 as

$$\begin{cases} a_1 = r_0 \text{ and } b_1 = b_0 & \text{if } [a_0, r_0] \cap X \neq \emptyset \text{ and } [a_0, r_0] \cap Y \neq \emptyset \\ a_1 = a_0 \text{ and } b_1 = r_0 & \text{if } [r_0, b_0] \cap X \neq \emptyset \text{ and } [r_0, b_0] \cap Y \neq \emptyset \end{cases}$$

and set $r_1 = \frac{1}{2}(a_1 + b_1)$. By continuing this process, we either find an r_n such that $x \leq r_n \leq y$ for all $x \in X$ and $y \in Y$, and we set $z = r_n$, or we obtain sequences $(a_n)_{n=0}^\infty$, $(b_n)_{n=0}^\infty$, and $(r_n)_{n=0}^\infty$ with $|b_n - a_n| \leq 2^{-n}|a_0 - b_0|$ and

$$[a_n, b_n] \cap X \neq \emptyset \quad \text{and} \quad [a_n, b_n] \cap Y \neq \emptyset.$$

As the diligent reader can verify, this implies that $(a_n)_{n=0}^\infty$, $(b_n)_{n=0}^\infty$, and $(r_n)_{n=0}^\infty$ are all Cauchy sequences, and the real number

$$z = [(a_n)_{n=0}^\infty] = [(r_n)_{n=0}^\infty] = [(b_n)_{n=0}^\infty]$$

satisfies the inequalities $x \leq z \leq y$ for all $x \in X$ and $y \in Y$.

9.2 Topology of Metric Spaces

9.2.1 Open and closed sets

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DEFINITION 9.35: OPEN BALLS

Let (X, d) be a metric space, $x \in X$, and $r > 0$ a real number. In this context, we write

$$B(x, r) := \{y \in X \mid d(x, y) < r\}$$

and refer to the set $B(x, r)$ as the **open ball** with center x and radius r .

DEFINITION 9.36: OPEN AND CLOSED SETS

Let (X, d) be a metric space:

- A subset $E \subset X$ is called **open** if, for every $x \in E$, there exists $r > 0$ such that $B(x, r) \subset E$.
- The collection of all open sets, $\mathcal{T}_d = \{U \subset X \mid U \text{ open}\}$, is called the **topology** generated by d .
- A subset $E \subset X$ is called **closed** if $X \setminus E$ is open.

9.37. — In particular, $\emptyset$ and X are always both open and closed. In general, a subset U needs not to be neither open nor closed.

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It is not true in general that the only “clopen” sets in a space are the empty set $\emptyset$ and the space itself X . A set is termed “clopen” if it is both open and closed. For illustration, take the space $X = (0, 1) \cup (2, 3)$, equipped with the standard metric from $\mathbb{R}$. Here, the intervals $(0, 1)$ and $(2, 3)$ are clopen: they are open and closed in X . This example underscores the presence of other clopen sets beyond just $\emptyset$ and X . The significance of clopen sets will become more apparent in our discussions on connectedness. As we will see, connected spaces are precisely characterized by the absence of nontrivial (neither empty nor the whole space) clopen sets.

9.38. — Consider the set $X = [0, 2]$, equipped with the standard metric inherited from $\mathbb{R}$. In this context, the subset $[0, 1)$ is open within X (an exercise worth verifying). However, when considered as subset of the whole $\mathbb{R}$, $[0, 1)$ is neither open nor closed. This example illustrates that statements regarding the openness of a set like $[0, 1)$ require clarity about the ambient space (X, d) being considered. In practice, though, such nuances are often glossed over when the context is clear, and delving into these subtleties is usually unnecessary for typical discussions.

EXERCISE 9.39. — Let (X, d) be a metric space. Show that

- The open ball $B(x, r)$ is an open set.
- Every finite subset of X is closed.

PROPOSITION 9.40: ARBITRARY UNIONS AND FINITE INTERSECTIONS

Let $(U_i)_{i \in I}$ be any family of open subsets of X , then $\bigcup_{i \in I} U_i$ is also open. If I is a finite set, also $\bigcap_{i \in I} U_i$ is open.

Proof. Set

$$U = \bigcup_{i \in I} U_i$$

and let $x \in U$. Then there exists $i \in I$ with $x \in U_i$, and since U_i is open, there exists an $\varepsilon > 0$ such that $B(x, \varepsilon) \subset U_i$, implying $B(x, \varepsilon) \subset U$. Thus, U is open. Finally, let $(U_i)_{i \in I}$ be a finite family of open subsets of X . Set

$$U = \bigcap_{i \in I} U_i$$

and let $x \in U$. Then $x \in U_i$ for all $i \in I$, and for each $i \in I$, there exists $\varepsilon_i > 0$ such that $B(x, \varepsilon_i) \subset U_i$. For $\varepsilon := \min\{\varepsilon_i \mid i \in I\}$, we have $\varepsilon > 0$ and $B(x, \varepsilon) \subset U_i$ for all $i \in I$. Thus, $B(x, \varepsilon) \subset U$, completing the proof. $\square$

PROPOSITION 9.41: ARBITRARY INTERSECTIONS AND FINITE UNIONS

Let $(A_i)_{i \in I}$ be any family of closed subsets of X , then $\bigcap_{i \in I} A_i$ is also closed. If I is a finite set, also $\bigcup_{i \in I} A_i$ is closed.

Proof. Apply Proposition 9.40 to the (open) complements of the closed sets. $\square$

EXAMPLE 9.42. — The intersection of infinitely many open sets may not be open. Take for example $\mathbb{R}$ with the standard metric. The intersection of the family of open sets $\{(-1/n, 1/n) \mid n \in \mathbb{N}\}$ gives $\{0\}$, which is not open. Taking complements you obtain an example where an infinite union of closed sets is not closed.

DEFINITION 9.43:

Let X be a metric space and $E \subset X$. We define

- the **interior** $E^\circ := \bigcup \{U \subset E : U \text{ is open}\}$, which is the largest open set contained in E .
- the **closure** $\overline{E} := \bigcap \{A \supset E : A \text{ is closed}\}$, which is the smallest closed set containing E .
- the **(topological) boundary** $\partial E := \overline{E} \setminus E^\circ$.

EXERCISE 9.44. — Using Proposition 9.40, prove that E° is always open while $\overline{E}$ and ∂E are always closed.

EXERCISE 9.45. — For balls $B(x, r)$ prove that $\overline{B(x, r)} = \{y \in X \mid d(x, y) \leq r\}$ and deduce $\partial B(x, r) = \{y \in X \mid d(x, y) = r\}$.

LEMMA 9.46: OPEN AND CLOSED TROUGH SEQUENCES

Let (X, d) be a metric space.

- (1) A subset $U \subset X$ is open if and only if, for every convergent sequence in X with a limit in U , the sequence eventually lies in U .
- (2) A subset $A \subset X$ is closed if and only if, for every convergent sequence $(x_n)_{n=0}^\infty$ in X with $x_n \in A$ for all $n \in \mathbb{N}$, the limit also lies in A . In other words, if and only if A coincides with the set of all its accumulation points.

Proof. Let $U \subset X$ be an open subset of X , and let $(x_n)_{n=0}^\infty$ be a sequence in X with a limit x in U . Then, there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subset U$, and since $(x_n)_{n=0}^\infty$ converges to x , there exists an $N \in \mathbb{N}$ such that $x_n \in B(x, \varepsilon)$ for all $n \geq N$. Conversely, let $V \subset X$ be a non-open subset. Then there exists a point $x \in V$ such that $B(x, \varepsilon) \setminus V \neq \emptyset$ for every $\varepsilon > 0$. For each $n \in \mathbb{N}$, we can find $x_n \in B(x, 2^{-n}) \setminus V$. The sequence $(x_n)_{n=0}^\infty$ in $X \setminus V$ converges to $x \in V$, and satisfies $x_n \notin V$ for every $n \in \mathbb{N}$. This completes the proof of the first statement.

Let $A \subset X$ be closed, and let $(x_n)_{n=0}^\infty$ be a convergent sequence in X with $x_n \in A$ for all $n \in \mathbb{N}$. Let x be the limit of the sequence $(x_n)_{n=0}^\infty$. Then, $U = X \setminus A$ is open and cannot contain the limit x of $(x_n)_{n=0}^\infty$, as otherwise almost all elements of the sequence $(x_n)_{n=0}^\infty$ would have to lie in U . Therefore, the limit x belongs to A . Finally, suppose $A \subset X$ is not closed. Then $U = X \setminus A$ is not open, and according to the previous argument, there exists a sequence $(x_n)_{n=0}^\infty$ in $A = X \setminus U$ with a limit $x \in U$. $\square$

EXERCISE 9.47. — Let (X, d) be a complete metric space and $E \subset X$ a closed subset. Show that E is complete as well.

PROPOSITION 9.48: TOPOLOGICAL NOTION OF CONVERGENCE

Let (X, d) be a metric space a sequence $(x_n)_{n \geq 0}$ converges to x if and only if for all U open containing x , x_n eventually lies in U .

Proof. Notice that we can rewrite the definition of convergence as follows: $x_n \rightarrow x$ if and only if for all $\varepsilon > 0$, x_n eventually lies in $B(x, \varepsilon)$. Now, if U is any open set containing x then by definition of open set there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subset U$ and hence $x_n \rightarrow x$ implies that x_n eventually lies in U , establishing the “only if” direction. For the “if” we just that for given $\varepsilon > 0$ we can take $U = B(x, \varepsilon)$ (open balls are open) and hence x_n eventually lies in $B(x, \varepsilon)$. $\square$

COROLLARY 9.49: DISTANCES WITH SAME CONVERGENT SEQUENCES

Let X be a set endowed with two different distances d_1 and d_2 . Then (X, d_1) and (X, d_2) have the same convergent sequences if and only if the topologies generated by d_1 and d_2 coincide.

Proof. By Corollary 9.48 the notion of convergent sequence only depends on the collection of open sets. That is, it only depends on the distance through the topology it generate. Hence distances generating the same open sets have the same convergent sequences. $\square$

EXERCISE 9.50. — Let (X, d) be a metric space, and $A \subset X$ a subset.

- (i) Assume X is complete and A is closed. Show that the subspace A is also complete.
- (ii) Assume A is complete. Show that $A \subset X$ is closed.

9.2.2 Continuity

We now aim to generalize the concept of continuity to functions defined between metric spaces.

DEFINITION 9.51: CONTINUITY

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \rightarrow Y$ be a function. We say that f is continuous if one of the following **equivalent** conditions hold:

- (1) We say f is **$\varepsilon - \delta$ continuous** if, for all $x \in X$ and $\varepsilon > 0$, there exists a $\delta > 0$ such that if $d_X(x, x') < \delta$, $x' \in X \implies d_Y(f(x), f(x')) < \varepsilon$. In other words, $B(x, \delta) \subset B(f(x), \varepsilon)$.
- (2) We say f is **sequentially continuous** if, for every convergent sequence $(x_n)_n$ in X with limit $x = \lim_{n \rightarrow \infty} x_n$, the sequence $(f(x_n))_n$ converges in Y , with $f(x) = \lim_{n \rightarrow \infty} f(x_n)$.
- (3) We say f is **topologically continuous** if, for every open subset $U \subset Y$, the preimage $f^{-1}(U) = \{x \in X \mid f(x) \in U\}$ is open in X .

PROPOSITION 9.52: THE THREE FACES OF CONTINUITY

Let X and Y be metric spaces, and let $f : X \rightarrow Y$ be a function. The following conditions are equivalent:

- (1) The function f is $\varepsilon - \delta$ continuous.
- (2) The function f is sequentially continuous.
- (3) The function f is topologically continuous.

Proof. (1) $\implies$ (2): Let $(x_n)_{n=0}^\infty$ be a convergent sequence in X with limit $x \in X$, and let $\varepsilon > 0$. There exists a $\delta > 0$ such that $f(x') \in B(f(x), \varepsilon)$ for all $x' \in B(x, \delta)$. Since $(x_n)_{n=0}^\infty$ converges to x , there exists an $N \in \mathbb{N}$ such that $x_n \in B(x, \delta)$ for all $n \geq N$. In particular, for $n \geq N$, $f(x_n) \in B(f(x), \varepsilon)$. Since $\varepsilon > 0$ was arbitrary, it follows that $\lim_{n \rightarrow \infty} f(x_n) = f(x)$, and thus f is sequentially continuous as claimed.

$\neg(3) \implies \neg(1)$: Assume f is not topologically continuous. Then exists $U \subset Y$ open such that $f^{-1}(U)$ is not open. Therefore, there is $x \in f^{-1}(U)$ and a sequence $(x_n)_{n \geq 0} \subset X \setminus f^{-1}(U)$ with $x_n \rightarrow x$. But then $f(x) \in U$ and $f(x_n) \in Y \setminus U$ for all n which contradicts. Since U we have found a sequence such that $x_n \rightarrow x$ but $f(x_n)$ does not converge to $f(x)$.

(3) $\implies$ (1): Let $x \in X$ and $\varepsilon > 0$. The preimage $f^{-1}(B(f(x), \varepsilon))$ contains the point x and is open by assumption, as $B(f(x), \varepsilon) \subset Y$ is open. Thus, there exists a $\delta > 0$ such that $B(x, \delta) \subset f^{-1}(B(f(x), \varepsilon))$. Therefore, f is ε - δ -continuous as claimed. $\square$

DEFINITION 9.53:

Let (X, d_X) and (Y, d_Y) be metric spaces. We say that $f: X \rightarrow Y$ is

- **Uniformly continuous** if for every $\varepsilon > 0$ there is $\delta > 0$ such that $d_Y(f(x), f(x')) < \varepsilon$ whenever $d_X(x, x') < \delta$.
- **Lipschitz continuous** if there is a constant $L > 0$ such that

$$d_Y(f(x), f(x')) \leq L d_X(x, x') \quad \text{for all } x, x' \in X.$$

The constant L is called **Lipschitz constant** of f

EXERCISE 9.54. — Let (X, d) be a metric space, and let $E \subset X$ be a non-empty subset. For $x \in X$, define

$$f_E(x) = \inf\{d(x, z) \mid z \in E\}.$$

Show that the function $f_E: X \rightarrow \mathbb{R}$ is 1-Lipschitz continuous, and that $E \subset X$ is closed if and only if $E = \{x \in X \mid f_E(x) = 0\}$.

EXERCISE 9.55. — Let (X, d_X) and (Y, d_Y) be metric spaces. Assume that Y is complete. Show that if $E \subset X$ and $f: E \rightarrow Y$ is a uniformly continuous function defined on a subset then there is a unique continuous extension $\bar{f}: \bar{E} \rightarrow Y$, which is also uniformly continuous.

9.2.3 Banach's Fixed-Point Theorem

Fixed-point theorems investigate when a mapping $f: X \rightarrow X$ possess a fixed point, i.e., a point $x \in X$ for which $f(x) = x$. They can be rather powerful tools to prove existence theorems.

We prove in this section Banach's Fixed-Point Theorem, which will be used later to prove the Implicit Function Theorem ?? and the existence of solutions to ODEs.

THEOREM 9.56: BANACH'S FIXED-POINT THEOREM

Let (X, d) be a **complete** metric space and let $T: X \rightarrow X$ be a Lipschitz map with Lipschitz constant $\lambda < 1$. In other words, assume that for some $\lambda \in (0, 1)$ it holds

$$d(T(x), T(x')) \leq \lambda d(x, x') \quad \text{for all } x, x' \in X.$$

Then, there **exists a unique** element $a \in X$ such that $T(a) = a$.

The function T is a **Lipschitz contraction**. A point $x \in X$ with $T(x) = x$ is called a **fixed point** of the mapping T , and the theorem states that a Lipschitz contraction has a unique fixed point, provided the ambient space is complete (as in Definition ??).

Proof. First, we show uniqueness of a putative fixed point. Let $a \in X$ and $b \in X$ be fixed points of T . Then,

$$d(a, b) = d(T(a), T(b)) \leq \lambda d(a, b),$$

which, since $\lambda < 1$, implies $d(a, b) = 0$ and thus $a = b$.

We turn to prove the existence of a fixed point. Choose any $x_0 \in X$ and define a sequence $(x_n)_{n=0}^\infty$ recursively by $x_{n+1} = T(x_n)$, for $n \geq 0$. We claim that the sequence $(x_n)_{n=0}^\infty$ is a Cauchy sequence. Iterating the contractivity assumption we find that, for all integers $p \geq 0$,

$$\begin{aligned} d(x_{n+1}, x_n) &= d(T(x_n), T(x_{n-1})) \leq \lambda d(x_n, x_{n-1}) \\ &\leq \lambda^2 d(x_{n-1}, x_{n-2}) \leq \dots \leq \lambda^n d(x_1, x_0). \end{aligned}$$

Pick now any integers $m \geq n \geq N$, then using this observation and the triangular inequality we find

$$\begin{aligned} d(x_n, x_m) &\leq \sum_{p=n}^{m-1} d(x_p, x_{p+1}) = \sum_{p=n}^{m-1} d(T^p(x_0), T^p(x_1)) \\ &\leq \sum_{p=n}^{m-1} \lambda^p d(x_0, x_1) \leq d(x_0, x_1) \sum_{p \geq N}^\infty \lambda^p = \frac{\lambda^N}{1 - \lambda} d(x_0, x_1). \end{aligned}$$

We crucially used that $\lambda < 1$ to sum the geometric series. Now given any $\varepsilon > 0$ we can find some N so large that $\frac{\lambda^N}{1 - \lambda} d(x_0, x_1) < \varepsilon$, thus proving that $(x_n)_{n=0}^\infty$ is Cauchy (this estimate is uniform in n, m as long as they are larger than N !).

Now we use the completeness assumption to infer that $x_n \rightarrow a$ for some $a \in X$. Since T is continuous, we have

$$T(a) = \lim_{n \rightarrow \infty} T(x_n) = \lim_{n \rightarrow \infty} x_{n+1} = \lim_{n \rightarrow \infty} x_n = a$$

which shows that a is a fixed point of T . □

9.57. — We remark that the proof is constructive and in concrete situation can be implemented in an algorithm to find approximate fixed points.

EXERCISE 9.58. — Find examples for:

- (1) A Lipschitz contraction $T : X \rightarrow X$ on a non-complete metric space X that has no fixed point.
- (2) A complete metric space (X, d) and an isometry (i.e., a mapping $T : X \rightarrow X$ with $d(T(x_1), T(x_2)) = d(x_1, x_2)$) that has no fixed point, and an isometry that has exactly 13 fixed points.

9.2.4 Compactness

A closed and bounded interval of the real line is called a **compact** interval, as we saw in Analysis I. We proved some fundamental properties of continuous functions on compact intervals: boundedness, existence of maxima and minima, and uniform continuity. We intend to investigate these and other properties in the broader context of metric spaces. We start giving a general definition of compactness that works in metric spaces.

Let us immediately clarify a possible source of confusion.

ACHTUNG! 9.59: CLOSED & BOUNDED VS COMPACT

- In a general metric space, it is **not** necessarily true that a closed and bounded set is compact.
- Nevertheless in $\mathbb{R}^n$, which is the main focus of this course, it will turn out that a closed and bounded set **is** indeed compact, and viceversa.

INTERLUDE: “COVER”

Let $E \subset X$ and let $\mathcal{U} = \{U_i\}_{i \in I}$ be a family of subsets of X , where I is some set of indices. We say that $\mathcal{U}$ **covers** E if

$$E \subset \bigcup_{U \in \mathcal{U}} U = \bigcup_{i \in I} U_i.$$

Any family $\mathcal{V} \subset \mathcal{U}$ that still covers X is called a **subcover** of $\mathcal{U}$.

DEFINITION 9.60: COMPACTNESS

A metric space (X, d) is **compact** if one of the following equivalent conditions hold:

- (1) X is **topologically compact**: every family of open sets $\{U_i\}_{i \in I}$ that cover X , has a finite sub-family that still covers X .
- (2) X is **sequentially compact**: every sequence $(x_n)_{n \in \mathbb{N}}$ in X has a subsequence that is convergent in X .
- (3) X is **complete** and **totally bounded**: for every $r > 0$, there exist finitely many $x_1, \dots, x_n \in X$ such that the balls $B(x_1, r), \dots, B(x_n, r)$ cover X .

A subset $E \subset X$ is **compact in X** if the metric space $(E, d|_{E \times E})$ is compact.

9.61. — The definition of topological compactness does not explicitly use the distance function d , but it is formulated only in terms of the collection of open sets (i.e., the topology). For this reason it is called “topological”.

9.62. — The Bolzano-Weierstrass Theorem ensures that a closed and bounded interval of $\mathbb{R}$ is sequentially compact.

EXAMPLE 9.63. — $\mathbb{Q} \cap [0, 2]$, endowed with the standard distance, is not topologically compact. Consider the covering

$$\mathbb{Q} \cap [0, 2] = (\mathbb{Q} \cap [0, \sqrt{2})) \cup \bigcup_{p \in \mathbb{Q}, p > \sqrt{2}} (\mathbb{Q} \cap (p, 2]).$$

Any finite subcover will miss some rationals $> \sqrt{2}$.

EXERCISE 9.64. — Let X be a metric space. Show that if X is totally bounded, then it is bounded, i.e., $\sup_{x, x' \in X} d(x, x') < +\infty$.

EXAMPLE 9.65. — The half-open interval $X = (0, 1] \subset \mathbb{R}$ is not compact. Indeed, the open cover $\mathcal{U} = \{(2^{-n}, 1] \mid n \in \mathbb{N}\}$ has no finite subcover.

The main result of this section is Theorem 9.67, which shows that the definition of compact metric space is indeed well-posed.

EXERCISE 9.66. — Show that a totally bounded metric space is bounded, and find an example of a bounded metric space that is not totally bounded.

THEOREM 9.67: THE THREE FACES OF COMPACTNESS

Let (X, d) be a metric space, the following statements are equivalent:

- (1) *X is topologically compact.*
- (2) *X is sequentially compact.*
- (3) *X is totally bounded and complete (in the sense of Cauchy sequences).*

We will prove that $(1) \implies (3) \implies (2) \implies (1)$.

In order to prove the first of these implications we single out a rephrasing of (1) which is sometimes useful to keep in mind.

LEMMA 9.68: NESTING PRINCIPLE

(X, d) is topologically compact if and only if has the following property. For every collection $\mathcal{A}$ of closed subsets of X , it holds:

$$\text{Every intersection of finitely many elements of } \mathcal{A} \text{ is non-empty} \implies \bigcap_{A \in \mathcal{A}} A \neq \emptyset.$$

Proof of Lemma 9.68. Assume X is compact, and let $\mathcal{A}$ be a collection of closed subsets of X with an empty intersection. The collection of complements $\mathcal{U} = \{X \setminus A \mid A \in \mathcal{A}\}$ is then an

open cover of X , and there exists a finite subcover $X = U_1 \cup \dots \cup U_n$ of it. Set $A_i = X \setminus U_i$. Then $A_1 \cap \dots \cap A_n \neq \emptyset$. Thus, X satisfies the Nesting Principle.

Now, assume X satisfies the Nesting Principle, and show that X is compact. Let $\mathcal{U}$ be an open cover of X . Then $\mathcal{A} = \{X \setminus U \mid U \in \mathcal{U}\}$ is a collection of closed subsets with an empty intersection. According to the Nesting Principle, there exist $A_1, \dots, A_n \in \mathcal{A}$ with an empty intersection. Set $U_i = X \setminus A_i$. Then $X = U_1 \cup \dots \cup U_n$ is a finite subcover. Since the cover $\mathcal{U}$ was arbitrary, this shows that X is compact. $\square$

Proof that (1) $\implies$ (3). We first prove that X is totally bounded. Pick any $r > 0$, consider the open covering $\mathcal{U} := \{B(x, r) : x \in X\}$ and extract a finite subcover $\{B(x_1, r), \dots, B(x_N, r)\}$.

Now we prove that X must be complete, hence we pick a Cauchy sequence $(x_n)_{n \in \mathbb{N}}$ and show that it has a limit point. For each $k \geq 0$ there is $n(k)$ so large that

$$n, m \geq n(k) \implies d(x_n, x_m) < 2^{-k}.$$

For each k , consider the closed balls $A_k := \overline{B}(x_{n(k)}, 2^{-k})$; any finite intersection of them is nonempty, indeed for every $k_1, \dots, k_N$ one has that $x_m \in A_{k_1} \cap \dots \cap A_{k_N}$, provided $m \geq \max\{k_1, \dots, k_N\}$. Hence we can apply the Nesting Principle (see Proposition 9.68) and find some $z \in \bigcap_{k \geq 0} A_k$. We claim that $x_n \rightarrow z$. Indeed if $m \geq n(k)$ it holds

$$d(x_m, z) \leq d(x_m, x_{n(k)}) + d(x_{n(k)}, z) \leq 2^{1-k},$$

and 2^{1-k} is arbitrarily small. $\square$

Before continuing the proof we need an auxiliary Lemma

LEMMA 9.69: DIAGONAL ARGUMENT

Let $\mathbb{N} \supset N_0 \supset N_1 \supset N_2 \supset \dots$ be an infinite family of nested sets. Assume further that each N_k has infinite many elements. Then there exists $f: \mathbb{N} \rightarrow \mathbb{N}$ strictly increasing such that $f(k) \in N_k$ for all $k \geq 0$.

Proof of Lemma 9.69. Set $f(0)$ equal to an arbitrary element of N_0 . Then set inductively $f(k) := \min\{m \in N_k : m > f(k-1)\}$, this set is non-empty because each N_k is infinite. $\square$

Proof that (3) $\implies$ (2). We pick any sequence $(x_n)_{n \in \mathbb{N}}$ and show that admits a Cauchy subsequence, by completeness, this will prove (2).

By assumption, we can cover X by a finite number of balls of radius 1; it follows that $(x_n)_{n \in \mathbb{N}}$ will frequently lie in one of these balls, let this ball be $B(z_1, 1)$ for some $z_1 \in X$. Accordingly, we define the set of indices $N_0 := \{j \in \mathbb{N} : x_j \in B(z_1, 1)\}$, which is infinite.

Now we proceed to do same thing to the restricted sequence $(x_n)_{n \in N_0}$, but we shorten the radius of from 1 to $1/2$. Accordingly, we find a ball $B(z_2, 1/2)$ such that the set $N_1 := \{j \in N_0 : x_j \in B(z_2, 1/2)\}$ is infinite.

We proceed inductively, halving the radius each time, and construct a descending family of infinite sets $\mathbb{N} \supset N_0 \supset N_1 \supset N_2 \supset \dots$ with the property that

$$\forall k \geq 0, \exists z_k \in X, j \in N_k \implies d(z_k, x_j) < 2^{-k}.$$

We apply to these sets Lemma 9.69 and find $f: \mathbb{N} \rightarrow \mathbb{N}$ strictly increasing such that $f(k) \in N_k$ for all $k \geq 0$. Then, the subsequence $(x_{f(k)})_{k \in \mathbb{N}}$ is Cauchy: for $n, m \geq k$ it holds

$$\begin{aligned} f(n) \in N_n, f(m) \in N_m &\implies f(n), f(m) \in N_k \\ &\implies x_{f(n)}, x_{f(m)} \in B(z_k, 2^{-k}) \implies d(x_{f(n)}, x_{f(m)}) < 2^{1-k}. \end{aligned}$$

□

We conclude the chain of implications thus proving Theorem 9.67.

Proof that (2) $\implies$ (1). Assume $\mathcal{U}$ is an open covering of X , we want to construct a finite open subcover out of it.

We start with a preliminary observation. Consider the function

$$r(x) := 1 \wedge \sup\{r > 0 : B(x, r) \subset U \text{ for some } U \in \mathcal{U}\}, \text{ for all } x \in X.$$

Notice that $r(x) > 0$ since $\mathcal{U}$ is open cover. For each $x \in X$ we choose — once and for all — some $U(x) \in \mathcal{U}$ which is almost optimal in the sense that $B(x, r(x)/2) \subset U(x)$.

Let us now proceed with the construction of our finite subcover. Pick any $U_0 \in \mathcal{U}$. If $X \subset U_0$ then we are done, otherwise there is some $x_1 \in X \setminus U_0$. In this case we set $U_1 := U(x_1)$.

Now we check if $X \subset U_0 \cup U_1$, in which case we have found our finite subcover. If not, there is some $x_2 \in X \setminus (U_0 \cup U_1)$ and we set $U_2 := U(x_2)$.

Now we check again if $X \subset U_0 \cup U_1 \cup U_2$, in which case we have found our finite subcover. If not, there is some $x_3 \in X \setminus (U_0 \cup U_1 \cup U_2)$ and we set $U_3 := U(x_3)$.

If this procedure stops at a certain point, it means that we have found our finite open subcover. So let's assume that it goes on indefinitely, and find a contradiction. We obtain a sequence $(x_n)_{n \in \mathbb{N}}$ with the property that

$$x_m \notin U_0 \cup \dots \cup U_n \text{ for all } m > n \geq 0.$$

By assumption (2), we have $x_{f(n)} \rightarrow z$ for a suitable subsequence. On the other hand, for each $n \geq 0$ we have $z \notin U_{f(n)}$, and in particular $z \notin B(x_{f(n)}, r(x_{f(n)})/2)$. Combining this information with $x_{f(n)} \rightarrow z$, we find $r(x_{f(n)}) \rightarrow 0$ and in particular

$$r(x_{f(n)}) = \sup\{r > 0 : B(x_{f(n)}, r) \subset U \text{ for some } U \in \mathcal{U}\} \in (0, 1).$$

This is a contradiction as $r(z) > 0$, so for n large enough we would have

$$B(x_{f(n)}, 5r(x_{f(n)})) \subset B(z, r(z)/2) \subset U(z) \in \mathcal{U},$$

which, by definition of $r(x_{f(n)})$, implies $5r(x_{f(n)}) \leq r(x_{f(n)})$, impossible. $\square$

COROLLARY 9.70: PRODUCT OF COMPACTS IS COMPACT

Let (X, d_X) and (Y, d_Y) be two compact metric spaces. Endow $X \times Y$ with the distance $d_{X \times Y}$ defined as

$$d_{X \times Y}((x, y), (x', y')) := d_X(x, x') + d_Y(y, y').$$

Then also $(X \times Y, d_{X \times Y})$ is compact.

Proof. Use the sequential characterization. **TODO: proof + add exercise in PS2** $\square$

COROLLARY 9.71: CLOSED IN COMPACT IS COMPACT

Let (X, d) be compact and let $E \subset X$ be closed. Then E is compact.

Proof. We check that E is sequentially compact: take any sequence $(x_n)_{n \in \mathbb{N}} \subset E$, by compactness of X , it has a converging subsequence $x_{f(n)} \rightarrow x$, for some $x \in X$. Since E is closed it contains its accumulation points, so in fact $x \in E$. $\square$

COROLLARY 9.72: COMPACT IS ALWAYS CLOSED

If E is compact in (X, d) , then E is closed.

Proof. Take any sequence in E which is convergent to some $x \in X$. By compactness a suitable subsequence is converging to some $x' \in E$, by uniqueness of the limit $x = x'$, so E is closed (see (2) in Lemma 9.46). $\square$

We also easily get the following version of the Heine Borel theorem in $\mathbb{R}^n$

THEOREM 9.73: COMPACT SUBSETS OF $\mathbb{R}^n$ (HEINE-BOREL)

A subset $K \subset \mathbb{R}^n$ is compact if and only if it is closed and bounded.

Proof. If K is compact, then it is closed by Corollary 9.72; and bounded, because it is totally bounded.

To show the converse, we show that K is complete and totally bounded. Since $\mathbb{R}^n$ is complete and K is closed, then K is complete as well.

Given $r > 0$, take some large integer $N \geq 2n/r$ and consider all the closed cubes that have sidelength $1/N$, corners with coordinates of the form m/N for some $m \in \mathbb{Z}$ and intersect K . Such cubes are only a finite number, because K is bounded. Furthermore, each of them is contained in the ball of radius r and center one of the edges, thus we have found our finite set of balls that cover K . $\square$

EXAMPLE 9.74. — We stress that the Heine-Borel Theorem **fails** for general metric spaces: take $\mathbb{R}$ with the bounded distance $d(x, y) := \arctan|x - y|$ (see Exercise 9.7). Then in this metric space the set $\mathbb{N}$ would be closed and bounded, but it is trivial to construct sequences that do not converge.

EXERCISE 9.75. — Show that an open set $U \subset \mathbb{R}^n$ is complete if and only if $U = \mathbb{R}^n$.

9.2.5 Compactness and continuity

THEOREM 9.76: CONTINUOUS FUNCTIONS PRESERVE COMPACTNESS

Let X and Y be metric spaces, let $f : X \rightarrow Y$ be a continuous function, and let $K \subset X$ be a compact subset. Then, $f(K)$ is a compact subset of Y .

Proof. Let $\mathcal{U}$ be an open cover of $f(A)$. For each $U \in \mathcal{U}$, the set $f^{-1}(U) \subset X$ is open due to the continuity of f . The collection $\{f^{-1}(U) \mid U \in \mathcal{U}\}$ is an open cover of A . Since A is compact, there exist $U_1, \dots, U_n \in \mathcal{U}$ such that $\{f^{-1}(U_i) \mid 1 \leq i \leq n\}$ is a cover of A . This implies that $\{U_i \mid 1 \leq i \leq n\}$ is a cover of $f(A)$, and since $\mathcal{U}$ was arbitrary, it shows that $f(A)$ is compact. $\square$

PROPOSITION 9.77: CONTINUOUS IN A COMPACT IS UNIFORMLY CONTINUOUS

Let X and Y be metric spaces, and $f : X \rightarrow Y$ a continuous function. If X is compact, then f is uniformly continuous.

Proof. Let $\varepsilon > 0$. Due to the continuity of f , for each $x \in X$, there exists $\delta_x > 0$ such that $f(B(x, \delta_x)) \subset B(f(x), \frac{\varepsilon}{2})$. The collection $\{B(x, \frac{1}{2}\delta_x) \mid x \in X\}$ forms an open cover of X . Since X is compact by assumption, there exists a finite subcover of this collection. This implies the existence of $x_1, \dots, x_n \in X$ such that

$$X = B(x_1, \frac{1}{2}\delta_{x_1}) \cup \dots \cup B(x_n, \frac{1}{2}\delta_{x_n}).$$

Let $\delta = \frac{1}{2} \min\{\delta_{x_1}, \dots, \delta_{x_n}\}$. For $x, x' \in X$ with $d(x, x') < \delta$, there exists k such that $x \in B(x_k, \frac{1}{2}\delta_{x_k})$. This implies $x' \in B(x_k, \delta_{x_k})$, leading to

$$d(f(x), f(x')) \leq d(f(x), f(x_k)) + d(f(x_k), f(x')) \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

which completes the proof. $\square$

COROLLARY 9.78: WEIERSTRASS

Let X be a compact metric space and $f : X \rightarrow \mathbb{R}$ be a continuous function, then f admits a maximum point, i.e., there exist $\bar{x} \in X$ such that $f(\bar{x}) = \sup_X f$. An analogous statement holds for the minimum. In particular, f must be bounded.

Proof. $f(X) \subset \mathbb{R}$ is compact by Theorem 9.76 and nonempty, so $\sup f(X) \in f(X)$, any element $\bar{x} \in f^{-1}(\sup f(X))$ works.

The fact that $\sup f(X) \in f(X)$ is readily proved: by definition of supremum there is a sequence $(s_n) \subset f(X)$ such that $s_n \rightarrow \sup f(X)$, but then $\sup f(X) \in f(X)$ since $f(X)$ is closed and so it contains its accumulation points. $\square$

9.2.6 Connectedness

DEFINITION 9.79: CONNECTEDNESS

Let (X, d) be a metric space. We call X **connected** if, apart from $\emptyset \subset X$ and $X \subset X$, there are no subsets of X that are both open and closed.

A subset $E \subset X$ is called **connected in X** if $(E, d|_{E \times E})$ is connected as an independent metric space.

A subset $Y \subset X$ is called a **connected component of X** if Y is non-empty, open, closed, and connected.

LEMMA 9.80:

Let (X, d) be a metric space, and let Y_1 and Y_2 be connected subspaces. If the intersection $Y_1 \cap Y_2$ is non-empty, then the union $Y_1 \cup Y_2$ is connected.

Proof. Let A be a non-empty open and closed subset of $Y_1 \cup Y_2$. Then, $A \cap Y_j$ is an open and closed subset of Y_j for $j = 1, 2$. Since A is non-empty, one of these intersections is non-empty; let's say $A \cap Y_1$ is non-empty. As Y_1 is connected, we have $A \cap Y_1 = Y_1$, implying $Y_1 \subset A$. Since $Y_1 \cap Y_2 \neq \emptyset$, we have $A \cap Y_2 \neq \emptyset$, and similarly, $Y_2 \subset A$. In summary, $A = Y_1 \cup Y_2$. This proves that $Y_1 \cup Y_2$ is connected. $\square$

EXERCISE 9.81. — Generalize Lemma 9.80 for arbitrary unions.

PROPOSITION 9.82: CONNECTED SUBSETS OF $\mathbb{R}$

A non-empty subset $X \subset \mathbb{R}$ is connected in $\mathbb{R}$ if and only if X is an interval.

Proof. Assume $X \subset \mathbb{R}$ is not an interval. Then there exist real numbers $x_1 < y < x_2$ with $x_1, x_2 \in X$ and $y \notin X$. Define the subsets

$$U_1 = (-\infty, y) \cap X = (-\infty, y] \cap X \quad \text{and} \quad U_2 = (y, \infty) \cap X = [y, \infty) \cap X.$$

These sets are open and closed (as subsets of X , not in $\mathbb{R}$!), non-empty, and satisfy $X = U_1 \cup U_2$. Thus, X is not connected.

Now, let $X \subset \mathbb{R}$ be an interval. Assume there exist non-empty, open, and closed subsets $Y_1 \subset X$ and $Y_2 \subset X$, with $Y_1 \cap Y_2 = \emptyset$ and $Y_1 \cup Y_2 = X$. Choose $x_1 \in Y_1$ and $x_2 \in Y_2$.

Without loss of generality, assume $x_1 = 0$ and $x_2 = 1$. Since X is an interval, it follows that $[0, 1] \subset X = Y_1 \cup Y_2$. Now let us define the “magic” number

$$t_* := \sup\{t \geq 0 : [0, t] \subset Y_1\},$$

since $1 \in Y_2$ we have $t_* \in [0, 1]$, so t_* must belong either to Y_1 or to Y_2 , but both eventualities are contradictory.

If $t_* \in Y_1$ then $t_* < 1$ and, since Y_1 is open, we could enlarge a bit t_* violating its very definition.

If $t_* \in Y_2$, since Y_2 is open, it would mean that a whole little neighbourhood of t_* is contained in Y_2 , but this is impossible because t_* must be an accumulation point of Y_1 . $\square$

PROPOSITION 9.83:

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \rightarrow Y$ be continuous. If X is connected, then the image $f(X)$ is a connected subspace of Y .

Proof. Without loss of generality, assume that $f : X \rightarrow Y$ is surjective, replacing $f : X \rightarrow Y$ by the continuous surjective map $X \rightarrow f(X)$, $x \mapsto f(x)$ if necessary. Suppose Y is not connected. Then there exists an open and closed subset $A \subset Y$ with $A \neq \emptyset$ and $A \neq Y$. The preimage $f^{-1}(A)$ of A under f is also open and closed due to the continuity of f . Since X is connected, either $f^{-1}(A) = \emptyset$ or $f^{-1}(A) = X$. However, this implies $A = \emptyset$ or $A = Y$ due to the surjectivity of f , contradicting the assumption on A . Therefore, $f(X)$ is connected. $\square$

COROLLARY 9.84: INTERMEDIATE VALUE THEOREM

Let $I \subset \mathbb{R}$ be an interval, $f : I \rightarrow \mathbb{R}$ a continuous function, and $a, b \in I$. For every $c \in \mathbb{R}$ between $f(a)$ and $f(b)$, there exists an $x \in I$ between a and b such that $f(x) = c$.

Proof. Without loss of generality, we can assume $a < b$. We apply Proposition 9.83 to the continuous function $f|_{[a,b]} : [a, b] \rightarrow \mathbb{R}$. Consequently, $f([a, b])$ is connected, since by Proposition 9.82, the interval $[a, b]$ in $\mathbb{R}$ is connected. Again, according to Proposition 9.82, $f([a, b]) \subset \mathbb{R}$ must be an interval. As $f(a), f(b) \in f([a, b])$, all values between $f(a)$ and $f(b)$ lie in the image of $f|_{[a,b]}$. $\square$

EXERCISE 9.85. — Show the following generalization of the Intermediate Value Theorem: Let X be a connected topological space, and $f : X \rightarrow \mathbb{R}$ be a continuous function. Let $a, b \in X$. Then, for every $c \in \mathbb{R}$ between $f(a)$ and $f(b)$, there exists $x \in X$ such that $f(x) = c$.

INTERLUDE: PATHS AND CURVES

Let X be a metric space. A **path** or **curve** in X is a continuous function $\gamma : [0, 1] \rightarrow X$. We call $\gamma(0)$ the **starting point** and $\gamma(1)$ the **ending point**. We also say that γ is a path from $\gamma(a)$ to $\gamma(b)$. A path γ with $\gamma(0) = \gamma(1)$ is called **closed** or a **loop**.

If $s: [a, b] \rightarrow [0, 1]$ is a bijective continuous function with continuous inverse we say that $\gamma \circ s$ is a **re-parametrization** of γ . Furthermore, exactly one of the following happens

- either, $s(a) = 0, s(b) = 1$, then we say that s is **orientation-preserving**,
- or, $s(a) = 1, s(b) = 0$, then we say that s is **orientation-reversing**.

DEFINITION 9.86:

We call a topological space X **path-connected** if, for every two points $x, y \in X$, there exists a path $\gamma: [0, 1] \rightarrow X$ from $x = \gamma(0)$ to $y = \gamma(1)$.

LEMMA 9.87:

Every path-connected topological space is connected.

Proof. Let X be a disconnected topological space. Then there exist non-empty, open, and closed subsets U_1 and U_2 of X such that $U_1 \cap U_2 = \emptyset$ and $U_1 \cup U_2 = X$. Let $x_1 \in U_1$ and $x_2 \in U_2$. If X were path-connected, there would exist a path $\gamma: [0, 1] \rightarrow X$ from x_1 to x_2 . However, this implies that $V_1 = \gamma^{-1}(U_1)$ and $V_2 = \gamma^{-1}(U_2)$ are non-empty, open, and closed subsets of $[0, 1]$ with $V_1 \cap V_2 = \emptyset$ and $V_1 \cup V_2 = [0, 1]$, which is a contradiction since $[0, 1]$ is connected. $\square$

EXERCISE 9.88. — Sketch the subspace $X \subset \mathbb{R}^2$ given by

$$X = \{0\} \times [-1, 1] \cup \{(t, \sin(\frac{1}{t})) \mid t > 0\}$$

and show that X is connected but not path-connected.

PROPOSITION 9.89:

Let $U \subset \mathbb{R}^n$ be an open subset. Then U is path-connected if and only if U is connected.

Proof. If U is path-connected, then U is also connected according to Lemma 9.87. Now, assume U is connected, non-empty, and $x_0 \in U$ is a fixed point. We define the set

$$G = \{x \in U \mid \text{there exists a path in } U \text{ from } x_0 \text{ to } x\}$$

and want to show that $G = U$. Since U is connected and G is non-empty, it suffices to show that G is both open and closed.

Let $x \in G$ and $\gamma: [0, 1] \rightarrow U$ be a path from x_0 to x . Since U is open, there exists $r > 0$ such that $B(x, r) \subset U$. For any $y \in B(x, r)$, the straight path $t \mapsto (1 - t)x + ty$, connecting x and y , lies in U . Concatenating these paths yields the path

$$t \mapsto \begin{cases} \gamma(2t) & \text{if } 0 \leq t \leq \frac{1}{2} \\ (2 - 2t)x + (2t - 1)y & \text{if } \frac{1}{2} < t \leq 1 \end{cases}$$

from x_0 to y . Thus, $y \in G$, and since y was arbitrary, we have $B(x, r) \subset G$. This shows that G is open. Using a similar argument, we can show that $U \setminus G$ is open. If $x \notin G$ and $r > 0$ with $B(x, r) \subset U$, then all points in $B(x, r)$ are not in G . If $y \in G \cap B(x, r)$, a concatenation of paths as above would connect x to x_0 . Therefore, $B(x, r) \subset U \setminus G$, and $U \setminus G$ is open. Thus, G is closed. □

COROLLARY 9.90: $\mathbb{R}^n$ AND ITS BALLS ARE CONNECTED

For all $n \geq 1$ and $r > 0$, the metric space $\mathbb{R}^n$ (with the standard distance) and the subsets $B(x, r)$ and $\overline{B(x, r)}$ of $\mathbb{R}^n$ are connected.

9.3 Normed vector spaces

9.3.1 Definition of Normed Vector spaces

A **norm** on a real vector space V is a function from V to $\mathbb{R}$ that assigns to each vector a non-negative number, informally its **length**. In general there are many different norms on a vector space, and we can use any of them to construct a distance and turn V into a metric space. A particularly interesting class of norms is obtained from **scalar products**. Many notions of this sections would carry out to vector spaces over $\mathbb{C}$ with little modifications, but we will stick to real vector spaces.

DEFINITION 9.91:

Let V be a vector space over $\mathbb{R}$. A **norm** on V is a mapping $\|\cdot\| : V \rightarrow [0, \infty)$ that satisfies the following three properties.

- (1) (Definiteness) For all $v \in V$, $\|v\| = 0 \iff v = 0$.
- (2) (Homogeneity) For all $v \in V$ and all $\alpha \in \mathbb{R}$, $\|\alpha v\| = |\alpha|\|v\|$.
- (3) (Triangle Inequality) For all $v, w \in V$, $\|v + w\| \leq \|v\| + \|w\|$.

The pair $(V, \|\cdot\|)$ is called a **normed vector space**.

EXAMPLE 9.92. — Let $n \in \mathbb{N}$. The **maximum norm** or **infinity norm** $\|\cdot\|_\infty$, and the **1-norm** $\|\cdot\|_1$ on $\mathbb{R}^n$ are defined by

$$\|v\|_\infty = \max\{|v_1|, |v_2|, \dots, |v_n|\} \quad \text{and} \quad \|v\|_1 = \sum_{j=1}^n |v_j|$$

for $v = (v_1, \dots, v_n) \in \mathbb{R}^n$. The properties of definiteness and homogeneity, as well as the triangle inequality, can be verified by exercise.

EXAMPLE 9.93. — If V is the vector space of continuous $\mathbb{R}$ -valued functions on $[0, 1]$, we define analogously the 1-norm and the infinity norm as

$$\|f\|_1 = \int_0^1 |f| dx \quad \text{and} \quad \|f\|_\infty = \sup\{|f(x)| \mid x \in [0, 1]\}.$$

We immediately observe that Normed Vector Spaces are “automatically” Metric Spaces.

LEMMA 9.94: NORMED VECTOR SPACES ARE NATURALLY METRIC SPACES

Let V be a vector space over $\mathbb{R}$ and $\|\cdot\|$ be a norm on V . Define the function

$$d: V \times V \rightarrow [0, \infty), \quad d(v, w) := \|v - w\|,$$

then (V, d) is a metric space.

Proof. We check definiteness, symmetry, and the triangle inequality in the definition of a metric 9.3. For $v, w \in V$, we have $d(v, w) = \|v - w\| \geq 0$, and

$$d(v, w) = 0 \iff \|v - w\| = 0 \iff v - w = 0 \iff v = w$$

by the definiteness of the norm. Using homogeneity of the norm for $\alpha = -1$, we have for $v, w \in V$,

$$d(v, w) = \|v - w\| = \|(-1)(v - w)\| = \|w - v\| = d(w, v)$$

thus establishing the symmetry of d . Finally, using the triangle inequality of the norm, we obtain

$$d(u, w) = \|u - w\| = \|(u - v) + (v - w)\| \leq \|u - v\| + \|v - w\| = d(u, v) + d(v, w)$$

for all $u, v, w \in V$. This shows the triangle inequality for d , so d is indeed a metric on V . $\square$

We have seen that a normed space is naturally a metric space, now we check that the norm is indeed continuous. We show sequential continuity.

LEMMA 9.95: THE NORM IS CONTINUOUS WITH RESPECT TO ITS OWN DISTANCE

Let V be a $\mathbb{R}$ -vector space, and let $\|\cdot\|$ be a norm on V . Let $(v_n)_{n=0}^\infty$ be a sequence in V converging with respect to the norm $\|\cdot\|$ to a limit $w \in V$. Then,

$$\lim_{n \rightarrow \infty} \|v_n\| = \|w\|.$$

Proof. Let $\varepsilon > 0$. Since $(v_n)_{n=0}^\infty$ converges to w , there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, the estimate $\|v_n - w\| < \varepsilon$ holds. The sequence $(\|v_n - w\|)_{n=0}^\infty$ converges to 0. Using the triangle inequality, we get

$$\|w\| - \|v_n - w\| \leq \|v_n\| \leq \|v_n - w\| + \|w\|$$

and the lemma follows from the sandwich lemma for sequences of real numbers. $\square$

9.3.2 Inner product spaces

DEFINITION 9.96:

Let V be a vector space over $\mathbb{R}$. An **inner product** on V is a map

$$\langle -, - \rangle : V \times V \rightarrow \mathbb{R}$$

that satisfies the following properties for all $u, v, w \in V$ and $\alpha, \beta \in \mathbb{R}$:

- (1) (Bilinearity) $\langle \alpha u + \beta v, w \rangle = \alpha \langle u, w \rangle + \beta \langle v, w \rangle$.
- (2) (Symmetry) $\langle v, w \rangle = \langle w, v \rangle$.
- (3) (Definiteness) $\langle v, v \rangle \geq 0$ and $\langle v, v \rangle = 0 \iff v = 0$.

9.97. — An important example of an inner product is the **Euclidean inner product** or **standard inner product** on $\mathbb{R}^d$. It is given by

$$\langle -, - \rangle : V \times V \rightarrow \mathbb{R} \quad \langle v, w \rangle = \sum_{k=1}^d v_k \overline{w_k}$$

for $v = (v_1, \dots, v_d)$ and $w = (w_1, \dots, w_d)$. The proof of bilinearity and symmetry is left as an exercise. We verify definiteness. Let $v = (v_1, \dots, v_d) \in \mathbb{R}^d$. Then,

$$\langle v, v \rangle = \sum_{k=1}^d v_k v_k = \sum_{k=1}^d |v_k|^2 \geq 0$$

is a non-negative real number. If $v = 0$, then $\langle v, v \rangle = 0$. If $\langle v, v \rangle = 0$, then each term $|v_k|^2$ must be zero, and thus $v_k = 0$ for all k , implying $v = 0$.

PROPOSITION 9.98: CAUCHY-SCHWARZ INEQUALITY

Let V be a vector space over $\mathbb{R}$, let $\langle -, - \rangle$ be an inner product on V , and let $\|\cdot\| : V \rightarrow \mathbb{R}$ be given by $\|v\| = \sqrt{\langle v, v \rangle}$. Then the inequality holds

$$|\langle v, w \rangle| \leq \|v\| \|w\| \tag{9.2}$$

for all $v, w \in V$. Furthermore, equality in (9.2) holds if and only if v and w are linearly dependent.

Proof. If $v = 0$ or $w = 0$, then both sides of (9.2) are zero, and the vectors v, w are linearly dependent. So, we assume that $v \neq 0$ and $w \neq 0$. Then, for $\alpha = \langle v, w \rangle \|w\|^{-2}$, we have

$$\begin{aligned} \|v - \alpha w\|^2 &= \langle v - \alpha w, v - \alpha w \rangle = \langle v, v - \alpha w \rangle - \alpha \langle w, v - \alpha w \rangle \\ &= \langle v, v \rangle - \overline{\alpha} \langle v, w \rangle - \alpha \langle w, v \rangle + \alpha \overline{\alpha} \langle w, w \rangle = \|v\|^2 - \overline{\alpha} \langle v, w \rangle - \alpha \overline{\langle v, w \rangle} + |\alpha|^2 \|w\|^2 \end{aligned}$$

$$= \|v\|^2 - 2 \frac{|\langle v, w \rangle|^2}{\|w\|^2} + \frac{|\langle v, w \rangle|^2}{\|w\|^4} \|w\|^2 = \|v\|^2 - \frac{|\langle v, w \rangle|^2}{\|w\|^2}.$$

The real number $\|v - \alpha w\|^2$ is non-negative, and it follows

$$\|v\|^2 \|w\|^2 \geq |\langle v, w \rangle|^2$$

implying the desired inequality (9.2). Equality holds if and only if $\|v - \alpha w\| = 0$, implying $v = \alpha w$. $\square$

EXERCISE 9.99. — Prove the Cauchy-Schwarz inequality by following these steps: Let $a > 0$. Show that for all $v, w \in \mathbb{R}^n$, the inequality

$$|\langle v, w \rangle| \leq \frac{a^2}{2} \|v\|^2 + \frac{1}{2a^2} \|w\|^2$$

holds and conclude from it the Cauchy-Schwarz inequality.

We prove that the norm induced by an inner product is indeed a norm on V .

COROLLARY 9.100:

Let V be a vector space over $\mathbb{R}$, let $\langle -, - \rangle$ be an inner product on V . The map defined by (9.3)

$$\| \cdot \| : V \rightarrow \mathbb{R}, \quad \|v\| = \sqrt{\langle v, v \rangle}$$

satisfies the triangular inequality and is a norm.

Proof. Definiteness and homogeneity follow directly from the definiteness and sesquilinearity of the inner product. We only need to prove the triangle inequality. Let $v, w \in V$. Using the Cauchy-Schwarz inequality, we have the estimate

$$\begin{aligned} \|v + w\|^2 &= \langle v + w, v + w \rangle = \|v\|^2 + \langle v, w \rangle + \langle w, v \rangle + \|w\|^2 \\ &= \|v\|^2 + 2 \operatorname{Re}(\langle v, w \rangle) + \|w\|^2 \leq \|v\|^2 + 2|\langle v, w \rangle| + \|w\|^2 \\ &\leq \|v\|^2 + 2\|v\|\|w\| + \|w\|^2 = (\|v\| + \|w\|)^2, \end{aligned}$$

which implies the desired result after taking the square root. $\square$

9.101. — Let V be a vector space over $\mathbb{R}$. If $\langle -, - \rangle$ is an inner product on V , we call the norm treated in Corollary 9.100

$$\| \cdot \| : V \rightarrow \mathbb{R}, \quad \|v\| = \sqrt{\langle v, v \rangle} \tag{9.3}$$

the **induced norm** by $\langle -, - \rangle$. In particular, from the Euclidean inner product on $V = \mathbb{R}^n$, we can define a norm on $V = \mathbb{R}^n$. The **Euclidean norm** on $V = \mathbb{R}^n$ is given by

$$\|x\| = \sqrt{\langle x, x \rangle} = \sqrt{\sum_{k=1}^n |v_k|^2}$$

for all $v = (v_1, \dots, v_n) \in \mathbb{R}^n$.

9.102. — From now on, in order to keep the notation, simple, we will denote the Euclidean norm of a vector x in $\mathbb{R}^n$ by $|x|$, instead of $\|x\|$, that we will reserve for (less standard) norms. Notice that this notation does not create any ambiguity or collision with previously introduced notations. Indeed:

- If $n = 1$ the Euclidean norm coincides with the absolute value.
- If $n = 2$ and we are identifying the $\mathbb{R}^2$ and $\mathbb{C}$ via the usual map $(x_1, x_2) \mapsto x_1 + ix_2$, then Euclidean norm coincides with the complex absolute value.

9.103. — The Euclidean norm on $\mathbb{R}^n$ holds a special position among all norms on $\mathbb{R}^n$. On $\mathbb{R}^2$ or $\mathbb{R}^3$, it measures the “physical” length of vectors. However, many other norms $\|\cdot\|$ confer $\mathbb{R}^n$ the structure of normed vector space. A standard family of norms is given by

$$\|x\|_p := \left(\sum_{i=1}^n |x_i|^p \right)^{1/p},$$

where $p \in [1, +\infty)$ is a given number. Notice that the Euclidean norm $|x|$ corresponds to the $p = 2$ case.

One can check (exercise) that for any given $x \in \mathbb{R}^n$

$$\lim_{p \rightarrow +\infty} \|x\|_p = \max_{1 \leq i \leq n} |x_i|$$

is also a norm. That is why the maximum norm is commonly called infinity norm and denoted $\|\cdot\|_\infty$.

9.3.3 Equivalence of norms in finite dimensional normed spaces

INTERLUDE: FINITE AND INFINITE DIMENSIONAL VECTOR SPACES

A **basis** for a vector space V is a subset $\mathcal{B} = \{e_i\}_{i \in I} \subset V$ such that

- every $v \in V$ can be written as $v = \sum_{i \in I} v_i e_i$ for some coefficients $\{v_i\}_{i \in I} \subset \mathbb{R}$, only finitely many of which are nonzero (so that the previous sum always makes sense, it is not a series!)

- the coefficients $\{v_i\}_{i \in I}$ are uniquely determined, in other words the following implication holds:

$$\sum_{i \in I} v_i e_i = 0, \text{ with finitely many non-zero } v_i \in \mathbb{R} \implies c_i = 0, \quad \forall i \in I.$$

For every vector space we can obtain a sequence of linearly independent vectors $e_1, e_2, e_3, \dots$. In if this sequence necessarily stops we obtain a finite basis and we say that the vector space is finite dimensional. If the sequence can be continued indefinitely we say that the vector space is infinite dimensional.

All bases of a finite dimensional vector space have the same number of vectors. This number is called the **dimension** of V .

EXERCISE 9.104. — Show that the space of polynomials with real coefficients $\mathbb{R}[x]$ is an infinite dimensional vector space and find a basis.

9.105. — If V has finite dimension $n \in \mathbb{N}$ and we fix a basis $\mathcal{B} = \{e_i\}_{1 \leq i \leq n}$ then map

$$\iota_{\mathcal{B}}(x_1, \dots, x_n) \mapsto x_1 e_1 + \dots + x_n e_n$$

is a (vector space) isomorphism and allows to treat V as $\mathbb{R}^n$ for most practical tasks. In particular, if $(V, \|\cdot\|)$ is a normed space then $\iota_{\mathcal{B}}$ induces a norm in $\mathbb{R}^n$ defined as

$$\|x\| = \|\iota_{\mathcal{B}}(x)\|.$$

This is a motivation to prove results for $\mathbb{R}^n$ equipped with norms different from the Euclidean one. Indeed, all the concepts and results that can be stated for general norms will automatically hold in “abstract” finite dimensional normed vector space. We see next an important instance of this.

DEFINITION 9.106: EQUIVALENT (I.E., COMPARABLE) NORMS

Let V be a vector space over $\mathbb{R}$ and $\|\cdot\|_1$ and $\|\cdot\|_2$ be two norms on V . We call $\|\cdot\|_1$ and $\|\cdot\|_2$ **equivalent** if there are constants $A > 0$ and $B > 0$ such that

$$\|v\|_1 \leq A\|v\|_2 \quad \text{and} \quad \|v\|_2 \leq B\|v\|_1 \quad \text{for all } v \in V.$$

EXAMPLE 9.107. — Let $n \in \mathbb{N}$. The 1-norm $\|\cdot\|_1$ and the maximum norm $\|\cdot\|_{\infty}$ given in Example 9.92 are equivalent, as the inequalities

$$\|v\|_{\infty} \leq \|v\|_1 \quad \text{and} \quad \|v\|_1 \leq n\|v\|_{\infty}$$

hold for all $v \in \mathbb{R}^n$. As we will show in Theorem 9.108, all norms on a finite-dimensional

vector space over $\mathbb{R}$ are equivalent to each other. This is not the case for infinite-dimensional vector spaces. For example, the norms given in 9.92 on the space of continuous functions on $[0, 1]$ are not equivalent.

THEOREM 9.108:

All norms on $\mathbb{R}^n$ are equivalent (i.e., comparable).

Proof. Let $\|\cdot\|$ be any norm on $\mathbb{R}^n$, and let $\|\cdot\|_1$ denote the 1-norm on $\mathbb{R}^n$ given in Example 9.92. We show that $\|\cdot\|$ and $\|\cdot\|_1$ are equivalent, which proves the Theorem.

Let $e_1, \dots, e_n$ denote the standard basis of $\mathbb{R}^n$, and let $A = \max\{\|e_1\|, \|e_2\|, \dots, \|e_n\|\}$. For any vector $v = x_1e_1 + \dots + x_ne_n \in V$, we have

$$\|v\| \leq \sum_{k=1}^n |x_k| \cdot \|e_k\| \leq A \cdot \sum_{k=1}^n |x_k| = A\|v\|_1$$

which already shows one of the two required estimates. For the second estimate, consider the set

$$S = \{v \in V \mid \|v\|_1 = 1\}$$

and the real number $B = \inf\{\|v\| \mid v \in S\}$. There exists a sequence $(v_n)_{n=0}^\infty$ in S such that the sequence $(\|v_n\|)_{n=0}^\infty$ converges to B . Since $(v_n)_{n=0}^\infty$ is bounded for the 1-norm, it contains a convergent subsequence by the Heine-Borel theorem. By replacing $(v_n)_{n=0}^\infty$ with such a subsequence, we can ensure that $(v_n)_{n=0}^\infty$ converges to $w \in V$ with respect to the 1-norm. For any $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that

$$n \geq N \implies \|w - v_n\|_1 \leq A^{-1}\varepsilon \implies \|w - v_n\| \leq \varepsilon$$

We deduce that the sequence $(v_n)_{n=0}^\infty$ also converges to w with respect to the norm $\|\cdot\|$. Thus, by Lemma 9.95,

$$\|w\|_1 = 1 \quad \text{and} \quad \|w\| = B,$$

and, in particular, $w \neq 0$ and $B > 0$. For any vector $v \neq 0$ in V , $\frac{1}{\|v\|_1}v$ is an element of S , and it satisfies

$$\frac{\|v\|}{\|v\|_1} = \left\| \frac{1}{\|v\|_1}v \right\| \geq B$$

Thus, $\|v\|_1 \leq B^{-1}\|v\|$ for all $v \neq 0$ in V . Since this estimate also holds for $v = 0$, it is shown that the norms $\|\cdot\|$ and $\|\cdot\|_1$ are equivalent. $\square$

9.109. — Notice that as a simple consequence of Theorem 9.108, in every finite dimensional vector space V all normed are equivalent. Indeed, we can fix a basis $\mathcal{B} = \{e_1, \dots, e_n\}$ of V use the inclusion map $\iota_{\mathcal{B}}$ defined in 9.105 to export the result for $\mathbb{R}^n$ to V .

9.3.4 The space of bounded continuous functions with values in $\mathbb{R}^m$

If (X, d) is a metric space we denote by $C(X, \mathbb{R}^m)$ the vector space of continuous functions defined on X . We denote by $C_b(X, \mathbb{R}^m)$ the vector space of continuous functions which are also bounded, that is, $\sup_{x \in X} |f| < +\infty$. By

PROPOSITION 9.110:

For all $f \in C_b(X, \mathbb{R})$ set $\|f\| := \sup_{x \in X} |f(x)|$. Then $(C_b(X, \mathbb{R}), \|\cdot\|)$ is a complete normed vector space. Furthermore, $f_n \rightarrow f$ in this space if and only if the functions (f_n) converge uniformly to f , meaning that

$$\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall x \in X, \forall n \geq N : |f(x) - f_n(x)| < \varepsilon.$$

The norm $\|\cdot\|_\infty$ is called the **supremum norm**.

Proof. First of all, $\|f\|_\infty < \infty$ by assumption. Let us check the properties of the norm

- **Zero norm implies zero function:** If $\|f\| = 0$, then $|f(x)| = 0$ for all $x \in X$, that is $f(x) = 0$ for all $x \in X$. Hence f is the zero element of the vector space $C_b(X, \mathbb{R})$.
- **Homogeneity:** For any $\lambda \in \mathbb{R}, x \in X$, we have $|\lambda f(x)| = |\lambda| |f(x)|$. Taking the sup over all $x \in X$, we get $\|\lambda f\| = |\lambda| \|f\|$.
- **Triangle inequality:** For any $f, g \in C_b(X, \mathbb{R})$ and $x \in X$ we have

$$|f(x) + g(x)| \leq |f(x)| + |g(x)| \leq \|f\| + \|g\|.$$

Taking the sup over all $x \in X$, we get $\|f + g\| \leq \|f\| + \|g\|$.

Now, let $(f_n)_{n=0}^\infty$ be a Cauchy sequence in $C_b(X, \mathbb{R})$ with respect to $\|\cdot\|$. For each $x \in X$, the sequence $(f_n(x))_{n=0}^\infty$ is a Cauchy sequence in $\mathbb{R}$, and therefore converges to some limit $f_x \in \mathbb{R}$. Define the function $f : X \rightarrow \mathbb{R}$ by $f(x) : x \mapsto f_x$, we claim that $f \in C_b(X, \mathbb{R})$ and that $(f_n)_{n=0}^\infty$ converges to f uniformly.

- **Continuity of f :** Fix $x_0 \in X$ and let $\epsilon > 0$. Since $(f_n)_{n=0}^\infty$ is a Cauchy sequence with respect to $\|\cdot\|$, there exists $N \in \mathbb{N}$ such that for all $n, m \geq N$ and all $x \in X$, we have $|f_n(x) - f_m(x)| < \epsilon$. In particular, for $n \geq N$, we have

$$|f_n(x) - f_N(x)| < \epsilon$$

for all $x \in X$. Fix $n \geq N$. Since $|f_n(x_0) - f_N(x_0)| < \epsilon$ for all $x_0 \in X$, the sequence $(f_n(x))_{n=0}^\infty$ is a Cauchy sequence in $\mathbb{R}$, and therefore converges to some limit $y_{x_0} \in \mathbb{R}$. This shows that the function $f : X \rightarrow \mathbb{R}$ is indeed continuous.

- **Uniform convergence:** Let $\epsilon > 0$. Since $(f_n)_{n=0}^\infty$ is a Cauchy sequence with respect to $\|\cdot\|$, there exists $N \in \mathbb{N}$ such that for all $n, m \geq N$ and all $x \in X$, we have

$|f_n(x) - f_m(x)| < \epsilon$. Fix $n \geq N$. Then, for all $x \in X$, we have

$$|f(x) - f_n(x)| = \lim_{m \rightarrow \infty} |f_m(x) - f_n(x)| \leq \epsilon.$$

This shows that $(f_n)_{n=0}^\infty$ converges uniformly to f .

- **Boundedness:** Taking $\varepsilon = 1$ in the uniform convergence definition, we find $f_N \in C_b$ such that

$$|f(x)| \leq |f_N(x)| + |f(x) - f_N(x)| \leq \|f_N\| + 1,$$

taking the supremum for $x \in X$ we find that f is bounded.

Conversely, suppose $(f_n)_{n=0}^\infty$ converges uniformly to f . Let $\epsilon > 0$. Since $(f_n)_{n=0}^\infty$ converges uniformly to f , there exists $N \in \mathbb{N}$ such that for all $n \geq N$ and all $x \in X$, we have $|f_n(x) - f(x)| < \epsilon$. This implies that

$$\sup\{|f_n(x) - f(x)| \mid x \in X\} = \|f_n - f\| < \epsilon.$$

Since $\epsilon > 0$ was arbitrary, this shows that $(f_n)_{n=0}^\infty$ converges to f with respect to $\|\cdot\|$. This completes the proof. $\square$

EXERCISE 9.111. — Show that if $f_j \rightarrow f$ uniformly and $x_j \rightarrow x$, then $f_j(x_j) \rightarrow f(x)$. (F)

By Weierstrass theorem (Corollary 9.78), $C(X, \mathbb{R}^m) = C_b(X, \mathbb{R}^m)$, whenever X is a compact metric space.

THEOREM 9.112: ASCOLI–ARZELÀ (F)

Let (X, d) be a compact metric space and let $\mathcal{F} \subset C(X, \mathbb{R}^m)$ be a subset of functions which is

(1) *equi-bounded*, meaning $C := \sup_{f \in \mathcal{F}, x \in X} |f(x)| < \infty$,

(2) *equi-continuous*, meaning

$$\forall \varepsilon > 0, \exists \delta > 0, \forall f \in \mathcal{F}, (d(x, y) \leq \delta \implies |f(x) - f(y)| \leq \varepsilon). \quad (9.4)$$

Then $\overline{\mathcal{F}}$ is compact in $C(X, \mathbb{R}^m)$. Conversely, any compact subset of $C(X, \mathbb{R}^m)$ is equi-bounded and equi-continuous.

TODO: perhaps the proof with the diagonal argument is neater?

Proof of $(\implies)$. (F) We show that $\overline{\mathcal{F}}$ is complete and totally bounded. For the sake of simplicity, we work with $m = 1$, the general case follows arguing coordinate-by-coordinate.

Since $\overline{\mathcal{F}}$ is closed (by definition) and $C(X)$ is complete (by Proposition 9.110), then $\overline{\mathcal{F}}$ is complete.

Let us fix $\varepsilon > 0$, and let $\delta > 0$ be given by (9.4). Since X is compact, it is also totally bounded, so we find point $x_1, \dots, x_N$ in X such that

$$X = B(x_1, \delta) \cup \dots \cup B(x_N, \delta).$$

Now divide the interval $[-C, C]$ in intervals shorter than ε , so that

$$[-C, C] = I_1 \cup \dots \cup I_M, \quad M \sim 2C/\varepsilon.$$

Consider the finite set of choices

$$\Sigma := \{\sigma: \{1, \dots, N\} \rightarrow \{1, \dots, M\} : \exists f_\sigma \in \mathcal{F} \text{ with } f_\sigma(x_i) \in I_{\sigma(i)} \forall i = 1, \dots, N\},$$

and we claim that the family of balls $\{B(f_\sigma, 4\varepsilon)\}_{\sigma \in \Sigma}$ covers $\mathcal{F}$, that is

$$\forall g \in \mathcal{F}, \quad \exists \sigma \in \Sigma, \quad \forall x \in X : |f_\sigma(x) - g(x)| < 4\varepsilon.$$

To see this define $\rho: \{1, \dots, N\} \rightarrow \{1, \dots, M\}$ by requiring $g(x_i) \in I_{\rho(i)}$, for all $i = 1, \dots, N$. By definition $\rho \in \Sigma$. Now for any $x \in X$ take j in such a way that $x \in B(x_j, \delta)$ and we can bound

$$|f_\rho(x) - g(x)| \leq |f_\rho(x) - f_\rho(x_j)| + |f_\rho(x_j) - g(x_j)| + |g(x_j) - g(x)| \leq 3\varepsilon,$$

where the first and the third terms are controlled by equicontinuity ($d(x, x_j) < \delta$), and the second using that $f_\rho(x_j)$ and $g(x_j)$ both lies in I_j , which is shorter than ε .

Finally we conclude observing that if $\{B(f_\sigma, 4\varepsilon)\}_{\sigma \in \Sigma}$ covers $\mathcal{F}$, then $\{B(f_\sigma, 5\varepsilon)\}_{\sigma \in \Sigma}$ must cover $\overline{\mathcal{F}}$ (small exercise). $\square$

Proof of ($\Leftarrow$). (F) Assume $\mathcal{F}$ is compact in $C(X)$. Then the product metric space $X \times \mathcal{F}$ is compact as well by Corollary 9.70. Furthermore, the function $\Phi: X \times \mathcal{F} \rightarrow \mathbb{R}$, defined as

$$\Phi: (x, f) \mapsto f(x),$$

is continuous (recall Exercise 9.111). By Weierstarss Theorem, Φ is bounded, which means that $\mathcal{F}$ is equi-bounded. By Heine-Cantor Theorem, Φ is uniformly continuous, which means that $\mathcal{F}$ is equi-continuous. $\square$

EXERCISE 9.113. — Generalize Theorem 9.76. The image of a compact set trough a family of equibounded and equicontinuous functions is compact in $\mathbb{R}^m$. (F)

EXERCISE 9.114. — In a compact domain, pointwise convergence and equicontinuity imply uniform convergence.

EXERCISE 9.115. — Let $u_j: [0, 1] \rightarrow \mathbb{R}$ be a sequence of functions which is 2-lipschitz and such that $u_j(1) = 0$. Show that $\{u_j\}_{j \in \mathbb{N}}$ has a uniformly convergent subsequence. (F)

In practice, Ascoli-Arzelà Theorem is often employed in its sequential version, which is

COROLLARY 9.116: ASCOLI–ARZELÀ FOR SEQUENCES (F)

Let (X, d) be a compact metric space and let $\{f_k\}$ be a bounded sequence in $C(X, \mathbb{R}^m)$. Assume that $\{f_k\}$ is equi-continuous, meaning that

$$\forall \varepsilon > 0, \exists \delta > 0, \forall k \in \mathbb{N}, (d(x, y) \leq \delta \implies |f_k(x) - f_k(y)| \leq \varepsilon).$$

Then $\{f_k\}$ has a subsequence that converges uniformly to some $f \in C(X, \mathbb{R}^m)$.

9.3.5 The Length of a Curve

DEFINITION 9.117: LENGTH OF A C^1 CURVE

Let $\gamma \in C^1([a, b], \mathbb{R}^n)$ be a curve. We define the length of γ as

$$L(\gamma) := \int_a^b \|\gamma'(t)\| dt,$$

where the vector $\gamma'(t) = (\gamma'_1(t), \dots, \gamma'_n(t))$ is the **velocity** and the number $\|\gamma'(t)\|$ is the **speed** of the path at time t .

EXERCISE 9.118. — Show that if $s: [0, 1] \rightarrow [a, b]$ is a C^1 bijective map with C^1 inverse, then $L(\gamma \circ s) = L(\gamma)$.

The main result of this section is

THEOREM 9.119: INTRINSIC FORMULA FOR THE LENGTH (F)

Let $\gamma \in C^1([a, b], \mathbb{R}^n)$ be a curve, then

$$\int_0^1 \|\gamma'(s)\| ds = \sup \left\{ \sum_{j=0}^N \|\gamma(t_{j+1}) - \gamma(t_j)\| : N \in \mathbb{N}, 0 = t_0 \leq t_1 \leq \dots \leq t_{N-1} \leq t_N = 1 \right\}.$$

Proof. We start showing $\text{LHS} \geq \text{RHS}$. Denote $\gamma(t) = (\gamma_1(t), \dots, \gamma_n(t))$, fix $N \in \mathbb{N}$, some partition $0 = t_0 \leq t_1 \leq \dots \leq t_{N-1} \leq t_N = 1$ and any set of N unit vectors in $\nu_1, \dots, \nu_N \in \mathbb{R}^n$:

$$\begin{aligned} \int_0^1 \|\gamma'(s)\| ds &= \sum_{j=0}^N \int_{t_j}^{t_{j+1}} \|\gamma'(s)\| ds \geq \sum_{j=0}^N \int_{t_j}^{t_{j+1}} \nu_j \cdot \gamma'(s) ds \\ &= \sum_{j=0}^N \nu_j \cdot \int_{t_j}^{t_{j+1}} \gamma'(s) ds = \sum_{j=0}^N \nu_j \cdot (\gamma(t_{j+1}) - \gamma(t_j)). \end{aligned}$$

With the choice $\nu_j := \frac{\gamma(t_{j+1}) - \gamma(t_j)}{\|\gamma(t_{j+1}) - \gamma(t_j)\|}$ the last term becomes

$$\sum_{j=0}^N \nu_j \cdot \int_{t_j}^{t_{j+1}} \gamma'(s) ds = \sum_{j=0}^N \|\gamma(t_{j+1}) - \gamma(t_j)\|,$$

concluding the proof.

We turn to the proof of $\text{LHS} \leq \text{RHS} + \varepsilon$, so we fix some error threshold $\varepsilon > 0$ small. By uniform continuity of the maps

$$t \mapsto \frac{\gamma'(t)}{\max\{\varepsilon, \|\gamma'(t)\|\}} \quad \text{and} \quad t \mapsto \gamma'(t),$$

from $[0, 1]$ to $\mathbb{R}^n$, we find $\delta > 0$ such that

$$\begin{aligned} [a, b] \subset [0, 1], |b - a| \leq \delta \implies & \left\| \frac{\gamma'(a)}{\max\{\varepsilon, \|\gamma'(a)\|\}} - \frac{\gamma'(b)}{\max\{\varepsilon, \|\gamma'(b)\|\}} \right\| < \varepsilon \\ & \text{and } \|\gamma'(a) - \gamma'(b)\| < \varepsilon. \end{aligned} \quad (9.5)$$

Now divide $[0, 1]$ in N equal intervals of length less than δ and consider one of them, let it be $[t_j, t_{j+1}]$. We distinguish two cases.

Case 1. If $\|\gamma'(s)\| \leq 2\varepsilon$ for all $s \in [t_j, t_{j+1}]$, then we bound directly

$$\int_{[t_j, t_{j+1}]} \|\gamma'(s)\| ds \leq 2\varepsilon(t_j - t_{j+1}).$$

Case 2. If there is $\bar{s} \in [t_j, t_{j+1}]$ such that $\|\gamma'(\bar{s})\| > 2\varepsilon$ then, thanks to (9.5), we have

$$\|\gamma'(s)\| \geq \|\gamma'(\bar{s})\| - \|\gamma'(s) - \gamma'(\bar{s})\| \geq 2\varepsilon - \varepsilon = \varepsilon, \quad \text{for all } s \in [t_j, t_{j+1}],$$

and thus $\max\{\varepsilon, \|\gamma'\|\} = \|\gamma'\|$ in this interval. Then by (9.5) we find

$$\|\gamma'(s)\| = \gamma'(s) \cdot \frac{\gamma'(t_j)}{\|\gamma'(t_j)\|} + \gamma'(s) \cdot \left(\frac{\gamma'(s)}{\|\gamma'(s)\|} - \frac{\gamma'(t_j)}{\|\gamma'(t_j)\|} \right) \leq \gamma'(s) \cdot \frac{\gamma'(t_j)}{\|\gamma'(t_j)\|} + \varepsilon \|\gamma'(s)\|.$$

Thus, integrating this inequality in $[t_j, t_{j+1}]$, and using Cauchy-Schwarz, we find

$$\begin{aligned} (1 - \varepsilon) \int_{[t_j, t_{j+1}]} \|\gamma'(s)\| ds & \leq \frac{\gamma'(t_j)}{\|\gamma'(t_j)\|} \cdot \int_{[t_j, t_{j+1}]} \gamma'(s) ds \\ & = \frac{\gamma'(t_j)}{\|\gamma'(t_j)\|} \cdot (\gamma(t_{j+1}) - \gamma(t_j)) \leq \|\gamma(t_{j+1}) - \gamma(t_j)\|. \end{aligned}$$

So we proved that both in Case 1 and 2 we have

$$(1 - \varepsilon) \int_{[t_j, t_{j+1}]} \|\gamma'(s)\| ds \leq \|\gamma(t_{j+1}) - \gamma(t_j)\| + 2\varepsilon(t_j - t_{j+1}).$$

Summing in j we find that

$$\begin{aligned} (1 - \varepsilon) \text{ LHS} &= (1 - \varepsilon) \int_0^1 \|\gamma'(s)\| ds = (1 - \varepsilon) \sum_{j=0}^N \int_{t_j}^{t_{j+1}} \|\gamma'(s)\| ds \\ &\leq \sum_{j=0}^N \|\gamma(t_{j+1} - \gamma(t_j))\| + 2\varepsilon(t_j - t_{j+1}) \leq \text{RHS} + 2\varepsilon, \end{aligned}$$

we conclude letting $\varepsilon \downarrow 0$.

We remark that the idea is simply to take intervals so short that in each of them $\gamma'(s)$ is approximately constant. The use of ε is instrumental to handle the case where γ' is in fact short, but not pointing in a well-defined direction. In other words, the map $t \mapsto \frac{\gamma'(t)}{\|\gamma'(t)\|}$ is not necessarily continuous, so we regularized it with ε . $\square$

EXERCISE 9.120. — Let $U \subset \mathbb{R}^n$ be open and connected. A path $\gamma : [0, 1] \rightarrow U$ is called piecewise differentiable if there exist finitely many points $0 = t_0 < t_1 < t_2 < \dots < t_n = 1$ in $[0, 1]$ such that for all $k = 1, 2, \dots, n$, the restriction $\gamma|_{[t_{k-1}, t_k]}$ is continuously differentiable.

1. Show that for any two points $x, y \in U$, there exists a piecewise differentiable path from x to y .
2. The length of a piecewise differentiable path as above is defined as

$$L(\gamma) = \sum_{k=1}^K \int_{t_{k-1}}^{t_k} |\gamma'(s)| ds$$

Verify that the path metric $d_{\text{path}}(x, y)$ for $x, y \in U$, defined by

$$d_{\text{path}}(x, y) = \inf\{L(\gamma) \mid \gamma \text{ is a piecewise differentiable path from } x \text{ to } y\},$$

is indeed a metric.

3. Show that the path metric induces the standard topology on $U \subset \mathbb{R}^n$.
4. Let $f : U \rightarrow \mathbb{R}$ be continuously differentiable with a bounded derivative. Show that f is Lipschitz continuous when equipped with the path metric.
5. Find an example of a connected set $U \subset \mathbb{R}^n$ and a differentiable function $f : U \rightarrow \mathbb{R}$ with a bounded derivative that is not Lipschitz continuous with respect to the Euclidean metric.

Chapter 10

Multidimensional Differentiation

In this chapter, we extend the concept of derivatives to functions defined on open subsets $U \subset \mathbb{R}^n$ and taking values in $\mathbb{R}^m$. We will not impose any restrictions on the positive integers n, m .

10.1 The Differential

10.1.1 Definitions

The derivative of a real-variable function $f : \mathbb{R} \rightarrow \mathbb{R}$ at some point $x_0 \in \mathbb{R}$ has various equivalent interpretations. Of course each of these interpretations provides the same number $f'(x_0)$, but the “meaning” we attach to this number is slightly different in each case. (F) Let us recall three important ones:

- **Slope of tangent line to the graph.** We look at the graph of f , i.e. the curve $\{y = f(x)\} \subset \mathbb{R}^2$ and write the tangent line to the graph at $(x_0, f(x_0))$ in the form $y = ax + b$. Then we have $a = f'(x_0)$.
- **Coefficient in infinitesimal linear approximation.** The linear part of the best polynomial that approximates f near x_0 . More precisely, assume we want to approximate $f(x) \sim a + bx$ around $x \sim x_0$, for some real numbers a, b . Then, choosing $b = f'(x_0)$, we get the approximation that gets better and better as $x \rightarrow x_0$;
- **Stretching Factor.** Look at a short interval I around x_0 and the corresponding interval $f(I)$ around $f(x_0)$. These two intervals are related through a “stretching factor” which tends to be $f'(x_0)$ as I is taken shorter. Look here for an animation. More rigorously, the family of functions

$$Tf_r := [-1, 1] \ni y \mapsto \frac{f(x_0 + ry) - f(x_0)}{r},$$

as $r \downarrow 0$, converges uniformly in $[-1, 1]$ to the linear map $y \mapsto f'(x_0)y$.

To generalize derivatives to functions $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ we will start from the second point of view. As it will be clear later on, all the three viewpoints (conveniently reinterpreted) are valid also in the several variables case.

DEFINITION 10.1: DIFFERENTIAL OF A FUNCTION

Let $U \subset \mathbb{R}^n$ be open and $f: U \rightarrow \mathbb{R}^m$ be a function. Then f is called **differentiable** at $x_0 \in U$ if there exists a *linear map* $L: \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

$$\lim_{x \rightarrow 0} \frac{\|f(x_0 + x) - f(x_0) - L(x)\|}{\|x\|} = 0$$

holds. If such L exists it is unique (exercise) and it is called the **differential** of f at the point x_0 , and we denote it as

$$L = Df_{x_0}.$$

The function f is called **differentiable in U** if it is differentiable at every point in U .

TODO: I think it would be quite nice to show them that this is equivalent to the fact that the blow-ups of f are linear maps. They have seen uniform convergence and it is geometrically as insightful as the tangent plane approx (and more appropriate in some cases).

10.2. — If $f: U \rightarrow \mathbb{R}^m$ is differentiable at the point $x_0 \in U$, with differential $L = Df_{x_0}$, we can express f as

$$f(x_0 + x) = f(x_0) + L(x) + R(x)$$

Here we recognize the affine-linear approximation $x \mapsto f(x_0) + L(x)$ to f , and a remainder term $R(x)$, for which, according to the definition of the total derivative, $R(x) = o(\|x\|)$ holds for $x \rightarrow 0$. An notations often found in the literature for differential of f at the point x_0 is df_{x_0}

10.3. — For functions $f: \mathbb{R} \rightarrow \mathbb{R}$ the derivative $f'(x_0)$ is a real number. Notice that in this case $L(y) = Df_{x_0}(y) = f'(x_0)y$.

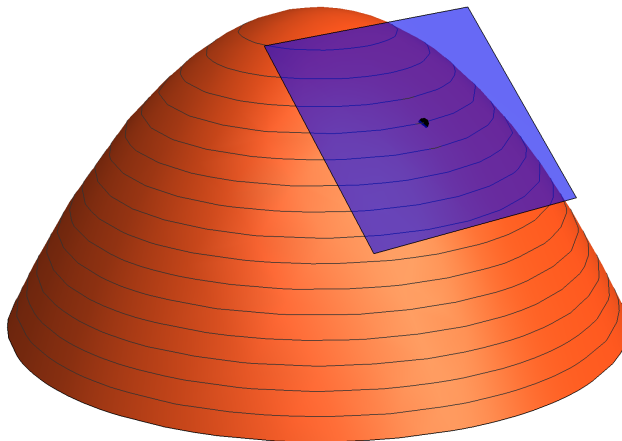


Figure 10.1: For a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, the best affine approximation corresponds to the tangent plane of the graph in $\mathbb{R}^3$.

Applet 10.4 ([Tangent Plane](#)). As shown in the above image, we depict the tangent planes for the graphs of two functions $f : \mathbb{R}^2 \rightarrow \mathbb{R}$. Additionally, we visualize the partial derivatives and directional derivatives in Definition 10.5. Is there a directional derivative that vanishes at every point?

DEFINITION 10.5: DIRECTIONAL DERIVATIVE

Let $U \subset \mathbb{R}^n$ be an open subset, $x_0 \in U$, $v \in \mathbb{R}^n$ and $f : U \rightarrow \mathbb{R}^m$. The **directional derivative of f in the direction v at x_0** is

$$\partial_v f(x_0) := \left. \frac{d}{ds} \right|_{s=0} f(x_0 + sv) = \lim_{s \rightarrow 0} \frac{f(x_0 + sv) - f(x_0)}{s} \in \mathbb{R}^m,$$

provided that the limit exists. If $v = e_j$, for some $j \in \{1, \dots, n\}$, we may denote $\partial_{e_j} f(x_0)$ with

$$D_j f(x_0), \quad \partial_j f(x_0), \quad \frac{\partial f(x_0)}{\partial x_j}.$$

Of course, if the partial derivative in the j -th coordinate exists at every point in U , we obtain a function $\partial_j f : U \rightarrow \mathbb{R}^m$, which we call the **j^{th} directional derivative of f** .

10.6. — The partial derivative and the directional derivative along any vector are the derivatives with respect to one of the independent variables, considering all other variables as constants. For example, for the function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ given by $f(x, y, z) = x(y^2 + \sin(z))$, the partial derivatives with respect to all coordinate directions are given by

$$\partial_x f(x, y, z) = y^2 + \sin(z)$$

$$\partial_y f(x, y, z) = 2xy$$

$$\partial_z f(x, y, z) = x \cos(z)$$

for all $(x, y, z) \in \mathbb{R}^3$, as we can apply all known rules from Analysis I. If the total derivative exists, we can connect it with partial derivatives and derivatives along arbitrary vectors using the following proposition.

PROPOSITION 10.7: DIFFERENTIABLE IMPLIES LINEAR DIRECTIONAL DERIVATIVES

Let $U \subset \mathbb{R}^n$ be open and let $f: U \rightarrow \mathbb{R}^m$ be differentiable at $x_0 \in U$. Then, for each $v \in \mathbb{R}^n$, the derivative of f in the direction v exists, and we have

$$\partial_v f(x_0) = Df_{x_0}(v) \in \mathbb{R}^m.$$

In particular $\partial_{v+\alpha w} f(x_0) = \partial_v f(x_0) + \alpha \partial_w f(x_0)$, for all $\alpha \in \mathbb{R}$ and $v, w \in \mathbb{R}^n$.

Proof. Assuming the total derivative $Df(x_0)$ exists, according to the definition of the derivative, $f(x_0 + h) = f(x_0) + Df(x_0)(h) + o(\|h\|)$ holds for $h \rightarrow 0$. Choosing $h = sv$ for $s \rightarrow 0$ and $v \in \mathbb{R}^n$, we get

$$\partial_v f(x_0) = \lim_{s \rightarrow 0} \frac{f(x_0 + sv) - f(x_0)}{s} = \lim_{s \rightarrow 0} (Df(x_0)(v) + o(1)) = Df(x_0)(v)$$

which concludes the proof. $\square$

EXERCISE 10.8. — Let $U \subset \mathbb{R}^n$ be open, and let $f_1, f_2 : U \rightarrow \mathbb{R}$ be functions. Assume that f_1 and f_2 are differentiable at $x_0 \in U$. Show that the functions $f_1 + f_2$ and $f_1 f_2$ are differentiable at x_0 and that

$$\begin{aligned} D(f_1 + f_2)(x_0) &= Df_1(x_0) + Df_2(x_0) \\ D(f_1 f_2)(x_0) &= f_1(x_0) Df_2(x_0) + f_2(x_0) Df_1(x_0) \end{aligned}$$

Formulate and prove analogous statements for directional derivatives in the direction of a fixed vector $v \in \mathbb{R}^n$.

LEMMA 10.9: MORE VARIABLES IN THE TARGET ARE NOT A PROBLEM

Let $U \subset \mathbb{R}^n$ be open, and let $f : U \rightarrow \mathbb{R}^m$ be a function. Denote $\pi_j : \mathbb{R}^m \rightarrow \mathbb{R}$ as the projection onto the j -th component. Then, f is differentiable at $x_0 \in U$ if and only if the components $f_j = \pi_j \circ f$ for each $j \in \{1, \dots, m\}$ are differentiable at x_0 . In this case, we have

$$\pi_j \circ Df(x_0) = D(\pi_j \circ f)(x_0).$$

Proof. Assume that $f_j := \pi_j \circ f$ is differentiable at x_0 for every j . There exists a linear function $L_j : \mathbb{R}^n \rightarrow \mathbb{R}$ and a remainder term $R_j : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$f_j(x_0 + x) = f_j(x_0) + L_j(x) + R_j(x)$$

with $R_j(x) = o(\|x\|)$ for $x \rightarrow 0$. We can summarize

$$f(x_0 + h) = \begin{pmatrix} f_1(x_0 + h) \\ \vdots \\ f_m(x_0 + h) \end{pmatrix} = \begin{pmatrix} f_1(x_0) \\ \vdots \\ f_m(x_0) \end{pmatrix} + \begin{pmatrix} L_1(x) \\ \vdots \\ L_m(x) \end{pmatrix} + \begin{pmatrix} R_1(x) \\ \vdots \\ R_m(x) \end{pmatrix}$$

In summary, it can be written as $f(x_0 + x) = f(x_0) + L(x) + R(x)$. In this expression, L is linear, and it holds that $R_j(x) = o(\|x\|)$ for $x \rightarrow 0$. Thus, f is differentiable, and the claimed formula for $Df(x_0)$ holds. Conversely, if f is differentiable at the point x_0 , then the claimed formula follows through the same calculation. $\square$

THEOREM 10.10: SUFFICIENT CONDITION FOR DIFFERENTIABILITY

Let $U \subset \mathbb{R}^n$ be open, and $f : U \rightarrow \mathbb{R}^m$ be a function. If for every $j \in \{1, \dots, n\}$ the partial derivative $\partial_j f$ exists on the entire U and defines a continuous function, then f is differentiable on the entire U .

Proof. Due to Lemma 10.9, we can assume $m = 1$. Let's fix $x_0 \in U$, and we need to show that f is differentiable at x_0 . By replacing f with $x \mapsto f(x + x_0) - f(x_0)$, we can also assume that $x_0 = 0$ and $f(0) = 0$. For $x = (x_1, \dots, x_n) \in U$, we then have

$$\begin{aligned} f(x) &= f(x_1, x_2, x_3, \dots, x_n) - f(0, x_2, x_3, \dots, x_n) \\ &\quad + f(0, x_2, x_3, \dots, x_n) - f(0, 0, x_3, \dots, x_n) \\ &\quad + f(0, 0, x_3, \dots, x_n) - \dots \\ &\quad + \dots - f(0, 0, \dots, 0, x_n) \\ &\quad + f(0, 0, \dots, 0, x_n) - f(0, 0, \dots, 0, 0). \end{aligned}$$

The function $[0, x_j] \rightarrow \mathbb{R}$ defined by $t \mapsto f(0, 0, \dots, 0, t, x_{j+1}, \dots, x_n)$ is continuously differentiable by hypothesis. Its derivative is given by the j -th partial derivative of f . Therefore, by the Mean Value Theorem, there exists an intermediate point $\xi_j \in [0, x_j]$ such that

$$\partial_j f(0, \dots, 0, \xi_j, x_{j+1}, \dots, x_n) x_j = f(0, \dots, 0, x_j, x_{j+1}, \dots, x_n) - f(0, \dots, 0, 0, x_{j+1}, \dots, x_n)$$

holds. For any choice of such intermediate points $\xi_j \in [0, x_j]$, we obtain

$$\begin{aligned} f(x) &= \partial_1 f(\xi_1, x_2, x_3, \dots, x_n) x_1 \\ &\quad + \partial_2 f(0, \xi_2, x_3, \dots, x_n) x_2 \\ &\quad + \dots \\ &\quad + \partial_n f(0, 0, \dots, 0, \xi_n) x_n. \end{aligned}$$

To show that the linear function $L : (v_1, \dots, v_n) \mapsto \partial_1 f(0) v_1 + \dots + \partial_n f(0) v_n$ is the derivative $Df(0)$, we need to estimate the difference $R(x) := f(x) - L(x) = f(0 + x) - f(0) - L(x)$.

$$R(x) = (\partial_1 f(\xi_1, x_2, x_3, \dots, x_n) - \partial_1 f(0)) x_1$$

$$\begin{aligned}
& + (\partial_2 f(0, \xi_2, x_3, \dots, x_n) - \partial_2 f(0))x_2 \\
& + \dots \\
& + (\partial_n f(0, 0, \dots, 0, \xi_n) - \partial_n f(0))x_n
\end{aligned}$$

According to the assumptions of the theorem and because $\frac{|x_j|}{\|x\|} \leq 1$ for all $x \in \mathbb{R}^n$, the asymptotics

$$\lim_{x \rightarrow 0} \frac{R(x)}{\|x\|} = 0$$

holds, demonstrating that f is differentiable at $x_0 = 0$ with the derivative $Df(0) = L$. $\square$

The following exercise shows that the existence of partial derivatives of a function f , without the continuity assumption, does not necessarily imply that the function is differentiable.

EXERCISE 10.11. — Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) = \begin{cases} \frac{xy}{\sqrt{x^2+y^2}} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0), \end{cases}$$

for $(x, y) \in \mathbb{R}^2$. Show that the partial derivatives $\partial_x f$ and $\partial_y f$ exist everywhere in $\mathbb{R}^2$, but f is not differentiable at $(0, 0)$.

DEFINITION 10.12: C^1 FUNCTIONS

We call a function $f : U \rightarrow \mathbb{R}^m$ on an open subset $U \subset \mathbb{R}^n$ **continuously differentiable** if all the partial derivatives $\partial_i f$, $i = 1, \dots, n$ exist and continuous in U . We will write in this case $f \in C^1(U, \mathbb{R}^m)$ and often $f \in C^1(U)$ when $m = 1$.

Extend definition to closed sets as well

10.13. — Notice that $f \in C^1(U, \mathbb{R}^m)$ if and only if the map $Df : U \rightarrow \text{Hom}(\mathbb{R}^n, \mathbb{R}^m)$ mapping $x \mapsto Df_x$ is continuous.

10.14. — From Proposition 10.7, it follows in particular that the total derivative (when it exists) $Df(x_0)$ is uniquely determined by the partial derivatives. Specifically, for $v = a_1 e_1 + \dots + a_n e_n \in \mathbb{R}^n$,

$$Df_{x_0}(v) = \sum_{i=1}^n a_i Df_{x_0}(e_i) = \sum_{i=1}^n a_i \partial_i f(x_0).$$

The $m \times n$ matrix of the linear map $Df_{x_0} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is thus given with respect to the canonical bases by the matrix $n \times m$ matrix

$$(\partial_1 f(x_0), \partial_2 f(x_0), \dots, \partial_n f(x_0)) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1}(x_0) & \frac{\partial f_1}{\partial x_2}(x_0) & \cdots & \frac{\partial f_1}{\partial x_n}(x_0) \\ \frac{\partial f_2}{\partial x_1}(x_0) & \frac{\partial f_2}{\partial x_2}(x_0) & \cdots & \frac{\partial f_2}{\partial x_n}(x_0) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(x_0) & \frac{\partial f_m}{\partial x_2}(x_0) & \cdots & \frac{\partial f_m}{\partial x_n}(x_0) \end{bmatrix}$$

This matrix is referred to as the **Jacobian matrix** of f evaluated at the point x_0 , commonly denoted by $Jf(x_0)$. An alternative notation for the Jacobian matrix is $Df(x_0)$; however, to prevent confusion with Df_{x_0} , which distinguishes between a linear map and a matrix representation, we will not use this notation yet. While the difference between a matrix and a linear map might appear to be a minor detail, it becomes more significant in future applications such as in Differential Geometry or Physics.

Notice that, given $U \subset \mathbb{R}^n$ open, $f \in C^1(U, \mathbb{R}^m)$ if and only if the maps

$$Jf : U \rightarrow \text{Mat}_{m,n}(\mathbb{R}) \cong \mathbb{R}^{m \times n}$$

defined by $x \mapsto Jf(x)$ is a continuous .

TODO: exercise where they compute the derivative tangential/normal to a circle of the function $(x, y) \mapsto x^2 + y^2$

TODO: example of computation of Jf for a particular $f : \mathbb{R}^3 \rightarrow \mathbb{R}^2$

EXAMPLE 10.15. — Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $f(x, y) = (x^2 - \cos(xy), y^4 - \exp(x))$. The Jacobian matrix of f at (x, y) is then

$$\begin{pmatrix} 2x + \sin(xy)y & \sin(xy)x \\ -\exp(x) & 4y^3 \end{pmatrix},$$

which is continuous as a function of $(x, y) \in \mathbb{R}^2$.

10.1.2 The Chain Rule

THEOREM 10.16: CHAIN RULE (POINTWISE VERSION)

Let $k, m, n \geq 1$, and let $U \subset \mathbb{R}^n$ and $V \subset \mathbb{R}^m$ be open. If $f : U \rightarrow V$ is differentiable at x_0 and $g : V \rightarrow \mathbb{R}^k$ is differentiable at $f(x_0)$, then $g \circ f$ is differentiable at x_0 , and the differential of $(g \circ f)$ at x_0 is given by

$$D(g \circ f)_{x_0} = Dg_{f(x_0)} \circ Df_{x_0} \quad (10.1)$$

In other words, at the level of matrices

$$J(g \circ f)(x_0) = Jg(f(x_0)) \cdot Jf(x_0)$$

or equivalently:

$$\frac{\partial(g \circ f)}{\partial x_i}(x_0) = \sum_{j=1}^m \frac{\partial g}{\partial x_j}(f(x_0)) \frac{\partial f_j}{\partial x_i}(x_0)$$

Proof. By the definition of differentiability of f at x_0 and g at $y_0 = f(x_0)$, we have

$$f(x_0 + x) = f(x_0) + L(x) + R(x) \text{ and } g(y_0 + y) = g(y_0) + M(y) + S(y)$$

with $L = Df(x_0)$ and $R(x) = o(\|x\|)$ as $x \rightarrow 0$, and $M = Dg(y_0)$ and $S(y) = o(\|y\|)$ as $y \rightarrow 0$. Together, for $x \in \mathbb{R}^n$ small enough and $y = f(x_0 + x) - f(x_0) = L(x) + R(x)$, we obtain the equation

$$\begin{aligned} g(f(x_0 + x)) &= g(y_0 + y) = g(y_0) + M(y) + S(y) = \\ &= g(f(x_0)) + M(L(x)) + \underbrace{M(R(x)) + S(L(x) + R(x))}_{T(x)} \end{aligned}$$

and we want to show that $T(x) = o(\|x\|)$ as $x \rightarrow 0$. Since $R(x) = o(\|x\|)$ as $x \rightarrow 0$, we also have $\|M(R(x))\| \leq \|M\|_{\text{op}} \|R(x)\| = o(\|x\|)$ as $x \rightarrow 0$. It remains to show that $S(L(x) + R(x)) = o(\|x\|)$ as $x \rightarrow 0$. The fact that $S(y) = o(\|y\|)$ as $y \rightarrow 0$ means that for every $\epsilon > 0$, there exists $\delta > 0$ such that for all $y \in \mathbb{R}^m$

$$\|y\| < \delta \implies \|S(y)\| \leq \epsilon \|y\| \quad (10.2)$$

After the differentiability of f at x_0 , for $y = L(x) + R(x)$, we have the estimate

$$\|y\| \leq \|L(x)\| + \|R(x)\| \leq \|L\|_{\text{op}} \|x\| + o(x)$$

as $x \rightarrow 0$. Let $C = \|L\|_{\text{op}} + 1$; then, there exists $\eta > 0$ such that for all $x \in \mathbb{R}^n$

$$\|x\| < \eta \implies \|L(x) + R(x)\| < C\|x\|$$

For $x \in \mathbb{R}^n$ with $\|x\| < \min\{\eta, \delta C^{-1}\}$, we have $\|L(x) + R(x)\| < \delta$, and using (10.2), we also have

$$\|S(L(x) + R(x))\| \leq \epsilon \|L(x) + R(x)\| \leq C\epsilon \|x\|.$$

Thus, we conclude the differentiability of $g \circ f$ at x_0 and the equation (10.1). $\square$

COROLLARY 10.17: CHAIN RULE

Let $k, m, n \geq 1$, and let $U \subset \mathbb{R}^n$ and $V \subset \mathbb{R}^m$ be open. If $f \in C^1(U, \mathbb{R}^n)$, $g \in C^1(V, \mathbb{R}^k)$, and $f(U) \subset V$ then $g \circ f \in C^1(U, \mathbb{R}^k)$ and

$$\partial_i(g \circ f) = \sum_{j=1}^m (\partial_j g) \circ f \partial_i f_j \quad (10.3)$$

TODO: add a lot of examples, they are usually not able to use correctly the chain rule in exams....

EXERCISE 10.18. — Euler's identity for homogeneous functions (F) Assume $f \in C^1(\mathbb{R}^n \setminus \{0\})$ is positively homogeneous of degree $\lambda \in \mathbb{R}$, that is to say

$$f(rx) = r^\lambda f(x) \text{ for all } r > 0, x \neq 0.$$

Show that $\sum_{j=1}^n x_j \partial_j f(x) = \lambda f(x)$.

EXAMPLE 10.19. — (F) Let $u: \mathbb{R}^n \rightarrow \mathbb{R}$, check that

$$\begin{aligned} D_i(\arctan(u)) &= \frac{D_i u}{1+u^2}, \quad D_i(1/u) = -\frac{D_i u}{u^2}, \quad D_i(|Du|^2) = 2 \sum_j D_j u D_{ij} u, \\ \sum_i D_{ii}(|Du|^2) &= 2 \sum_{i,j} (D_{ij} u)^2 + 2 \sum_i D_i u D_i \left(\sum_j D_{jj} u \right). \end{aligned}$$

If $A \in O(n)$ and $v(x) := u(Ax)$ then

$$D_i v(x) = \sum_j A_{i,j} D_j u(Ax) \text{ and } \sum_i D_{ii} v(x) = \sum_j (D_{jj} u)(Ax).$$

10.20. — Let's consider the special case $n = 1$ for the chain rule. Suppose $I \subset \mathbb{R}$ is an open interval, and $\gamma: I \rightarrow V \subset \mathbb{R}^m$ is a differentiable function with values in an open subset $V \subset \mathbb{R}^m$. Further, let $f: V \rightarrow \mathbb{R}^k$ be differentiable. Then, the chain rule implies that $f \circ \gamma$ is differentiable, and the formula

$$(f \circ \gamma)'(t) = Df_{\gamma(t)}(\gamma'(t))$$

holds for all $t \in I$. If additionally $k = 1$, then $f \circ \gamma: I \rightarrow \mathbb{R}$ and $Df_{\gamma(t)}\gamma'(t)$, expressed in matrix form, is the matrix product of the $1 \times m$ matrix $Df(\gamma(t))$ with the $m \times 1$ matrix $\gamma'(t)$.

In this case, we interpret $Df(x)$ for $x \in V$ as the column vector

$$\operatorname{grad} f(x) = \nabla f(x) = (Df(x))^T \in \mathbb{R}^m$$

and refer to it as the **gradient of the function** f at the point x . Using this notation, we obtain the formula

$$(f \circ \gamma)'(t) = Df(\gamma(t)) \cdot \gamma'(t) = \langle \nabla f(\gamma(t)), \gamma'(t) \rangle \quad (10.4)$$

for all $t \in I$.

10.21. — The concept of directional derivatives and the case of equality in the Cauchy-Schwarz inequality allow us to provide a geometric interpretation of the gradient of a function. If $f : U \rightarrow \mathbb{R}$ is a differentiable function on an open subset $U \subset \mathbb{R}^n$, then, according to Proposition 10.7 and the Cauchy-Schwarz inequality, Proposition,

$$\partial_v f(x) = Df_x(v) = \langle \nabla f(x), v \rangle \leq |\nabla f(x)| |v|$$

for any vector $v \in \mathbb{R}^n$, with equality if and only if $\nabla f(x)$ and v are linearly dependent. This implies that the gradient of f at every point points in the direction of the greatest directional derivative, indicating the direction of the steepest ascent of f around x . Furthermore, $\|\nabla f(x)\|$ gives the slope in that direction.

10.1.3 The Mean Value Theorem

We formulate a generalization of the Mean Value Theorem for real-valued differentiable functions on an open set $U \subset \mathbb{R}^n$. To do this, we consider a given function f along a straight segment in the open set.

THEOREM 10.22: MEAN VALUE THEOREM

Let $U \subset \mathbb{R}^n$ be open, and $f : U \rightarrow \mathbb{R}$ be differentiable. Let $x_0 \in U$ and $h \in \mathbb{R}^n$ such that $x_0 + th \in U$ for all $t \in [0, 1]$. Then, there exists $t \in (0, 1)$ such that for $\xi = x_0 + th$, the equation

$$f(x_0 + h) - f(x_0) = Df_\xi(h) = \partial_h f(\xi)$$

is satisfied.

Proof. The derivative of the straight path $\gamma : t \mapsto x_0 + th$ for fixed $x_0, h \in \mathbb{R}^n$ is given by $\gamma'(t) = h$. Therefore, the function $g = f \circ \gamma : [0, 1] \rightarrow \mathbb{R}$ satisfies all the conditions of the one-dimensional Mean Value Theorem due to the chain rule in Theorem 10.16. Hence, there exists $t \in (0, 1)$ with $g(1) - g(0) = g'(t) = Df(x_0 + th)(h)$ according to the chain rule, and thus

$$f(x_0 + h) - f(x_0) = g(1) - g(0) = g'(t) = Df(\xi)(h)$$

for $\xi = x_0 + th$. □

COROLLARY 10.23: SOLUTIONS OF $Df = 0$

Let $U \subset \mathbb{R}^n$ be open and let $f : U \rightarrow \mathbb{R}^m$ be differentiable with $Df(x) = 0$ for all $x \in U$. Then, f is constant on each connected component of U .

Proof. It suffices to consider the case $m = 1$. Assuming that U is non-empty, we choose $x_0 \in U$ and consider the subset

$$U' = \{x \in U \mid f(x) = f(x_0)\}$$

of U . Since f is continuous, $U' \subset U$ is a closed subset of U . From the assumption and the mean value theorem, it follows that U' is open: Indeed, for $x \in U'$, there exists $\epsilon > 0$ such that $B(x, \epsilon) \subset U$, and since every point $y \in B(x, \epsilon)$ can be connected by a straight path to x , it follows from Theorem 10.22 that $f(y) = f(x) = f(x_0)$. Thus, $y \in U'$, and since $y \in B(x, \epsilon)$ was arbitrary, $B(x, \epsilon) \subset U'$. However, since U is connected, we have $U' = U$, and the corollary follows. □

TODO: Exercise: what if only a partial derivative is zero? What if U not connected?

DEFINITION 10.24: LOCAL LIPSCHITZ CONTINUITY

A function $f : X \rightarrow Y$ between metric spaces X, Y is called **locally Lipschitz continuous** if, for every $x_0 \in X$, there exists $\epsilon > 0$ such that $f|_{B(x_0, \epsilon)}$ is Lipschitz continuous.

COROLLARY 10.25: DIFFERENTIABILITY VS LIPSCHITZ CONTINUITY

Let $U \subset \mathbb{R}^n$ be open and let $f \in C^1(U, \mathbb{R}^m)$. Then, f is locally Lipschitz continuous. If U is additionally convex, and the differential is bounded (as a matrix), then f is Lipschitz continuous.

Proof. It suffices to consider the case $m = 1$. First, assuming that U is convex and the derivative is bounded. There exists $M \geq 0$ such that $\|Df(\xi)\|_{\text{op}} \leq M$ for all $\xi \in U$. From the mean value theorem 10.22, it follows for $x, y \in U$

$$\|f(x) - f(y)\| = \|Df(\xi)(x - y)\| \leq M\|x - y\|$$

for some $\xi \in U$, since U is convex and thus contains the straight segment between x and y . This proves the second statement in the corollary. The first statement follows from the second applied to the ball $U_0 = B(x_0, \epsilon)$ where $\epsilon > 0$ is chosen such that $\overline{B(x_0, \epsilon)} \subset U$. Indeed, U_0 is convex, and the mapping $\xi \mapsto Df(\xi)$ is a continuous function on the compact set $\overline{B(x_0, \epsilon)}$, implying the boundedness of the derivative on $B(x_0, \epsilon)$. □

10.2 Higher Derivatives

10.2.1 Definition and basic properties

Recall that for functions $f : \mathbb{R} \rightarrow \mathbb{R}$ we defined second and higher order derivatives recursively as follows $f^{(k+1)} = (f^{(k)})'$, $k \geq 0$, where $f^{(0)} = f$. Next we will introduce higher order derivatives for function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$.

DEFINITION 10.26:

Let $U \subset \mathbb{R}^n$ be open, $f : U \rightarrow \mathbb{R}^m$ be a function, and $d \geq 1$. We say that f is k **times continuously differentiable** if, for all $j_1, \dots, j_d$ in $\{1, \dots, n\}$, the partial derivative

$$\partial_{j_1} \partial_{j_2} \cdots \partial_{j_d} f(x)$$

exists at every point $x \in U$ and, depending on $x \in U$, defines a continuous function on U . We write

$$C^k(U, \mathbb{R}^m) = \{f : U \rightarrow \mathbb{R}^m \mid f \text{ is } k\text{-times continuously differentiable}\}$$

for the vector space of d times continuously differentiable $\mathbb{R}^m$ -valued functions on U . We call the function f **smooth** if f is continuously differentiable an arbitrary number of times. We write

$$C^\infty(U, \mathbb{R}^m) = \{f : U \rightarrow \mathbb{R}^m \mid f \text{ is } d\text{-times continuously differentiable for all } d \geq 1\}$$

for the vector space of smooth $\mathbb{R}^m$ -valued functions on U .

PROPOSITION 10.27: HIGHER REGULARITY OF SUMS, PRODUCTS AND COMPOSITIONS

Let $f \in C^k(\mathbb{R}^n)$, $g \in C^\ell(\mathbb{R}^n)$ and $\phi \in C^m(\mathbb{R}^n, \mathbb{R}^n)$, then

(1) $f + g, f \cdot g$ are of class $C^{\min\{k, \ell\}}$

(2) $f \circ \phi$ is of class $C^{\min\{k, m\}}$

Proof. We prove only the second item and leave the first as exercise. We do an induction on $\min\{k, m\}$. Since composition of continuous functions is necessarily continuous the base case $\min\{k, m\} = 0$ is handled. Now assume the statement has been proved whenever $\min\{k, m\} \leq N$ and that we have f, ϕ, k, m with $\min\{k, m\} = N + 1$. For sure $\phi \in C^N$ (since $m \geq N + 1$) and $Df \in C^{k-1}$, so the inductive assumption applied to $\partial_i f \circ \phi$ gives

$$\partial_j(f \circ \phi) = \sum_{i=1}^n \underbrace{\partial_i f \circ \phi}_{C^{\min\{k-1, N\}}} \underbrace{\partial_j \phi^i}_{C^{m-1}}.$$

Thus by the first part $\partial_j(f \circ \phi)$ is of class $\min\{k-1, N, k-1\} \geq N$, for all j , which means that $f \circ \phi$ in C^{N+1} . $\square$

10.2.2 Schwartz's Theorem and Multi-indexes notation

THEOREM 10.28: SCHWARZ'S THEOREM

Let $U \subset \mathbb{R}^n$ be open and let $f \in C^2(U, \mathbb{R}^m)$, then

$$\partial_j \partial_i f(x) = \partial_i \partial_j f(x), \quad \text{for all } i, j \in \{1, \dots, n\}, x \in U.$$

Proof. It suffices to consider the case $n = 2$, $m = 1$ and $i = 1$, $j = 2$, the general case follows by renaming variables from the considered special case (applying it to the components $f_1, \dots, f_m$ of f). For $x \in U$ and a sufficiently small $h > 0$, we define a function F by

$$F(h) = f(x_1 + h, x_2 + h) - f(x_1 + h, x_2) - f(x_1, x_2 + h) + f(x_1, x_2).$$

Furthermore, for a sufficiently small but fixed $h > 0$, we consider the differentiable function $\varphi : [0, 1] \rightarrow \mathbb{R}$ given by $\varphi(t) = f(x_1 + th, x_2 + h) - f(x_1 + th, x_2)$ and obtain

$$F(h) = \varphi(1) - \varphi(0) = \varphi'(\xi_1) = (\partial_1 f(x_1 + \xi_1 h, x_2 + h) - \partial_1 f(x_1 + \xi_1 h, x_2))h$$

for some $\xi_1 \in (0, 1)$ by the one-dimensional Mean Value Theorem ??.

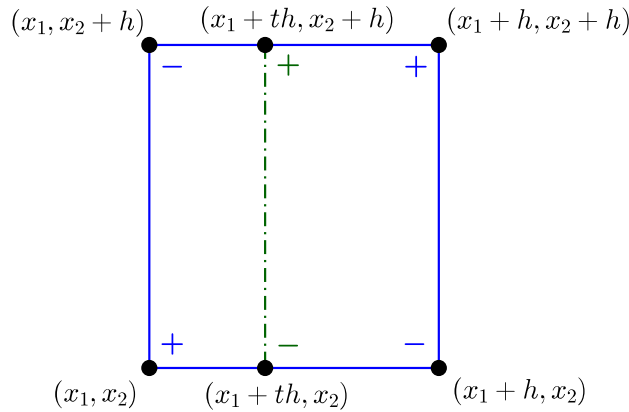


Figure 10.2: The function $h \mapsto F(h)$ is a signed sum of function values of f at the corners of a square (here marked by a solid line). The function $t \mapsto \varphi(t)$ corresponds to the difference of function values on a vertical segment through the square.

Applying the one-dimensional Mean Value Theorem again to $\psi : [0, 1] \rightarrow \mathbb{R}$ given by $\psi(t) = \partial_1 f(x_1 + \xi_1 h, x_2 + th)$ along with the chain rule, we obtain

$$F(h) = (\partial_1 f(x_1 + \xi_1 h, x_2 + h) - \partial_1 f(x_1 + \xi_1 h, x_2))h = \partial_2 \partial_1 f(x_1 + \xi_1 h, x_2 + \xi_2 h)h^2$$

for some intermediate point $\xi_2 \in (0, 1)$. Since both components were used symmetrically in the function $h \mapsto F(h)$, we can perform the argument again with the roles of the first and second components swapped. This yields similarly

$$F(h) = \partial_1 \partial_2 f(x_1 + \xi'_1 h, x_2 + \xi'_2 h) h^2.$$

for suitable $\xi'_1, \xi'_2 \in (0, 1)$. Dividing by $h^2 > 0$, we obtain

$$\partial_2 \partial_1 f(x_1 + \xi_1 h, x_2 + \xi_2 h) = \partial_1 \partial_2 f(x_1 + \xi'_1 h, x_2 + \xi'_2 h).$$

Since $\xi_1, \xi_2, \xi'_1, \xi'_2 \in (0, 1)$, the points $(\xi_1 h, \xi_2 h)$ and $(\xi'_1 h, \xi'_2 h)$ tend to $(0, 0)$ as h tends to 0. Therefore, due to the continuity of both partial derivatives, we conclude $\partial_2 \partial_1 f(x) = \partial_1 \partial_2 f(x)$. $\square$

TODO: along with the Hessian, perhaps define also the Laplacian?

DEFINITION 10.29: HESSIAN AND LAPLACIAN

The **Hessian matrix** of $f \in C^2(U)$ at $x \in U$ is the $n \times n$ matrix

$$H_{ij}(x) = \partial_i \partial_j f(x)$$

for $i, j \in \{1, \dots, n\}$. Schwarz's theorem 10.28 entails that $H(x)$ is a symmetric matrix.

The **Laplacian** is the trace of this matrix

$$\Delta f(x) := \operatorname{tr} H(x) = \sum_{i=1}^n \partial_{ii} f(x)$$

EXERCISE 10.30 (Polarisation formula). — Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be a smooth function. Prove that the map

$$\mathbb{R}^n \ni e \mapsto \left. \frac{d^2}{dt^2} \right|_{t=0} f(te) = D^2 f(0)(e, e),$$

determines all the second derivatives $\partial_{ij} f$.

INTERLUDE: MULTI-INDECES AND POLYNOMIALS OF SEVERAL VARIABLES

Any $\alpha \in \mathbb{N}^n$ is called a **multi-index of length n** , the length of a multi-index is

$$|\alpha| := \alpha_1 + \dots + \alpha_n.$$

We say that $\beta \leq \alpha$ if $\beta_i \leq \alpha_i$ for all $i = 1, \dots, n$, and define the factorial

$$\alpha! := \alpha_1! \dots \alpha_n!, \quad 0! = 1.$$

A polynomial of n variables $X_1, \dots, X_n$ of degree k can be uniquely (and compactly) expressed as

$$\sum_{\alpha \in \mathbb{N}^n, |\alpha| \leq k} c_\alpha X^\alpha := \sum_{|\alpha| \leq k} c_{(\alpha_1, \dots, \alpha_n)} X^{\alpha_1} \dots X^{\alpha_n}.$$

Many combinatorial formulas are simple when expressed in multi-index notation, such as the multi-nomial formula:

$$(X + Y)^\alpha = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} X^\beta Y^{\alpha - \beta}, \text{ where } \binom{\alpha}{\beta} := \binom{\alpha_1}{\beta_1} \dots \binom{\alpha_n}{\beta_n}.$$

EXERCISE 10.31. — Prove the identity

$$\frac{n^m}{m!} = \sum_{\alpha \in \mathbb{N}^n, |\alpha| = m} \frac{1}{\alpha!}.$$

Thanks to Schwartz's Theorem, multi-indexes are also useful to express higher order derivatives.

COROLLARY 10.32: SCHWARZ IN C^k

Let $f \in C^k(\mathbb{R}^n)$ and let $\alpha \in \mathbb{N}^n$ with $|\alpha| \leq k$. Then the derivative

$$\partial_\alpha f(x) = (\partial_{x_1})^{\alpha_1} \dots (\partial_{x_n})^{\alpha_n} f(x),$$

is independent of the precise order in which we take the derivatives.

Proof. Immediate using Schwartz's Theorem (exercise). $\square$

10.2.3 Multidimensional Taylor Approximation

THEOREM 10.33: TAYLOR'S THEOREM

Let $U \subset \mathbb{R}^n$ be open and let $f \in C^{k+1}(U)$, $k \geq 0$. Let $x_0 \in U$ and $h \in \mathbb{R}^n$ such that $x + th \in U$ for all $t \in [0, 1]$. Then, we have

$$f(x_0 + h) = \sum_{\alpha \in \mathbb{N}^n, |\alpha| \leq k} \partial^\alpha f(x_0) \frac{h^\alpha}{\alpha!} + R_{k+1}f(x_0, h)$$

where the reminder is given by

$$R_{k+1}f(x_0, h) := \int_0^1 (k+1)(1-t)^k \sum_{\alpha \in \mathbb{N}^n, |\alpha| = k+1} \partial^\alpha f(x_0 + th) \frac{h^\alpha}{\alpha!} dt = O(\|h\|^{k+1}).$$

Proof. Since U is open, there exists $\epsilon > 0$ such that $x + th \in U$ for all $t \in (-\epsilon, 1 + \epsilon)$. We apply the one-dimensional Taylor approximation to $\varphi : (-\epsilon, 1 + \epsilon) \rightarrow \mathbb{R}$ given by $\varphi(t) = f(x + th)$.

According to Taylor's Theorem, we obtain the Taylor approximation around 0 at 1

$$\varphi(1) = \sum_{m=0}^k \frac{\varphi^{(m)}(0)}{m!} + \int_0^1 \varphi^{(k+1)}(t) \frac{(1-t)^k}{k!} dt. \quad (10.5)$$

Applying the chain rule in Theorem 10.16 to φ , we get for $t \in (-\epsilon, 1 + \epsilon)$ the derivatives

$$\begin{aligned} \varphi'(t) &= \sum_{i=1}^n \partial_i f(x_0 + th) h_i, \\ \varphi''(t) &= \sum_{i,j=1}^n \partial_i \partial_j f(x_0 + th) h_i h_j, \\ \varphi'''(t) &= \sum_{i,j,\ell=1}^n \partial_i \partial_j \partial_\ell f(x_0 + th) h_i h_j h_\ell, \quad \dots \end{aligned}$$

So inductively for all $m \leq k+1$:

$$\varphi^{(m)}(t) = \sum_{i_1, \dots, i_m=1}^n \partial_{i_1 \dots i_m} f(x_0 + th) h_{i_1} \dots h_{i_m} = m! \sum_{|\alpha|=m} \partial_\alpha f(x_0 + th) \frac{h^\alpha}{\alpha!},$$

where we used a combinatorial count to re-write the last sum. Indeed unwrapping the definitions one has

$$h_{i_1} \dots h_{i_m} = h^\alpha \Leftrightarrow \alpha_r = \#\{j \in \{1, \dots, m\} : i_j = r\} \text{ for all } r = 1, \dots, n,$$

and so, given a n -multiindex α of length m , the equation $h_{i_1} \dots h_{i_m} = h^\alpha$ has $m!/\alpha!$ solutions: one has to choose α_1 elements among m to be sent to 1, then α_2 elements among the remaining $m - \alpha_1$ to be sent to 2, etc. Hence the total number of solution is

$$\binom{m}{\alpha_1} \binom{m - \alpha_1}{\alpha_2} \binom{m - \alpha_1 - \alpha_2}{\alpha_3} \dots \binom{m - \alpha_1 - \dots - \alpha_{n-1}}{\alpha_n} = \frac{m!}{\alpha_1! \dots \alpha_n!} = \frac{m!}{\alpha!}$$

Substituting this into (10.5), we obtain the theorem. $\square$

COROLLARY 10.34: PRACTICAL COMPUTATION OF HIGHER DERIVATIVES (F)

Let $x_0 \in U$, $f \in C^{k+1}(U)$ and $P(x)$ a polynomial of degree $k \geq 0$. Assume that

$$|f(x_0 + h) - P(h)| = o(|h|^k) \quad \text{as } h \rightarrow 0.$$

Then $D^j f(x_0)(h, \dots, h) = D^j P(h)$, for all $0 \leq j \leq k$.

Proof. By Taylor's theorem we immediately get

$$\left| P(h) - \sum_{k=0}^d \frac{1}{k!} D^k f(x_0)(h, \dots, h) \right| = o(|h|^k),$$

but two polynomials of degree k whose difference is $o(|h|^k)$ must have exactly the same coefficients. $\square$

This Corollary can be useful to compute Taylor polynomials for explicit functions, without having to care about factorials etc.

EXAMPLE 10.35. — (F) Let us compute the Taylor polynomial of degree 2 of $\sqrt{1+x-y^2}$ around the origin. From Analysis I, you know that

$$\sqrt{1+t} = 1 + \frac{t}{2} - \frac{t^2}{8} + O(t^3), \quad t \rightarrow 0.$$

Plugging in $t = x - y^2$ we find

$$\begin{aligned} \sqrt{1+x-y^2} &= 1 + \frac{x-y^2}{2} - \frac{(x-y^2)^2}{8} + O((x-y^2)^3) \\ &= 1 + \frac{1}{2}x - \frac{1}{8}x^2 - \frac{1}{2}y^2 + \frac{1}{4}xy^2 - \frac{1}{8}y^4 + O((x-y^2)^3). \end{aligned}$$

Now we want $(x, y) \rightarrow (0, 0)$, and a reminder which is $o(r^2)$ where $r := \sqrt{x^2 + y^2}$. Observing that

$$\frac{1}{4}xy^2 = O(r^3), \quad \frac{1}{8}y^4 = O(r^4), \quad O((x-y^2)^3) = O(r^3), \quad \text{as } r \downarrow 0,$$

we find out expansion

$$\sqrt{1+x-y^2} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 - \frac{1}{2}y^2 + O(r^3), \quad \text{as } r \downarrow 0.$$

EXERCISE 10.36. — (F) Compute the Taylor polynomials up to the quadratic order at $(0, 0)$ of the following functions in two variables

$$\sin(xy), \quad \sqrt{1+x+y^2}, \quad \exp \arctan(x-y), \quad \frac{1}{1-x^2-y^2}, \dots$$

Don't use the general formula!

EXERCISE 10.37. — (F) Prove the Taylor expansion of the determinant close to the identity is

$$\det(I + tX) = 1 + t \operatorname{tr}(X) + \frac{t^2}{2} (\operatorname{tr}(X)^2 - \operatorname{tr}(X^2)) + O(t^3),$$

and that the one of the inverse matrix function is

$$(I + tX)^{-1} = 1 - tX + t^2 X^2 + O(t^3).$$

10.38. — The formula in Theorem 10.33 is called the **Taylor expansion with remainder** of f at the point x . The main term

$$P(h) = f(x) + \sum_{k=1}^d \frac{1}{k!} D^k f(x)(h, \dots, h)$$

is exactly like in the one-dimensional Taylor approximation, a polynomial function of h , but this time in the d variables $h_1, \dots, h_n$. Here, $D^k f(x)(h, \dots, h)$ is precisely the homogeneous part of degree k . The integral

$$R(h) = \int_0^1 \varphi^{(d+1)}(t) \frac{(1-t)^d}{d!} dt$$

is called the **remainder**. The estimation $R(h) = O(\|h\|^{d+1})$ follows from the one-dimensional case.

Applet 10.39 ([Taylor Approximation](#)). We observe how the first, second, or third-order Taylor approximations approximate the function $f(x, y) = \sin(x) \cos(y) + 2$.

Chapter 11

Potentials, Optimization and Convexity

TODO: Euler identity for homogeneous functions

11.1 Optimization Problems

11.1.1 Critical Points

Let $U \subseteq \mathbb{R}^n$ be open and non-empty. We discuss the relationship between derivatives and extrema of real-valued functions $f : U \rightarrow \mathbb{R}$. As with functions in one variable, the vanishing of the derivative is a necessary, but not sufficient, condition for the presence of an extremum.

11.1. — Recall that an element $x_0 \in U$ is called a **local maximum** of f if there exists $\delta > 0$ such that $f(x) \leq f(x_0)$ for all $x \in B(x_0, \delta)$. We say x_0 is an **isolated local maximum** or **strict local maximum** if there exists $\delta > 0$ such that $f(x) < f(x_0)$ for all $x \in B(x_0, \delta)$ with $x \neq x_0$. The definition of a **local minimum** is analogous, and collectively, we refer to them as **local extrema**.

PROPOSITION 11.2: THE DIFFERENTIAL VANISH AT A LOCAL EXTREMUM

Let $U \subset \mathbb{R}^n$ be open, $f : U \rightarrow \mathbb{R}$ be a function, and let $x_0 \in U$ be a point where f is differentiable and assumes a local extremum. Then $Df(x_0) = 0$.

Proof. Without loss of generality, assume that f attains a local maximum at x_0 . For all $j \in \{1, \dots, n\}$ and sufficiently small $h > 0$, we have, by assumption,

$$f(x_0 + he_j) - f(x_0) \leq 0 \quad \text{and} \quad f(x_0 - he_j) - f(x_0) \leq 0$$

and, consequently, $\partial_j f(x_0) = 0$ due to

$$\partial_j f(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + he_j) - f(x_0)}{h} \leq 0, \quad \partial_j f(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 - he_j) - f(x_0)}{-h} \geq 0.$$

Since $j \in \{1, \dots, n\}$ was arbitrary, $Df(x_0) = 0$. □

INTERLUDE: THE SIGN OF A SQUARE SYMMETRIC MATRIX (F)

Let $A \in \mathbb{R}^{n \times n}$ be a symmetric matrix and let

$$q_A(x) := \sum_{i,j} A_{i,j} x_i x_j, \quad x \in \mathbb{R}^n,$$

be its associated quadratic polynomial. We say that

- (1) A is positive definite if $q_A(x) > 0$ for all $x \neq 0$.
- (2) A is negative definite if $q_A(x) < 0$ for all $x \neq 0$.
- (3) A is indefinite if q_A takes both positive and negative values.
- (4) A is degenerate if $\det A = 0$.

These conditions can be equivalently stated in terms of the eigenvalues of A (which can be diagonalised thanks to the Spectral Theorem).

- (1) All eigenvalues are positive.
- (2) All eigenvalues are negative.
- (3) At least one eigenvalue is positive and at least one is negative.
- (4) Zero is an eigenvalue.

PROPOSITION 11.3:

Let $U \subset \mathbb{R}^n$ be open, $f : U \rightarrow \mathbb{R}$ be twice continuously differentiable, and let $x_0 \in U$ with $Df(x_0) = 0$. Let $H(x_0)$ be the Hessian matrix of f at point x_0 .

- (1) If $H(x_0)$ is positive definite, then f has a strict local minimum at x_0 .
- (2) If $H(x_0)$ is negative definite, then f has a strict local maximum at x_0 .
- (3) If $H(x_0)$ is indefinite and non-degenerate, then f has no local extremum at x_0 .
- (4) If $H(x_0)$ is degenerate (i.e., has zero determinant), then x_0 might, or might not, be an extremum point: the Hessian test is inconclusive.

Proof. The Hessian matrix $H(x_0)$ is the matrix of the second derivative of f at x_0 as a symmetric bilinear form $D^2f(x_0) : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$. Let Q denote the associated quadratic form.

$$Q(h) = Df(x_0)(h, h) = \langle h, H(x_0)h \rangle$$

By Taylor Theorem (Theorem 10.33), we have

$$f(x_0 + h) - f(x_0) = \frac{1}{2}\|h\|^2 \left(Q\left(\frac{h}{\|h\|}\right) + \alpha(x_0, h) \right) \quad (11.1)$$

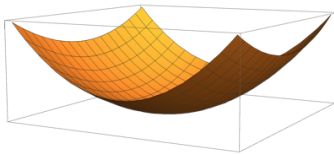
where $\alpha(x_0, h) = o(1)$ as $h \rightarrow 0$. If Q is positive definite, then $Q(w) > 0$ for all $w \in \mathbb{S}^{n-1} = \{v \in \mathbb{R}^n \mid \|v\| = 1\}$. Since $\mathbb{S}^{n-1}$ is compact by the Heine-Borel Theorem ??, and Q is continuous, by the Compactness of Metric Spaces Theorem ??, there exists $c > 0$ such that $Q(w) \geq c$ for all $w \in \mathbb{S}^{n-1}$. Furthermore, there exists $\delta > 0$ such that the error term $\alpha(x_0, h)$ is smaller in absolute value than $\frac{c}{2}$ for $h \in \mathbb{R}^n$ with $\|h\| < \delta$. It follows that

$$f(x_0 + h) - f(x_0) \geq \frac{1}{2}\|h\|^2 \left(Q\left(\frac{h}{\|h\|}\right) - \frac{c}{2} \right) \geq \frac{c}{4}\|h\|^2 > 0$$

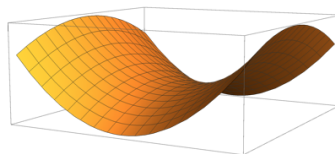
for all $h \in B(0, \delta) \setminus \{0\}$, implying that f has a strict local minimum at x_0 . If Q is negative definite, we replace f with $-f$, which replaces Q with $-Q$. However, the quadratic form $-Q$ is positive definite, and thus, $-f$ has a strict local minimum at x_0 , proving the second statement of the corollary.

If Q is indefinite but non-degenerate, then there exist $w_-, w_+ \in \mathbb{S}^{n-1}$ with $Q(w_-) < 0$ and $Q(w_+) > 0$. For sufficiently small $s \in \mathbb{R} \setminus \{0\}$, we have $|\alpha(x_0, sw_-)| < \frac{1}{2}|Q(w_-)|$ and $|\alpha(x_0, sw_+)| < \frac{1}{2}Q(w_+)$, and thus, $f(x_0 + sw_-) - f(x_0) < 0$ and $f(x_0 + sw_+) - f(x_0) > 0$ from (11.1). Therefore, f has neither a local minimum nor a local maximum at x_0 . $\square$

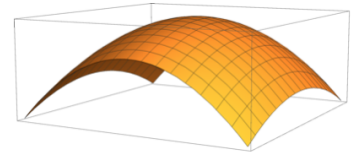
EXAMPLE 11.4. — The behavior of the following functions $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ at the point $0 \in \mathbb{R}^2$ illustrates the three cases in the corollary. In the indefinite case, it is also referred to as a **saddle point**.



$$f(x, y) = x^2 + y^2$$



$$f(x, y) = x^2 - y^2$$



$$f(x, y) = -x^2 - y^2$$

The corresponding Hessian matrices are $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$, $\begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$, and $\begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix}$. If the Hessian matrix is degenerate, i.e., if 0 is an eigenvalue of $H(x_0)$, then generally nothing can be said. The function $f(x, y) = ax^4 + by^4$ has a local maximum, a local minimum, or neither at 0, and the Hessian matrix at 0 is the zero matrix regardless of the choice of a and b .

EXAMPLE 11.5. — Let $a, b \in \mathbb{R}$ be fixed parameters. We define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $f(x, y) = x \sin(y) + ax^2 + by^2$ for $(x, y) \in \mathbb{R}^2$ and consider the point $0 \in \mathbb{R}^2$. We have $Df(0) = 0$, and the Hessian matrix of f at 0 is given by

$$H = \begin{pmatrix} 2a & 1 \\ 1 & 2b \end{pmatrix}$$

with $\det H = 4ab - 1$. We obtain the following cases.

- If $a > 0$ and $4ab - 1 > 0$, then H is positive definite, and f has a local minimum at 0.
- If $a < 0$ and $4ab - 1 > 0$, then H is negative definite, and f has a local maximum at 0.
- If $\det H = 4ab - 1 = 0$, then the Hessian matrix is degenerate.
- If $4ab - 1 < 0$, then H is indefinite, and 0 is a saddle point.

EXERCISE 11.6. — Let $\alpha \in \mathbb{R}$. Find all points $(x, y) \in \mathbb{R}^2$ where the derivative of the function given by $f(x, y) = x^3 - y^3 + 3\alpha xy$ vanishes. Determine whether each point is an extremum and, if so, whether it is a local minimum or maximum.

11.1.2 Convexity

11.1.3 Extrema with Constraints and Lagrange Multipliers

In this paragraph we give fairly concrete recepis to tackle the following problem

$$\min\{f(x) : x \in \overline{U}, g_1(x) = \dots = g_k(x) = 0\} \quad (11.2)$$

where

- $U \subset \mathbb{R}^n$ is open and bounded
- $f: \overline{U} \rightarrow \mathbb{R}$ is of class C^1
- for some $k \geq 0$, $g_1, \dots, g_k: U \rightarrow \mathbb{R}$ are C^1 functions, with the understanding that when $k = 0$ this constraint is always true.

We are going to prove that

- a minimum point exists $x_0 \in \overline{U}$
- if $x_0 \notin \partial U$, then there are coefficients $\lambda_0, \dots, \lambda_k$, not all zero, such that

$$\lambda_0 Df(x_0) + \lambda_1 Dg_1(x_1) + \dots + \lambda_k Dg_k(x_0) = 0,$$

in particular if $k = 0$ then we simply have $Df(x_0) = 0$.

PROPOSITION 11.7: LAGRANGE MULTIPLIERS

Let $U \subset \mathbb{R}^n$ be open, $B_r(x_0) \subset U$, $k \leq n$ and $f, g_1, \dots, g_k$ functions in $C^1(U, \mathbb{R})$. Assume that

$$X := \{x \in U : g_1(x) = \dots = g_k(x) = 0\} \neq \emptyset,$$

and that $f|_X$ has a local minimum at $x_0 \in X$:

$$f(x) \geq f(x_0) \text{ for all } x \in X \cap B_r(x_0).$$

Then there are real numbers $\lambda_0, \dots, \lambda_k$ such that

$$\lambda_0 Df(x_0) + \lambda_1 Dg_1(x_0) + \dots + \lambda_k Dg_k(x_0) = 0, \quad \lambda_0^2 + \dots + \lambda_k^2 = 1.$$

Proof “à la De Giorgi”. Replacing f with $f + |\cdot - x_0|^4$ we may assume that x_0 is a strict local minimum, without changing $Df(x_0)$.

Step 1. Consider $f_\varepsilon(x) := f(x) + \frac{1}{2\varepsilon}(g_1^2(x) + \dots + g_k(x)^2)$, for $x \in \overline{B}_r(x_0)$ and let x_ε be its minimum point by compactness.

Step 2. By contradiction one must have $f_\varepsilon(x_\varepsilon) \rightarrow f(x_0)$ and $x_\varepsilon \rightarrow x_0$, as $\varepsilon \rightarrow 0$. Indeed we have

$$g_1^2(x_\varepsilon) + \dots + g_k(x_\varepsilon)^2 \leq 2\varepsilon f_\varepsilon(x_\varepsilon) \leq 2\varepsilon f(x_0) \rightarrow 0,$$

so if $x_\varepsilon \rightarrow \bar{x}$, then $\bar{x} \in X$. Hence we have

$$f(x_0) \leq f(\bar{x}) = \lim_{\varepsilon} f(x_\varepsilon) \leq \liminf_{\varepsilon} f_\varepsilon(x_\varepsilon) \leq \limsup_{\varepsilon} f_\varepsilon(x_\varepsilon) \leq f(x_0),$$

which implies $\bar{x} = x_0$ (recall that x_0 is a strict local maximum). This proves that $f_\varepsilon(x_\varepsilon) \rightarrow f(x_0)$ and that $\{x_\varepsilon\}$ can only accumulate at x_0 . On the other hand $\{x_\varepsilon\} \subset \overline{B}_r(x_0)$ is bounded so it must have accumulation points. Hence we must have $x_\varepsilon \rightarrow x_0$ as $\varepsilon \downarrow 0$ (no need to take subsequences).

Step 3. In particular $x_\varepsilon \in B_r(x_0)$ eventually, so

$$0 = \varepsilon Df_\varepsilon(x_\varepsilon) = \varepsilon Df(x_\varepsilon) + g_1(x_\varepsilon)Dg_1(x_\varepsilon) + \dots + g_k(x_\varepsilon)Dg_k(x_\varepsilon).$$

This means that the $k+1$ vectors $\{Df(x_\varepsilon), Dg_1(x_\varepsilon), \dots, Dg_k(x_\varepsilon)\}$ are linearly dependent, hence there is a unit vector $\lambda^\varepsilon \in \mathbb{R}^{k+1}$ such that

$$0 = \lambda_0^\varepsilon Df(x_\varepsilon) + \lambda_1^\varepsilon Dg_1(x_\varepsilon) + \dots + \lambda_k^\varepsilon Dg_k(x_\varepsilon).$$

Step 4. Passing this equation to the limit and using the compactness of the k -sphere we conclude. □

11.8. — Often Proposition 11.7 is used in practice in the following way, under the extra assumption that

$$Dg(x) \text{ has rank } k \text{ for all } x \in U,$$

with $g = (g_1, \dots, g_k)$. In this situation we consider the so-called **Lagrangian function**

$$L : U \times \mathbb{R}^m \rightarrow \mathbb{R} \quad L(x, \lambda) = f(x) - \sum_{j=1}^k \lambda_j g_j(x).$$

The components of $\lambda \in \mathbb{R}^k$ are called **Lagrange multipliers**, then Proposition 11.7 guarantees that there exists $\lambda \in \mathbb{R}^k$ such that the equations

$$\partial_{x_i} L(p, \lambda) = 0 \quad \text{and} \quad \partial_{\lambda_j} L(p, \lambda) = 0$$

are satisfied for all $i \in \{1, \dots, n\}$ and $j \in \{1, \dots, k\}$.

EXAMPLE 11.9. — Consider the function $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $F(x, y) = y^3 - x^2$, and the compact set

$$K = \{(x, y) \in [-1, 1]^2 \mid F(x, y) = 0\}$$

and aim to find the minimum of the function $f(x, y) = 4y - 3x$ on K . The set $M = K \setminus \{(0, 0), (1, 1), (-1, 1)\}$ is a one-dimensional manifold. We have

$$f(0, 0) = 0, \quad f(1, 1) = 1, \quad f(-1, 1) = 7.$$

Now, using the method of Lagrange multipliers, we seek the local extremum values of $f|_M$. The Lagrange function associated with f and F is given by

$$L(x, y, \lambda) = 4y - 3x - \lambda(y^3 - x^2).$$

The partial derivatives are calculated as

$$\begin{aligned} \partial_x L(x, y, \lambda) &= -3 + 2\lambda x \\ \partial_y L(x, y, \lambda) &= 4 - 3\lambda y^2 \\ \partial_\lambda L(x, y, \lambda) &= -(y^3 - x^2). \end{aligned}$$

From $-3 + 2\lambda x = 0$, we deduce $\lambda \neq 0$ and $x \neq 0$ with $\lambda = \frac{3}{2x}$. Similarly, from $4 - 3\lambda y^2 = 0$, we conclude that $y \neq 0$ and $\lambda = \frac{4}{3y^2}$. Thus, $\frac{3}{2x} = \frac{4}{3y^2}$ or equivalently $x = \frac{9}{8}y^2$. Furthermore, $\partial_\lambda L(x, y, \lambda) = -(y^3 - x^2) = 0$. Substituting $x = \frac{9}{8}y^2$, we obtain

$$0 = y^3 - \left(\frac{9}{8}y^2\right)^2 = y^3 - \frac{9^2}{8^2}y^4 = y^3 \left(1 - \frac{9^2}{8^2}y\right).$$

Since $y \neq 0$, this yields $y = \frac{8^2}{9^2}$ and $x = \frac{9}{8}y^2 = \frac{8^3}{9^3}$. Therefore, using the Lagrange multipliers method, we find a single additional candidate for extremal values:

$$f\left(\frac{8^3}{9^3}, \frac{8^2}{9^2}\right) = 4\frac{8^2}{9^2} - 3\frac{8^3}{9^3} = 1.053\dots$$

The set of all points in K where f attains a local extremum on K is thus contained in

$$\{(0, 0), (1, 1), (1, -1), (\frac{8^3}{9^3}, \frac{8^2}{9^2})\}.$$

The global maximum of f is at the point $(1, 1)$ with a value of 7, and the global minimum is at the point $(0, 0)$ with a value of 0.

Applet 11.10 (Lagrange Multipliers and Normal Vectors). *In this applet, we illustrate Proposition ?? and Corollary ?? using a one-dimensional submanifold. Under this assumption, only one gradient vector ∇F is present, so Proposition ?? states that ∇F and ∇f should be parallel.*

11.2 Relevant examples

11.2.1 Operator norm of a matrix

11.11. — As a first simple application of the Heine-Borel theorem, we want to define a natural norm on the vector space $\text{Mat}_{m,n}(\mathbb{R})$ of real $m \times n$ matrices. First, note that $\text{Mat}_{m,n}(\mathbb{R})$ is isomorphic to $\mathbb{R}^{mn}$ as a real vector space. In particular, any norm on $\mathbb{R}^{mn}$ induces a norm on $\text{Mat}_{m,n}(\mathbb{R})$, and vice versa. All such norms are equivalent according to Theorem ??, so we obtain a canonical induced topology on $\text{Mat}_{m,n}(\mathbb{R})$ following Exercise ??.

EXERCISE 11.12. — Show that $\langle A, B \rangle = \text{tr}(AB^T)$ defines an inner product on $\text{Mat}_{n,n}(\mathbb{R})$. Identify the corresponding norm on $\mathbb{R}^{n^2}$.

DEFINITION 11.13: OPERATOR NORM

The **operator norm** of a matrix $A \in \text{Mat}_{m,n}(\mathbb{R})$ is defined by

$$\|A\|_{\text{op}} = \sup \{ \|Ax\|_2 \mid x \in \mathbb{R}^n \text{ with } \|x\|_2 \leq 1 \},$$

where $\|\cdot\|_2$ represents the Euclidean norm.

PROPOSITION 11.14:

For $m, n \in \mathbb{N}$, the operator norm $\|\cdot\|_{\text{op}}$ indeed defines a norm on the vector space $\text{Mat}_{m,n}(\mathbb{R})$. Furthermore, the following inequalities hold:

$$\|Ax\|_2 \leq \|A\|_{\text{op}} \|x\|_2 \quad \text{and} \quad \|AB\|_{\text{op}} \leq \|A\|_{\text{op}} \|B\|_{\text{op}}, \quad (11.3)$$

for all $x \in \mathbb{R}^n$ and all $A, B \in \text{Mat}_{m,n}(\mathbb{R})$.

Proof. For the sake of brevity, we write $\overline{B}_n = \{x \in \mathbb{R}^n \mid \|x\|_2 \leq 1\}$. According to the Heine-Borel theorem ??, $\overline{B}_n \subset \mathbb{R}^n$ is compact. Since the function $\overline{B}_n \rightarrow \mathbb{R}$ given by $x \mapsto \|Ax\|_2$ is continuous as a composition of continuous functions, by Theorem ??, it is bounded, implying $\|A\|_{\text{op}} < \infty$.

Definiteness, i.e., $\|A\|_{\text{op}} = 0 \iff A = 0$, and the relations

$$\|\lambda A\|_{\text{op}} = |\lambda| \|A\|_{\text{op}} \quad \text{and} \quad \|A + B\|_{\text{op}} \leq \|A\|_{\text{op}} + \|B\|_{\text{op}}$$

follow directly from the definition of the operator norm and the corresponding properties of the Euclidean norm.

For the left inequality in (11.3), it is noted that there is nothing to prove in the case of $x = 0$. If $x \neq 0$, then

$$\|Ax\|_2 = \|A \frac{x}{\|x\|_2}\|_2 \leq \|A\|_{\text{op}} \|x\|_2.$$

For the second inequality in (11.3), one calculates $\|ABx\|_2 \leq \|A\|_{\text{op}} \|Bx\|_2 \leq \|A\|_{\text{op}} \|B\|_{\text{op}} \|x\|_2$ for all $x \in \overline{B}_n$, proving the claim. $\square$

EXERCISE 11.15. — Show that for $B \in \text{Mat}_{m,m}(\mathbb{R})$ with $\|B\|_{\text{op}} < 1$, the matrix $I_m - B$ is invertible with

$$(I_m - B)^{-1} = \sum_{j=0}^{\infty} B^j.$$

Part of the exercise is to show that this sequence converges.

Upgrade your result showing that the set of invertible matrices in $\text{Mat}_{m,m}(\mathbb{R})$ is open and that for $B \in \text{GL}_m(\mathbb{R})$, any $C \in \text{Mat}_{m,m}(\mathbb{R})$ with $\|C - B\|_{\text{op}} < \|B^{-1}\|_{\text{op}}^{-1}$ is also invertible.

11.2.2 Fundamental Theorem of Algebra

11.16. — A second application of the Heine-Borel theorem is a proof of the Fundamental Theorem of Algebra. The **Fundamental Theorem of Algebra** states that every non-constant polynomial with complex coefficients has a complex root. This is equivalent to the statement that the field $\mathbb{C}$ is **algebraically closed**. Polynomial division with remainder and induction show that any polynomial $f \in \mathbb{C}[T]$ of degree $n > 0$ can be written as the product of a scalar and n linear factors, i.e.,

$$f(T) = a \prod_{j=1}^n (T - \alpha_j) \tag{11.4}$$

with $a \in \mathbb{C}^\times$ and $\alpha_1, \dots, \alpha_n \in \mathbb{C}$. There are various proofs of the Fundamental Theorem of Algebra; some rely on complex analysis, while others on algebraic topology. Here, we will use the asymptotic behavior of polynomials, the Heine-Borel theorem, and the polar decomposition of complex numbers.

A Fundamental Theorem can also be formulated for polynomials with real coefficients. We note that for every $f \in \mathbb{R}[x]$ with a root $\alpha \in \mathbb{C}$, $\bar{\alpha}$ is also a root. In the factorization of f into linear factors in (11.4), for every non-real root $\alpha \in \mathbb{C} \setminus \mathbb{R}$, we have the factor

$$(x - \alpha)(x - \bar{\alpha}) = x^2 - (\alpha + \bar{\alpha})x + |\alpha|^2 = x^2 - 2\text{Re}(\alpha)x + |\alpha|^2$$

standing. Applying this insight to every genuinely complex root of f shows that f can be written as a product of real polynomials of degree 1 and degree 2. We have implicitly used this in the discussion of the integration of rational functions.

THEOREM 11.17: FUNDAMENTAL THEOREM OF ALGEBRA

Every non-constant polynomial $f \in \mathbb{C}[T]$ has a root in $\mathbb{C}$.

Proof. Let $f \in \mathbb{C}[T]$ be a polynomial of degree $n > 0$. If $f(0) = 0$, we have already found a root. So, assume $M = 2|f(0)| > 0$. Since f is not constant, according to Proposition ??, there exists an $R \geq 1$ such that for all $z \in \mathbb{C}$ with $|z| \geq R$, $|f(z)| \geq M$ holds. According to the Heine-Borel theorem ??,

$$K = \{z \in \mathbb{C} \mid |z| \leq R\}$$

is a compact subset. We now apply the existence of the minimum value in Theorem ?? to the continuous function $z \mapsto |f(z)| \in \mathbb{R}$ and find $z_0 \in K$ with $|f(z_0)| = \min\{|f(z)| \mid z \in K\}$. Since $|f(0)| = \frac{M}{2}$, the inequality $|f(z_0)| < M$ holds. Since $|f(z)| \geq M$ for all $z \in \mathbb{C} \setminus K$, we obtain $|f(z)| \geq |f(z_0)|$ for all $z \in \mathbb{C}$.

We claim that z_0 is a root of f . Instead, assume that $|f(z_0)| > 0$, leading to a contradiction. We represent f as a power series around z_0 :

$$f(z) = \sum_{k=0}^n b_k (z - z_0)^k$$

with $b_0 = f(z_0)$, $b_1 = f'(z_0)$, $\dots$, $b_n = \frac{f^{(n)}(z_0)}{n!} \in \mathbb{C}$. The existence of this representation follows from polynomial division with remainder and induction on $n = \deg(f)$. By the assumption on z_0 , $b_0 \neq 0$. Let $\ell \geq 1$ be the smallest index ≥ 1 with $b_\ell \neq 0$. Set $z = z_0 + r \exp(i\varphi)$ for a fixed $\varphi \in \mathbb{R}$, which we will choose precisely later, and a varying $r > 0$. We have

$$f(z_0 + r e^{i\varphi}) = b_0 + b_\ell r^\ell e^{i\ell\varphi} + O(r^{\ell+1}) = b_0 \left(1 + \frac{b_\ell}{b_0} r^\ell e^{i\ell\varphi}\right) + O(r^{\ell+1})$$

as $r \rightarrow 0$. Write $\frac{b_\ell}{b_0} = s e^{i\psi}$ and choose $\varphi = \frac{-\psi + \pi}{\ell}$. Then, $e^{i(\ell\varphi + \psi)} = -1$ and

$$|f(z_0 + r e^{i\varphi})| = |b_0 (1 - s r^\ell) + O(r^{\ell+1})| \leq |b_0| (1 - s r^\ell) + O(r^{\ell+1})$$

for $r \rightarrow 0$. For sufficiently small $r > 0$, this upper bound is smaller than $|b_0|$, leading to a contradiction with $|f(z_0)| = |b_0| = \min\{|f(z)| \mid z \in K\}$. This proves that $f(z_0) = 0$ must hold. $\square$

EXERCISE 11.18. — Let $U \subset \mathbb{C}$ be open, and $f : U \rightarrow \mathbb{C}$ be a complex-valued function that can be locally represented by power series (an analytic function). More precisely, for every $x_0 \in U$, there exists an $r > 0$ such that $B(x_0, r) \subset U$, and f on $B(x_0, r)$ is equal to a power series around x_0 with a convergence radius greater than or equal to r . Mimicking the proof of Theorem 11.17, show that the function $z \mapsto |f(z)|$ does not assume a minimum value on U .

11.2.3 Diagonalizability of Symmetric Matrices

With the method of Lagrange multipliers, we can relatively easily prove the following important theorem from linear algebra.

THEOREM 11.19:

Every symmetric matrix $A \in \text{Mat}_{n,n}(\mathbb{R})$ is diagonalizable, and there exists an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of A .

LEMMA 11.20:

Let $n \geq 1$ and $A \in \text{Mat}_{n,n}(\mathbb{R})$ be a symmetric matrix. Then, A has a real eigenvector.

Proof. Consider the sphere $\mathbb{S} = \mathbb{S}^{n-1}$ as the zero set of the function $F : \mathbb{R}^n \rightarrow \mathbb{R}$, given by $F(x) = \|x\|^2 - 1$, and the real-valued function

$$f : \mathbb{R}^n \rightarrow \mathbb{R}, \quad f(x) = x^t A x.$$

Since $\mathbb{S}$ is compact, $f|_{\mathbb{S}}$ attains a maximum and a minimum. Assume $f|_{\mathbb{S}}$ has an extremum at $p \in \mathbb{S}$. Define

$$L : \mathbb{R}^n \times \mathbb{R}, \quad L(\lambda, x) = f(x) - \lambda F(x) = x^t A x - \lambda(\|x\|^2 - 1)$$

as the Lagrange function associated with $\mathbb{S}$ and f . According to Corollary ??, there exists $\lambda \in \mathbb{R}$ such that

$$\partial_{x_j} L(p, \lambda) = \partial_j f(p) - \lambda \partial_j F(p) = 0 \tag{11.5}$$

for all $j = 1, \dots, n$ and $\partial_\lambda L(p, \lambda) = F(p) = 0$. The latter simply implies $\|p\| = 1$, or in other words, $p \in \mathbb{S}$, as already known.

Now, let's compute the partial derivatives of F and f . For all $x \in \mathbb{R}^n$, we have

$$\begin{aligned} \partial_j F(x) &= \partial_j \left(\sum_{k=1}^n x_k^2 \right) = 2x_j \\ \partial_j f(x) &= \partial_j \left(\sum_{k,\ell=1}^n a_{k\ell} x_k x_\ell \right) = \sum_{\ell=1}^n a_{j\ell} x_\ell + \sum_{k=1}^n a_{kj} x_k \end{aligned}$$

by the product rule, and since $\partial_j(x_k)$ is zero unless $k = j$. Assuming A is symmetric, we obtain

$$\partial_j f(x) = 2 \sum_{\ell=1}^n a_{j\ell} x_\ell = 2(Ax)_j.$$

Thus, $(Ap)_j = \lambda p_j$ for all $j = 1, \dots, n$, or equivalently $Ap = \lambda p$. □

Proof of Theorem 11.19. In addition to Lemma 11.20, we need some more linear algebra for the proof, which we will now carry out by induction on n . For $n = 1$, there is nothing to prove. So, let $A \in \text{Mat}_{n,n}(\mathbb{R})$ be a symmetric matrix. According to Lemma 11.20, there exists a real eigenvector $v_1 \in \mathbb{S}^{n-1}$ corresponding to an eigenvalue $\lambda_1 \in \mathbb{R}$. Consider the orthogonal complement

$$W = v_1^\perp = \{w \in \mathbb{R}^n \mid \langle w, v_1 \rangle = 0\}$$

of v_1 . For $w \in W$, due to the symmetry of A , we have

$$\langle Aw, v_1 \rangle = \langle w, Av_1 \rangle = \lambda_1 \langle w, v_1 \rangle = 0$$

and it follows that $A(W) \subset W$. Let $w_1, \dots, w_{n-1}$ be an orthonormal basis of W with respect to $\langle \cdot, \cdot \rangle$ (which exists due to the Gram-Schmidt orthonormalization process). For $i, j \in \{1, \dots, n-1\}$, we now have

$$\langle Aw_j, w_i \rangle = \langle w_j, Aw_i \rangle = \langle Aw_i, w_j \rangle.$$

In other words, the basis representation B of $A|_W : W \rightarrow W$ with respect to the basis $w_1, \dots, w_{n-1}$ is symmetric again. By the induction hypothesis, there exists an orthonormal basis $v_2, \dots, v_n$ of eigenvectors of B . Since B (together with the standard basis of $\mathbb{R}^{n-1}$) corresponds exactly to $A|_W$ (together with the orthonormal basis $w_1, \dots, w_{n-1}$), there exists an orthonormal basis $v_2, \dots, v_n$ of eigenvectors of A .

Thus, $v_1, \dots, v_n$ forms an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of A . $\square$

EXERCISE 11.21. — For the sake of completeness, we present an elementary argument for the proof of Lemma 11.20 using the Fundamental Theorem. Let $n \geq 1$ and $A \in \text{Mat}_{n,n}(\mathbb{R})$ be a symmetric matrix.

- (i) Show that all complex eigenvalues of A are real.
- (ii) Prove Lemma 11.20 by showing that A has a complex eigenvector if and only if A has a real eigenvector.

The geometric understanding of the eigenvalues of A gained in our proof can also be utilized differently. As an example, one can prove a special case of the Courant-Fischer theorem.

- (iii) Show that the values

$$\min_{x \in \mathbb{S}^{n-1}} x^t A x, \quad \max_{x \in \mathbb{S}^{n-1}} x^t A x$$

represent the smallest and largest eigenvalue of A , respectively.

EXERCISE 11.22. — For $n \geq 2$, prove that two points $x, y \in \mathbb{S}^{n-1}$ have maximum distance if and only if $x = -y$. Consider the function $(x, y) \mapsto \|x - y\|^2$ on $\mathbb{S}^{n-1} \times \mathbb{S}^{n-1} \subset \mathbb{R}^{2n}$.

11.3 Potentials and the equation $Du = F$

11.3.1 The work of a Vector Field along a line

DEFINITION 11.23: NON-DEFINITION OF VECTOR FIELD

Let $U \subset \mathbb{R}^n$ be open. A **vector field** on U is a function $F : U \rightarrow \mathbb{R}^n$. Continuous, continuously differentiable, or smooth vector fields are functions $F : U \rightarrow \mathbb{R}^n$ with corresponding regularity. If we write

$$F(x) = (F_1(x), \dots, F_n(x)),$$

then functions $\{F_i\}$, from U to $\mathbb{R}^n$, are called the **components** of the vector field.

11.24. — We often visualize vector fields by drawing the vector $F(x)$ with the point $x \in U$. In physics, vector fields are often force fields or indicate the flow velocity of a medium.

DEFINITION 11.25: WORK OF A FIELD ALONG A PATH

Let $U \subset \mathbb{R}^n$ be an open subset, and let $F : U \rightarrow \mathbb{R}^n$ be a continuous vector field and $\gamma : [a, b] \rightarrow U$ a continuously differentiable path. We define the **work of F along γ** as

$$\int_{\gamma} F := \int_a^b F(\gamma(t)) \cdot \gamma'(t) dt.$$

If γ is piecewise continuously differentiable with respect to a partition $a = t_0 < t_1 < \dots < t_N = b$, the integral is interpreted as the sum of integrals over intervals $[t_{k-1}, t_k]$.

In Physics, one often use the notation

$$\int_{\gamma} F = \int_{\gamma} \vec{F} \cdot d\vec{\ell}.$$

LEMMA 11.26:

Let $U \subset \mathbb{R}^n$ be an open subset, $f : U \rightarrow \mathbb{R}^n$ be a continuous vector field, let $\gamma : [a, b] \rightarrow \mathbb{R}^d$ be a continuously differentiable path and let $\psi : [0, 1] \rightarrow [a, b]$ be a C^1 function such that $\psi(0) = a, \psi(1) = b$. Then

$$\int_{\gamma} F = \int_{\gamma \circ \psi} F.$$

Proof. This is a consequence of the change of variable formula of Analysis I and the chain rule:

$$\int_{\gamma \circ \psi} F dt = \int_0^1 F(\gamma(\psi(t))) \cdot (\gamma \circ \psi)'(t) dt = \int_0^1 F(\gamma(\psi(t))) \cdot \gamma'(\psi(t)) \psi'(t) dt$$

$$= \int_a^b F(\gamma(s)) \cdot \gamma'(s) ds = \int_{\gamma} F.$$

□

LEMMA 11.27: **TODO: IT SEEMS TO ME THAT IT IS NOT USED LATER...**

Let $U \subset \mathbb{R}^n$ be open, and let $\gamma : [0, 1] \rightarrow U$ be a piecewise continuously differentiable path. Then there exists a continuously differentiable reparametrization $\varphi = \gamma \circ \psi$. Moreover, $\varphi'(0) = \varphi'(1) = 0$ can be arranged.

Proof. Let $0 = s_0 < s_1 < \dots < s_N = 1$ be a suitable partition of the unit interval. There exists a smooth function $\beta : [0, 1] \rightarrow \mathbb{R}$ with non-negative values such that $\beta(s_k) = 0$ for $k = 0, 1, 2, \dots, N$, and

$$\int_{s_{k-1}}^{s_k} \beta(t) dt = s_k - s_{k-1}$$

for $k = 1, 2, \dots, N$. The function $\psi : [0, 1] \rightarrow [0, 1]$ given by

$$\psi(t) = \int_0^t \beta(s) ds$$

is continuously differentiable, monotonically increasing, and satisfies $\psi(s_k) = s_k$ and $\psi'(s_k) = 0$ for all k . The reparametrization $\varphi = \gamma \circ \psi$ is thus continuously differentiable, and $\varphi'(s_k) = 0$ for all k , in particular, $\varphi'(s_0) = \varphi'(0) = 0$ and $\varphi'(s_N) = \varphi'(1) = 0$. □

DEFINITION 11.28: **POTENTIAL OF A VECTOR FIELD**

Let $U \subset \mathbb{R}^n$ be open, and $F : U \rightarrow \mathbb{R}^n$ be a continuous vector field. A continuously differentiable function $f : U \rightarrow \mathbb{R}$ is called a **potential** for F , if $F = \text{grad}(f)$ holds. That is to say

$$D_i f(x) = F_i(x) \text{ for all } x \in U, 1 \leq i \leq n.$$

If F admits a potential f in U , we say that F is **conservative in U** .

In the next two propositions we show that a vector field F admits a potential if and only if the value of $\gamma \mapsto \int_{\gamma} F$ depends only on the endpoints of γ .

PROPOSITION 11.29:

Let $U \subset \mathbb{R}^n$ be open, and let $F : U \rightarrow \mathbb{R}^n$ be a continuous vector field. Suppose there exists a potential $f : U \rightarrow \mathbb{R}$ for F . Then

$$\int_{\gamma} F = f(\gamma(1)) - f(\gamma(0))$$

for any piecewise continuously differentiable path $\gamma : [0, 1] \rightarrow U$.

Proof. If $\gamma : [0, 1] \rightarrow U$ is a continuously differentiable path, then for $t \in [0, 1]$, $F(\gamma(t)) = \text{grad } f(\gamma(t)) = Df(\gamma(t))$, and thus

$$\int_{\gamma} F dt = \int_0^1 \langle F(\gamma(t)), \gamma'(t) \rangle dt = \int_0^1 Df(\gamma(t))(\gamma'(t)) dt = \int_0^1 (f \circ \gamma)'(t) dt = f(\gamma(1)) - f(\gamma(0))$$

by the chain rule. If γ is only piecewise continuously differentiable with respect to a partition $0 = s_0 < s_1 < \dots < s_N = 1$, the calculation can be applied to the subintervals $[s_{k-1}, s_k]$. This leads to a telescoping sum where all terms except $f(\gamma(1)) - f(\gamma(0))$ cancel. $\square$

11.30. — One of the many physical interpretations of such path integrals is the calculation of work along a path γ . Suppose $F(x)$ indicates the direction and strength of a force field at point $x \in U$. Then, $\langle F(\gamma(t)), \gamma(t + \delta) - \gamma(t) \rangle$ is approximately the work done when an object moves along the path γ from $\gamma(t)$ to $\gamma(t + \delta)$. This leads to the interpretation of $\int_{\gamma} F dt$ as the performed **work** along the path γ . This total work generally depends on the chosen path, not just the starting point $\gamma(a)$ and the endpoint $\gamma(b)$. However, the work done does not depend on the chosen parameterization of the path, as shown in Lemma 11.26.

EXAMPLE 11.31. — Consider the vector field F on $\mathbb{R}^2$ defined by $F(x, y) = (-y, x)$ and calculate the integral of F along different paths from $(0, 0)$ to $(1, 1)$.

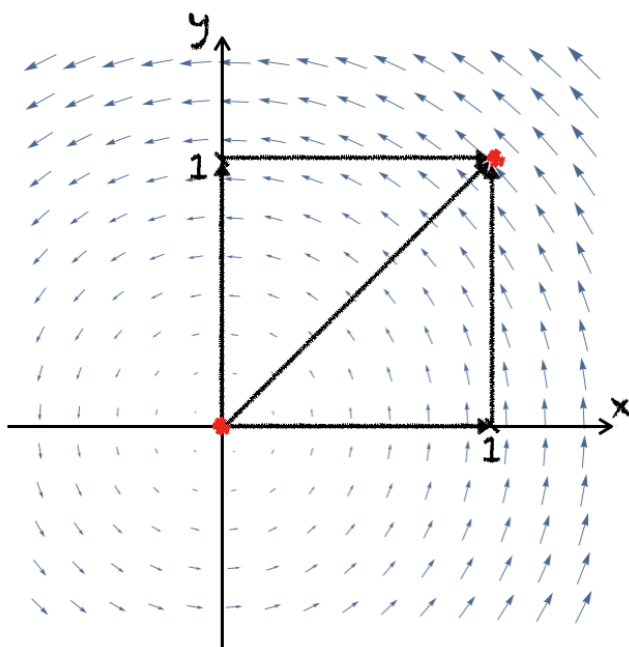


Figure 11.1: The vector field F - vector lengths are scaled by a factor of 0.1.

Let γ_0, γ_1 , and $\gamma_2 : [0, 1] \rightarrow \mathbb{R}^2$ be the paths from $(0, 0)$ to $(1, 1)$ given by $\gamma_0(t) = (t, t)$ and

$$\gamma_1(t) = \begin{cases} (2t, 0) & \text{if } t \in [0, \frac{1}{2}] \\ (1, 2t - 1) & \text{if } t \in [\frac{1}{2}, 1] \end{cases} \quad \gamma_2(t) = \begin{cases} (0, 2t) & \text{if } t \in [0, \frac{1}{2}] \\ (2t - 1, 1) & \text{if } t \in [\frac{1}{2}, 1] \end{cases}$$

Then,

$$\begin{aligned}\int_{\gamma_0} F &= \int_0^1 \langle F(\gamma_0(t)), \gamma_0'(t) \rangle dt = \int_0^1 \langle (-t, t), (1, 1) \rangle dt = 0 \\ \int_{\gamma_1} F &= \int_0^{\frac{1}{2}} \langle (0, 2t), (2, 0) \rangle dt + \int_{\frac{1}{2}}^1 \langle (1-2t, 1), (0, 2) \rangle dt = 1 \\ \int_{\gamma_2} F &= \int_0^{\frac{1}{2}} \langle (-2t, 0), (0, 2) \rangle dt + \int_{\frac{1}{2}}^1 \langle (-1, 2t-1), (2, 0) \rangle dt = -1\end{aligned}$$

We see that the work performed $\int_{\gamma} F$ depends on the chosen path γ . If one moves perpendicular to the vector field, no work is done. If one moves with the vector field, positive work is done, and if one moves against the vector field, negative work is done.

From these calculations, it follows in particular that the vector field F does not possess a potential.

EXERCISE 11.32. — Let $U \subset \mathbb{R}^n$ be open and connected, and let $x_0, x_1 \in U$. Show that x_0 and x_1 can be connected by a continuously differentiable path.

EXERCISE 11.33. — A **loop** in an open subset $U \subset \mathbb{R}^n$ is a path $\gamma : [0, 1] \rightarrow U$ with $\gamma(0) = \gamma(1)$. Show that a continuous vector field $F : U \rightarrow \mathbb{R}^n$ is conservative if and only if, for every piecewise continuously differentiable loop γ in U ,

$$\int_{\gamma} F = 0$$

holds.

PROPOSITION 11.34:

Let $U \subset \mathbb{R}^n$ be open, and $F : U \rightarrow \mathbb{R}^n$ be a continuous vector field. Assume that, for all piecewise continuously differentiable paths $\gamma : [0, 1] \rightarrow U$ and $\sigma : [0, 1] \rightarrow U$, with $\gamma(0) = \sigma(0), \gamma(1) = \sigma(1)$ it holds

$$\int_{\gamma} F = \int_{\sigma} F.$$

Then F is conservative, i.e., it has a continuously differentiable potential $f : U \rightarrow \mathbb{R}$.

Proof. Let $x_0 \in U$ be a fixed point. Since U is connected, there exists, according to Exercise 11.32, for every $x \in U$, a piecewise continuously differentiable path γ_x in U with initial point x_0 and endpoint x . We consider the function $f : U \rightarrow \mathbb{R}$ given by

$$f(x) = \int_{\gamma_x} F$$

which does not depend on the chosen paths γ_x by assumption. For $k = 1, 2, \dots, n$, we verify that $\partial_k f(x) = F_k(x)$, the k -th component of $F(x) \in \mathbb{R}^n$. Let $x \in U$ and $h \in \mathbb{R} \setminus \{0\}$ be small enough so that $x + the_k \in U$ for all $t \in [0, 1]$. Using the path $\gamma_x : [0, 1] \rightarrow U$ from x_0 to x , we can define a path φ_h from x_0 to $x + he_k$ as

$$\varphi_h(t) = \begin{cases} \gamma_x(2t) & \text{if } 0 \leq t \leq \frac{1}{2} \\ x + (2t - 1)he_k & \text{if } \frac{1}{2} \leq t \leq 1 \end{cases}$$

For the partial derivative $\partial_k f$, we obtain

$$\begin{aligned} \partial_k f(x) &= \lim_{h \rightarrow 0} \frac{f(x + he_k) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{\varphi_h} F - \int_{\gamma_x} F \right) \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \int_0^1 \langle F(x + the_k), he_k \rangle dt = \lim_{h \rightarrow 0} \int_0^1 F_k(x + she_k) ds = F_k(x) \end{aligned}$$

due to Theorem 14.80 and the continuity of F . Since this holds for all $x \in U$ and $k = 1, 2, \dots, n$, and $F_1, \dots, F_n$ are continuous by assumption, it follows from Theorem 10.10 that f is differentiable and $\text{grad } f(x) = F(x)$ for all $x \in U$. $\square$

COROLLARY 11.35:

Let $U \subset \mathbb{R}^n$ be open, and let F be a continuously differentiable conservative vector field on U , with components $F_1, \dots, F_n$. Then we necessarily have the **integrability conditions**:

$$\partial_j F_k = \partial_k F_j$$

for all pairs $j, k \in \{1, \dots, n\}$.

Proof. If F is conservative, then, according to Theorem 11.34, there exists a differentiable function f with $\text{grad } f = F$. Then, for $j, k \in \{1, \dots, n\}$, we have

$$\partial_j F_k = \partial_j \partial_k f = \partial_k \partial_j f = \partial_k F_j. \quad (11.6)$$

We are allowed to commute second derivatives because f is $C^2(U)$, since $\text{grad } f = F \in C^1(U, \mathbb{R}^n)$, hence Schwarz Lemma applies. $\square$

11.3.2 The Poincaré Lemma

For a continuously differentiable vector field F on $U \subset \mathbb{R}^n$ to satisfy the partial differential equations (11.6) established in Corollary 11.35 is a necessary condition for the existence of a potential for F . We call these differential equations **integrability conditions** and aim to investigate to what extent they are also sufficient for the existence of a potential for F . The following example shows that the equations 11.35 do not generally imply that F has a potential.

EXAMPLE 11.36. — Let $U = \mathbb{R}^2 \setminus \{0\}$, and consider the vector field $F : U \rightarrow \mathbb{R}^2$ given by

$$F(x, y) = \left(\frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right)$$

for $(x, y) \in U$. A direct calculation shows

$$\partial_1 F_2(x, y) = \partial_x \left(\frac{x}{x^2 + y^2} \right) = \frac{-x^2 + y^2}{(x^2 + y^2)^2} = \partial_y \left(\frac{-y}{x^2 + y^2} \right) = \partial_2 F_1(x, y)$$

thus satisfying the integrability conditions (11.6) throughout U . However, F is not conservative. Let $\gamma : [0, 1] \rightarrow U$ be the continuously differentiable loop defined by

$$\gamma(t) = (\cos(2\pi t), \sin(2\pi t))$$

which rotates once counterclockwise around the unit circle. Then,

$$\int_{\gamma} F dt = 2\pi \int_0^1 \langle (-\sin(2\pi t), \cos(2\pi t)), (-\sin(2\pi t), \cos(2\pi t)) \rangle dt = 2\pi,$$

even though γ is a closed path with $\gamma(0) = \gamma(1) = (1, 0)$.

In order to state the main theorem of this section we need to introduce the concept of *homotopy of curves*.

DEFINITION 11.37:

Let X be a metric space, and let γ_0 and γ_1 be paths in X with the same initial point $x_0 = \gamma_0(0) = \gamma_1(0)$ and the same endpoint $x_1 = \gamma_0(1) = \gamma_1(1)$.

A **homotopy** from γ_0 to γ_1 is a continuous function $H : [0, 1] \times [0, 1] \rightarrow X$ with the following properties:

$$H(0, t) = \gamma_0(t), \quad H(1, t) = \gamma_1(t) \quad \text{and} \quad H(s, 0) = x_0, \quad H(s, 1) = x_1$$

for all $t \in [0, 1]$ and all $s \in [0, 1]$. We say γ_1 is **homotopic to** γ_0 if there exists a homotopy from γ_0 to γ_1 .

11.38. — Let H be a homotopy from γ_0 to γ_1 as in the definition. For each fixed $s \in [0, 1]$, the function $\gamma_s : t \mapsto H(s, t)$ is a path from x_0 to x_1 . For $s = 0$ and $s = 1$, we obtain the given paths γ_0 and γ_1 . This way, we can view the homotopy H as a parametrized family of paths depending continuously on the parameter $s \in [0, 1]$.

EXERCISE 11.39. — Let X be a topological space, $x_0, x_1 \in X$, and $P(x_0, x_1)$ denote the set of all paths in X from x_0 to x_1 . Show that the relation

$$\gamma_0 \sim \gamma_1 \iff \gamma_1 \text{ is homotopic to } \gamma_0$$

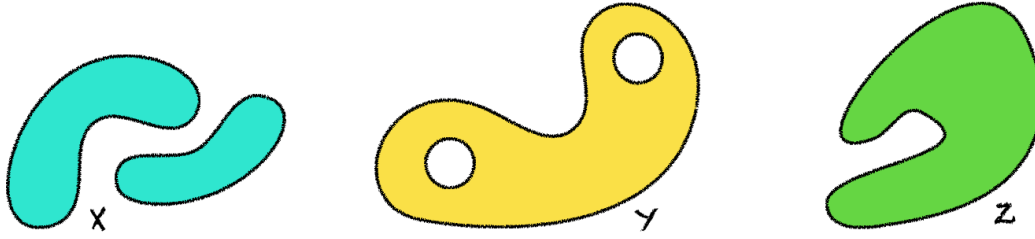


Figure 11.2: A non-connected space X , a connected but not simply connected space Y , and a simply connected space Z .

on the set $P(x_0, x_1)$ is an equivalence relation.

DEFINITION 11.40: SIMPLY CONNECTED SPACE

A metric space X is called **simply connected** if it is path-connected and if, for all $x_0, x_1 \in X$, all paths from x_0 to x_1 are homotopic to each other.

11.41. — A connected topological space X is called **simply connected** if every path γ from x_0 to x_1 in X can be continuously transformed into any other path from x_0 to x_1 .

EXERCISE 11.42. — Show that a simply connected open set $X \subset \mathbb{R}^n$ does not have “co-dimension two” holes. This is the precise meaning: every $f: \partial D \rightarrow X$ can be extended to some $F: D \rightarrow X$ where D denotes the closed bi-dimensional disk. In some sense, any (distorted) loop bounds a (distorted) disk in X .

EXERCISE 11.43. — Let X be a topological space, and let $\gamma: [0, 1] \rightarrow X$ be a path from $x_0 \in X$ to $x_1 \in X$.

- (i) (Reversing Paths) Show that $t \in [0, 1] \mapsto \gamma(1 - t)$ is a path from x_1 to x_0 .
- (ii) (Concatenation of Paths) Suppose $\tilde{\gamma}: [0, 1] \rightarrow X$ is a path from x_1 to $x_2 \in X$. Show that

$$t \mapsto \begin{cases} \gamma(2t) & \text{if } 0 \leq t \leq \frac{1}{2} \\ \tilde{\gamma}(2t - 1) & \text{if } \frac{1}{2} < t \leq 1 \end{cases}$$

defines a path from x_0 to x_2 .

- (iii) Repeat the proof of Proposition 9.89 using this.

INTERLUDE: DOMAINS IN ANALYSIS

Let $U \subset \mathbb{R}^n$ be a non-empty subset, we say that

- (1) is **connected**: The set U cannot be written as the disjoint union of two open, non-empty subsets of U (Definition 9.79).

- (2) is **path-connected**: For any two points $x_0, x_1 \in U$, there exists a path in U from x_0 to x_1 (Definition 9.86).
- (3) is **simply connected**: For any two points $x_0, x_1 \in U$, there exists a path in U from x_0 to x_1 , and between any two such paths, there exists a homotopy (Definition ??).
- (4) is **star-shaped**: There exists an $x_0 \in U$ such that for all $x_1 \in U$ and $t \in [0, 1]$, $(1 - t)x_0 + tx_1 \in U$.
- (5) is **convex**: For all $x_0, x_1 \in U$ and all $t \in [0, 1]$, $(1 - t)x_0 + tx_1$ is also in U .

EXERCISE 11.44. — Show that for a subset $U \subset \mathbb{R}^n$, the implications

$$(5) \implies (4) \implies (3) \implies (2) \implies (1)$$

hold among the properties listed above. Find examples of subsets in $\mathbb{R}^2$ that demonstrate the failure of each reverse implication.

THEOREM 11.45: POINCARÉ LEMMA

Let $U \subset \mathbb{R}^n$ be open, and let $F : U \rightarrow \mathbb{R}^n$ be a continuously differentiable vector field that satisfies the integrability conditions

$$\partial_k F_j = \partial_j F_k \tag{11.7}$$

for all $j, k \in \{1, \dots, n\}$. Let $\gamma_0 : [0, 1] \rightarrow U$ and $\gamma_1 : [0, 1] \rightarrow U$ be piecewise continuously differentiable paths with the same initial point x_0 and the same endpoint x_1 . If γ_0 and γ_1 are homotopic, then

$$\int_{\gamma_0} F = \int_{\gamma_1} F.$$

Theorem 11.45 is an example of a so-called **global integration theorem** because, it has something to do with the global nature of the domain U . This is evident in the following important

COROLLARY 11.46:

Let $U \subset \mathbb{R}^n$ be open and simply connected. A continuously differentiable vector field on U is conservative if and only if it satisfies the integrability conditions (11.7).

Proof. This directly follows from Theorem 11.45, Proposition 11.34 and Definition 11.40. $\square$

Before proving Theorem 11.45 we show a simpler proof under the additional assumption that U is convex.

LEMMA 11.47: POINCARÉ LEMMA FOR CONVEX DOMAINS

Let $U \subset \mathbb{R}^n$ be open and convex, and let $F : U \rightarrow \mathbb{R}^n$ be a continuously differentiable vector field that satisfies the integrability conditions (11.7). Then F is conservative in U .

Proof. The necessity of the integrability conditions was already proven in Corollary 11.35. For the converse, assume without loss of generality that $0 \in U$. We use the path integral of F along the straight line from 0 to $x \in U$ to define a function $f : U \rightarrow \mathbb{R}$ by

$$f(x) = \int_0^1 \langle F(tx), x \rangle dt$$

for $x \in U$. As per the proof of Theorem 11.34, f is a candidate for a potential of F . Fix $j \in \{1, \dots, n\}$ and consider, as a preparation for the computation of $\partial_j f$, for $h \in \mathbb{R}^n$

$$\partial_h F_j = \sum_{k=1}^n h_k \partial_k F_j = \sum_{k=1}^n h_k \partial_j F_k \quad (11.8)$$

by the assumed integrability conditions. According to the Theorem 14.80, $\partial_j f$ exists, and for $x \in U$ it holds

$$\partial_j f(x) = \partial_j \int_0^1 \left(\sum_{k=1}^n F_k(tx) x_k \right) dt = \int_0^1 \left(\sum_{k=1}^n (\partial_j F_k)(tx) t x_k + F_j(tx) \right) dt, \quad (11.9)$$

since only the term with $k = j$ requires the product rule, and the partial derivative of $x \mapsto F_k(tx)$ with respect to x_j is given by $t(\partial_j F_k)(tx)$ for $x \in U$, following from the chain rule. We use (11.8) for $h = x$ in (11.9) and obtain, by partial integration,

$$\partial_j f(x) = \int_0^1 t \underbrace{\partial_x F_j(tx)}_{\partial_j F_j(tx)} dt + \int_0^1 F_j(tx) dt = [t F_j(tx)]_0^1 - \int_0^1 F_j(tx) dt + \int_0^1 F_j(tx) dt = F_j(x)$$

where we have recognized the derivative with respect to t of $F_j(tx)$ in the underbraced term. Thus, $F = \nabla f$, and f is continuously differentiable by Theorem 10.10. $\square$

EXERCISE 11.48. — For which values of $\lambda \in \mathbb{R}$ is the vector field $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$F(x, y) = (\lambda x \exp(y), (y + 1 + x^2) \exp(y))$$

conservative? Determine a potential for f for these values.

11.49. — In rough terms, the proof of Theorem 11.45 goes as follows. Let $U \subset \mathbb{R}^n$ be open, and let $x_1, x_2 \in U$. We equip the set

$$\Omega := \{\gamma : [0, 1] \rightarrow U \mid \gamma(0) = x_0, \gamma(1) = x_1, \text{ and } \gamma \text{ is continuous}\}$$

with the distance given by $d(\gamma_0, \gamma_1) := \sup\{\|\gamma_0(t) - \gamma_1(t)\| \mid 0 \leq t \leq 1\}$, so that (Ω, d) is a metric space whose “points” are in fact paths. In particular concepts like continuous functions and connectedness applies to this “abstract” metric space. We are also interested in the subset

$$\Omega' := \{\gamma : [0, 1] \rightarrow U \mid \gamma(0) = x_0, \gamma(1) = x_1, \text{ and } \gamma \text{ is piecewise continuously differentiable}\}.$$

LEMMA 11.50:

Ω' is dense in Ω with respect to the distance d .

Proof. We need to prove that any continuous path γ can be approximated, in the distance d , by a piecewise continuously differentiable path θ having the same extrema. It is of course enough to approximate each component of γ .

Once we observe that each $\gamma_i : [0, 1] \rightarrow \mathbb{R}$ is uniformly continuous, it is sufficient to approximate it by linear interpolation of a sufficiently fine mesh. $\square$

LEMMA 11.51: HOMOTOPIES ARE “PATHS OF PATHS”

The paths $\gamma_0, \gamma_1 \in \Omega$ are homotopic if and only if there exists a continuous path $\varphi : [0, 1] \rightarrow \Omega$ with $\varphi(0) = \gamma_0$ and $\varphi(1) = \gamma_1$.

Proof. If such an homotopy H exists we set

$$\varphi(s) := ([0, 1] \ni t \mapsto H(s, t)), \quad s \in [0, 1],$$

and this function is continuous. Vice-versa, if such φ exists we set

$$H(s, t) := (\varphi(s))(t), \quad (s, t) \in [0, 1]^2.$$

Checking the continuity of these maps is an instructive exercise. $\square$

Proof of Theorem 11.45. Under the hypotheses of Theorem 11.45, let x_0 and x_1 be elements of U , and define Ω, Ω' as in 11.49. We consider the function

$$I : \Omega' \rightarrow \mathbb{R} \quad I(\gamma) = \int_{\gamma} F = \int_0^1 \langle F(\gamma(t)), \gamma'(t) \rangle dt.$$

Step 1. We claim that for every point $\sigma \in \Omega$, there exists an $\epsilon > 0$ such that I is constant on the open ball $B(\sigma, \epsilon) \cap \Omega'$ (of course this ball is taken with respect to the distance d).

Fix $\sigma \in \Omega$. The image $\sigma([0, 1]) \subset U$ is compact, $U \subset \mathbb{R}^n$ is open and therefore there exists an $\epsilon > 0$ such that

$$B(\sigma(t), 2\epsilon) \subset U \quad \text{for all } t \in [0, 1],$$

where $B(\sigma(t), \epsilon)$ here denotes the open ball in $\mathbb{R}^n$ centered at $\sigma(t)$ with radius ϵ .

For $\gamma, \gamma_0 \in B(\sigma, \epsilon) \cap \Omega'$ and all $s, t \in [0, 1]$, we define

$$H(s, t) := s\gamma(t) + (1 - s)\gamma_0(t) \in B(\sigma(t), 2\epsilon) \subset U.$$

Notice that $t \mapsto H(s, t)$ is a piecewise continuously differentiable path in $\mathbb{R}^n$ (i.e., belongs to Ω'), so we may consider the parameter integral

$$E(s) := I(s\gamma + (1 - s)\gamma_0) = \int_0^1 \langle F(s\gamma(t) + (1 - s)\gamma_0(t)), s\gamma'(t) + (1 - s)\gamma_0'(t) \rangle dt.$$

A direct but somewhat tedious calculation (we postpone it to the end of the proof) shows $E'(s) = 0$ and so

$$I(\gamma_0) = E(0) = E(1) = I(\gamma),$$

the claim is proved.

Step 2. We claim that the function $I : \Omega' \rightarrow \mathbb{R}$ admits a (unique) continuous extension $\bar{I}$ to the whole Ω .

We first define $\bar{I}$. For any $\sigma \in \Omega$ pick $\epsilon_\sigma > 0$ as in step 1 and some $\gamma_\sigma \in B(\sigma, \epsilon_\sigma) \cap \Omega'$, which exists thanks to Lemma 11.50, then set

$$\bar{I}(\sigma) := I(\gamma_\sigma).$$

Thanks to step 1, the value of $\bar{I}(\sigma)$ does not depend on the specific curve γ_σ that we picked.

We check that $\bar{I}$ is locally constant (and thus continuous) around any point σ_0 . Pick σ, σ_0 and $\epsilon_\sigma, \epsilon_{\sigma_0}$ as in step 1 and assume $d(\sigma, \sigma_0) < \epsilon_{\sigma_0}/10$. Then, by density, pick some

$$\gamma_\sigma \in B(\sigma_0, \epsilon_{\sigma_0}) \cap B(\sigma, \epsilon_\sigma) \cap \Omega'.$$

By definition γ_σ is sufficiently close to both of them to say that $\bar{I}(\sigma_0) = I(\gamma_\sigma) = \bar{I}(\sigma)$.

Step 3. We show that $I(\gamma_0) = I(\gamma_1)$. We proved that the set

$$\{\gamma \in \Omega \mid \bar{I}(\gamma) = \bar{I}(\gamma_0)\}$$

is open ($\bar{I}$ being locally constant) and closed ($\bar{I}$ being continuous). Thus it must contain the path-connected component of γ_0 . If $\gamma_0 \in \Omega$ and $\gamma_1 \in \Omega$ are homotopic paths as in the theorem, then, according to Lemma 11.51, γ_0 and γ_1 lie in the same path-connected component of Ω . It follows that $I(\gamma_0) = I(\gamma_1)$, completing the proof.

Step 4. We conclude checking that indeed $E'(s) = 0$. Observe that

$$\partial_t H(s, t) = s\gamma'(t) + (1 - s)\gamma_0'(t), \quad \partial_s H(s, t) = \gamma(t) - \gamma_0(t), \quad \partial_{st} H = \partial_{ts} H.$$

So we have

$$E'(s) = \sum_k \frac{d}{ds} \int_0^1 F_k(H(s, t))(s\gamma'_k(t) + (1 - s)\gamma'_{0,k}(t)) dt$$

$$\begin{aligned}
&= \sum_k \int_0^1 \partial_s \left(F_k(H(s, t)) \partial_t H_k(s, t) \right) dt \\
&= \sum_{k, \ell} \int_0^1 \partial_\ell F_k(H(s, t)) \partial_s H_\ell(s, t) \partial_t H_k(s, t) dt + \sum_k \int_0^1 F_k(H(s, t)) \partial_{ts} H_k(s, t) dt.
\end{aligned}$$

The trick here is to integrate by parts the second term to cancel the first one, exploiting both the integrability conditions and γ, γ_0 having the same ends.

$$\begin{aligned}
\sum_k \int_0^1 F_k(H(s, t)) \partial_{ts} H_k(s, t) dt &= \sum_k \int_0^1 F_k(H(s, t)) (\gamma'_k(t) - \gamma'_{0,k}(t)) \\
&= \sum_k \int_0^1 F_k(H(s, t)) \frac{d}{dt} (\gamma_k(t) - \gamma_{0,k}(t)) \\
&= - \sum_k \int_0^1 \frac{d}{dt} (F_k(H(s, t))) (\gamma_k(t) - \gamma_{0,k}(t)) \\
&= - \sum_{k, \ell} \int_0^1 \partial_\ell F_k(H(s, t)) \partial_t H_\ell(s, t) \partial_s H_k(s, t) dt
\end{aligned}$$

Now, exchanging the role of the indexes k, ℓ in the two sums, and using $\partial_k F_\ell = \partial_\ell F_k$, one finds

$$\sum_{k, \ell} \int_0^1 \partial_\ell F_k(H) \partial_s H_\ell \partial_t H_k dt = - \sum_{k, \ell} \int_0^1 \partial_\ell F_k(H) \partial_t H_\ell \partial_s H_k dt.$$

□

Applet 11.52 ([Integrability Conditions](#)). *What different values for the path integral can you obtain when considering closed paths? Why does the value of the path integral usually not change, but sometimes does when you move the middle three points?*

Chapter 12

Foundations of Differential Geometry

12.1 The inverse function Theorem (F)

LEMMA 12.1: SMALL LIPSCHITZ PERTURBATIONS OF THE IDENTITY (F)

Let $(V, \|\cdot\|)$ be a complete normed vector space, $B_1 \subset V$ be the unit ball in the $\|\cdot\|$ norm, and let $F: B_1 \rightarrow V$ a function of the form $F(x) = x + \phi(x)$, with $\text{Lip}(\phi) \leq \lambda < 1$. Then

- (1) $B_{1-\lambda}(F(0)) \subset F(B_1) \subset B_{1+\lambda}(F(0))$, in particular $F(B_1)$ is open,
- (2) F is injective and $F^{-1}: F(B_1) \rightarrow B_1$ is of the form $F^{-1}(y) = y + \psi(y)$, with $\text{Lip}(\psi) \leq \frac{\lambda}{1-\lambda}$.

Proof. (F) We start with the first inclusion in (1), which is the core of this proof. For $y \in B(F(0), 1 - \lambda)$ consider the recurrence

$$x_0 = 0, \quad x_{k+1} = y - \phi(x_k).$$

If we prove that $x_k \rightarrow \bar{x}$, then $\bar{x} = y - \phi(\bar{x})$, that is we solved $y = F(\bar{x})$. Using the triangular inequality and the contraction property of ϕ we find

$$|x_{k+1} - x_k| \leq |\phi(x_k) - \phi(x_{k-1})| \leq \lambda |x_{k-1} - x_k| \leq \dots \leq \lambda^k |x_1 - x_0| = \lambda^k |y - F(0)|,$$

which proves that $\{x_k\}$ is Cauchy. We are not done yet though, we have to check that $\{x_k\}$ is well defined, i.e., never escapes from B_1 , as we are computing ϕ on it. This is checked by the same computation, for all k it holds

$$|x_{k+1}| \leq \sum_{i=0}^k |x_{i+1} - x_i| = |y - F(0)| \sum_{i=0}^k \lambda^i \leq \frac{|y - F(0)|}{1 - \lambda} < 1.$$

Now, we prove that $F(B_1)$ is open. Let $B_r(x_0) \subset B_1$ and $y_0 = F(x_0)$. Then the map $F_{x_0,r}(x) := x + r\phi((x - x_0)/r)$, defined in B_1 , is again a λ -Lipschitz perturbation of the

identity. Applying what we just proved to $F_{x_0,r}$, we find

$$F(B_1) \supset F(B_r(x_0)) = F_{x_0,r}(B_1) \supset B_{1-\lambda}(F_{x_0,r}(0)) = B_{(1-\lambda)r}(F(x_0)) = B_{(1-\lambda)r}(y_0),$$

which means that $F(B_1)$ is open.

Finally, the second inclusion in (1) is readily checked

$$|F(x) - F(0)| = |x + \phi(x) - \phi(0)| \leq |x| + \lambda|x| < 1 + \lambda.$$

We turn to (2). First of all, if $F(x) = F(x')$, then $x - x' = \phi(x) - \phi(x')$, which is in contradiction with $\lambda < 1$. So F is injective and F^{-1} is a well-defined function. We call $\psi(y) := F^{-1}(y) - y, y \in F(B_1)$. Then ψ satisfies

$$y = F(F^{-1}(y)) = y + \psi(y) + \phi(y + \psi(y)), \quad y \in F(B_1).$$

This readily implies

$$\begin{aligned} |\psi(y) - \psi(y')| &\leq |\phi(y + \psi(y)) - \phi(y' + \psi(y'))| \leq \lambda|y - y' + \psi(y) - \psi(y')| \\ &\leq \lambda|y - y'| + \lambda|\psi(y) - \psi(y')|, \end{aligned}$$

which is what we wanted to prove (again, we use $\lambda < 1$). □

LEMMA 12.2: AUTOMATIC REGULARITY OF THE INVERSE

Let $U, V \subset \mathbb{R}^n$ be open sets and let $f: U \rightarrow V$ and $g: V \rightarrow U$ be bijective functions. Assume that f is differentiable at $\bar{x} \in U$ with $\det Df(\bar{x}) \neq 0$, that g is Lipschitz in V , and that

$$f(g(y)) = y, \quad \text{for all } y \in V.$$

Then g must be differentiable at $\bar{y} = f(\bar{x})$ and $Dg(\bar{y}) = [Df(\bar{x})]^{-1}$.

Proof. Let us set $A := Df(\bar{x})$. By the differentiability assumption, as $y \rightarrow \bar{y}$,

$$y = f(\bar{x} + (g(y) - g(\bar{y}))) = f(\bar{x}) + A(g(y) - g(\bar{y})) + o(|g(y) - g(\bar{y})|).$$

Using that A is invertible, we re-write this as

$$g(y) = g(\bar{y}) + A^{-1}(y - \bar{y}) + o(\|A^{-1}\|_{\text{op}}|g(y) - g(\bar{y})|). \quad (12.1)$$

We conclude noticing that, since g is Lipschitz,

$$o(\|A^{-1}\|_{\text{op}}|g(y) - g(\bar{y})|) = o\left(\frac{|g(y) - g(\bar{y})|}{|y - \bar{y}|}|y - \bar{y}|\right) = O(1)o(|y - \bar{y}|) = o(|y - \bar{y}|).$$

Thus (12.1) is saying that g is differentiable at $\bar{y}$, with differential A^{-1} . □

LEMMA 12.3: SMOOTHNESS OF THE INVERSE

Let $U \subset \mathbb{R}^{n \times n}$ be the set of invertible matrices, and let $\theta: U \rightarrow U$ be defined as

$$\theta: X \mapsto X^{-1}.$$

Then $\theta \in C^\infty$.

Proof. The formula for the inverse matrix expresses the (p, q) entry of X^{-1} as a polynomial of $\{X_{i,j}\}$ divided by the polynomial $\det X$ (which is nonzero in U). $\square$

THEOREM 12.4: INVERSE FUNCTION (F)

Let $U \subset \mathbb{R}^n$ be an open set and $\phi \in C^1(U, \mathbb{R}^n)$ with $\det D\phi(\bar{x}) \neq 0$ at some $\bar{x} \in U$. Then, for a sufficiently small $r > 0$,

- (1) ϕ is injective in $B_r(\bar{x})$,
- (2) the set $\phi(B_r(\bar{x}))$ is open,
- (3) the inverse function $\psi: \phi(B_r(\bar{x})) \rightarrow B_r(\bar{x})$ is of class C^1 and

$$D\psi(y) = [D\phi(\psi(y))]^{-1}, \quad \text{for all } y \in \phi(B_r(\bar{x})).$$

Furthermore, if $\phi \in C^k(U, \mathbb{R}^n)$ for some $k \geq 1$, then also $\psi \in C^k(\phi(B_r(\bar{x})), U)$.

Proof. Up to replacing $\phi(x)$ with $\phi(x - \bar{x}) - \phi(\bar{x})$, we can assume $0 = \bar{x} = \phi(\bar{x})$.

Let $A := D\phi(\bar{x})$, then we claim that $F(x) := A^{-1} \circ \phi(x)$ is a Lipschitz perturbation of the identity in B_r for $r > 0$ small. Indeed, as $x \rightarrow 0$, we have

$$\|D(F(x) - x)\|_{\text{op}} = \|DF(x) - \mathbf{1}_n\|_{\text{op}} = \|A^{-1} \circ (D\phi(x) - A)\|_{\text{op}} \leq \|A^{-1}\|_{\text{op}} \|D\phi(x) - A\|_{\text{op}} \rightarrow 0,$$

so choosing r small enough we have

$$\det D\phi(x) \neq 0 \text{ and } \|D(F(x) - x)\|_{\text{op}} \leq \frac{1}{2}, \quad \text{for all } x \in B_r.$$

By Corollary 10.25 we deduce that $\text{Lip}(F(x) - x, B_r) \leq 1/2$. By Lemma 12.1, we find that $F(B_r)$ is open, that $F|_{B_r}$ is injective and that the inverse function $G: F(B_r) \rightarrow B_r$ is Lipschitz.

Now (1) holds because $\phi = A \circ F$, which is a composition of injective functions.

For (2) we need to show that $\phi(B_r) = A(F(B_r))$ is open. But $F(B_r)$ is open and an invertible linear map sends open sets to open sets.

For (3) we set $\psi(y) := G(A^{-1}y)$ for $y \in \phi(B_r) = A(F(B_r))$. Then it holds

$$z = A^{-1}\phi(G(z)) \quad \text{for all } z \in F(B_r),$$

which means, setting $z := Ay$, that

$$y = \phi(\psi(y)), \text{ for all } y \in \phi(B_r).$$

Since ϕ is differentiable at every point of B_r , and ψ is Lipschitz, then Lemma 12.2 shows that ψ is differentiable for all $y \in \phi(B_r)$, and

$$D\psi(y) = [D\phi(\psi(y))]^{-1}, \quad \text{for all } y \in \phi(B_r(\bar{x})).$$

Finally, assume that $\phi \in C^k$ and that you know $\psi \in C^{k-1}$. The formula for $D\psi$ give it as the composition of the following functions:

$$\begin{aligned} \psi: y &\mapsto \psi(y) && \text{which is of class } C^{k-1}, \\ D\phi: x &\mapsto D\phi(x) && \text{which is of class } C^k, \\ \theta: X &\mapsto X^{-1} && \text{which is of class } C^\infty \text{ on the set of invertible matrices by Lemma 12.3.} \end{aligned}$$

Thus $D\psi \in C^{k-1}$, which means that $\psi \in C^k$ (bootstrap). This proves the last part of the statement by induction from the $k = 1$ case. $\square$

DEFINITION 12.5:

Let $U, V \subset \mathbb{R}^n$ be open. A bijective, smooth function $f: U \rightarrow V$ with a smooth inverse $f^{-1}: V \rightarrow U$ is called a **diffeomorphism**. If f and f^{-1} are both d -times continuously differentiable for $d \geq 1$, we call f a **C^d -diffeomorphism**.

12.2 Manifolds in $\mathbb{R}^n(\mathbf{F})$

12.2.1 Definition of Manifold

Let us introduce some rough intuition. There are three common ways to give a geometric shape of dimension k in $\mathbb{R}^n$:

- **Cartesian.** This means describing the set as the set of points that solves $n - k$ given equation(s).
- **Parametric.** This means identifying each point in the set with k numerical parameters, which lie in some range and don't necessarily have direct geometric meaning (think about generalized coordinates in Lagrangian mechanics).
- **Graphical.** Describe the coordinates of the points in the set prescribing how the last $n - k$ coordinates depend on the first k coordinates.

INTERLUDE: LINEAR SPACE IN $\mathbb{R}^n(\mathbf{F})$

A set $X \subset \mathbb{R}^n$ is a k -dimensional linear subspace if one of the following equivalent conditions holds:

- (1) There is a linear map $f: \mathbb{R}^n \rightarrow \mathbb{R}^{n-k}$, with maximal rank (i.e., with rank $n - k$), such that $X = \{f = 0\}$. That is, X is described as the set of solutions of a system of $n - k$ equations.
- (2) There is a linear map $f: \mathbb{R}^k \rightarrow \mathbb{R}^n$, with maximal rank (i.e., with rank k), such that $X = f(\mathbb{R}^k)$. That is, X is parametrised with a linear function of k variables.
- (3) up to reordering the coordinates, X coincides with the graph of a linear map $f: \mathbb{R}^k \rightarrow \mathbb{R}^{n-k}$. In other words, $(n - k)$ coordinates of every point in X can be expressed as a linear function of the other k coordinates.

Let us recall that proving the equivalence of these definitions is not trivial in the linear case, as one needs essentially to learn how to solve linear equations in order to pass from one representation of X to the other.

Now we aim to define a suitable concept of k -dimensional surface in $\mathbb{R}^n$, using C^1 maps rather than linear maps. This is indeed possible, paying the price of working “locally” around each point. In this framework, the Inverse Function Theorem is the key tool that allows us to “locally solve non-singular equations”.

DEFINITION 12.6: MANIFOLD

Let $0 < k < n$ be integers and $X \subset \mathbb{R}^n$ be a nonempty set. We say that **X is a k -dimensional manifold embedded in $\mathbb{R}^n$** if, for every $p \in X$, one of the following three equivalent conditions holds:

- (1) There is $f \in C^1(\mathbb{R}^n, \mathbb{R}^{n-k})$ such that the matrix $Df(p)$ has maximal rank (i.e., has rank $n - k$) and for $r > 0$ sufficiently small

$$X \cap B_r(p) = \{f = 0\} \cap B_r(p).$$

- (2) There is $f \in C^1(\mathbb{R}^k, \mathbb{R}^n)$ such that $f(0) = p$, the matrix $Df(0)$ has maximal rank (i.e., has rank k) and for $s, r > 0$ sufficiently small

$$X \cap B_r(p) = f(B_s) \cap B_r(p),$$

where $B_s \subset \mathbb{R}^k$.

- (3) There is $f \in C^1(\mathbb{R}^k, \mathbb{R}^{n-k})$ such that for $r > 0$ sufficiently small

$$X \cap B_r(p) = P(\text{graph}(f)) \cap B_r(p),$$

where $P: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is some permutation of the coordinates.

Finally, we define a 0-dimensional submanifold as a discrete set of points in $\mathbb{R}^n$ and a n -dimensional submanifold as an open subset of $\mathbb{R}^n$.

The fact that these three conditions are equivalent is not obvious and will be proved by the means of the Inverse Function Theorem.

We will prove that $(1) \implies (3) \implies (2) \implies (1)$.

We just record that $(3) \implies (2)$ is straightforward since the graph of a function $f: \mathbb{R}^k \rightarrow \mathbb{R}^{n-k}$ coincides, by definition, with the image of the function $\tilde{f}: \mathbb{R}^k \rightarrow \mathbb{R}^n$ given by $\tilde{f}(x) = (x, f(x))$.

THEOREM 12.7: (1) $\implies$ (3), A.K.A. THE IMPLICIT FUNCTION THEOREM

Let $U \subset \mathbb{R}^n \times \mathbb{R}^m$ be open and let $F \in C^1(U, \mathbb{R}^m)$. We write a point in $\mathbb{R}^n \times \mathbb{R}^m$ as (x, y) with $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$. Assume that the $m \times m$ matrix

$$D_y F(\bar{x}, \bar{y}) := \{\partial_{y_k} F_j(\bar{x}, \bar{y})\}_{j,k}$$

is invertible at a certain $(\bar{x}, \bar{y}) \in U$. Then, for sufficiently small $r, s > 0$, there is a function f from $B(\bar{x}, r) \subset \mathbb{R}^n$, to $B(\bar{y}, s) \subset \mathbb{R}^m$, such that, for all (x, y) in the cylinder $B(\bar{x}, r) \times B(\bar{y}, s) \subset U$, it holds

$$F(x, y) = F(\bar{x}, \bar{y}) \iff y = f(x),$$

and

$$Df(x) = -((D_y F)(x, f(x)))^{-1} \circ (D_x F)(x, f(x)).$$

Furthermore, if $F \in C^k(U, \mathbb{R}^m)$ for some $k \geq 1$, then also $f \in C^k(B_r(\bar{x}), U)$.

Proof. Consider the function $\phi \in C^1(U, \mathbb{R}^{n+m})$ given by

$$\phi(x, y) := (x, F(x, y)), \quad \text{for all } (x, y) \in U.$$

By assumption the matrix $D\phi(\bar{x}, \bar{y})$ — which has size $(n+m) \times (n+m)$ — has a block decomposition

$$D\phi(\bar{x}, \bar{y}) = \begin{pmatrix} \mathbf{1}_n & 0 \\ D_x F(\bar{x}, \bar{y}) & D_y F(\bar{x}, \bar{y}) \end{pmatrix}$$

so in particular it is invertible and we are under the assumptions of the Inverse function Theorem. Hence ϕ has a smooth inverse when restricted to a small cylinder $B_r(\bar{x}) \times B_s(\bar{y}) \subset U$, which is mapped to the open set $V := \phi(B_r(\bar{x}) \times B_s(\bar{y})) \subset \mathbb{R}^{n+m}$.

Let $\psi \in C^1(V, B_r(\bar{x}) \times B_s(\bar{y}))$ be the inverse. Notice that ψ must have the form

$$\psi(\zeta, \xi) = (\zeta, g(\xi, \zeta)), \quad \text{for all } (\zeta, \xi) \in V,$$

for some $g \in C^1(V, B_s(\bar{y}))$. Now by definition we have

$$(x, y) = (\psi \circ \phi)(x, y) = (x, g(x, F(x, y))) \text{ for all } (x, y) \in B_r(\bar{x}) \times B_s(\bar{y}),$$

thus we set $f(x) := g(x, F(\bar{x}, \bar{y}))$, and get

$$(x, y) \in B_r(\bar{x}) \times B_s(\bar{y}), F(x, y) = F(\bar{x}, \bar{y}) \implies y = f(x).$$

For the converse implication we notice that ψ is injective, so we have

$$(x, y) \in B_r(\bar{x}) \times B_s(\bar{y}), y = f(x) \implies g(x, F(x, y)) = g(x, F(\bar{x}, \bar{y})) \implies F(x, y) = F(\bar{x}, \bar{y}).$$

The formula for $Df(\bar{x})$ can be read off the matrix $D\psi(\bar{x}, \bar{y})$. Recall that

$$\begin{pmatrix} 1 & 0 \\ A & B \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 \\ -B^{-1}A & B^{-1} \end{pmatrix},$$

as can be readily checked. So we have

$$\begin{aligned} D\psi(\phi(x, y)) &= \begin{pmatrix} \mathbf{1}_n & 0 \\ D_x F(x, y) & D_y F(x, y) \end{pmatrix}^{-1} \\ &= \begin{pmatrix} \mathbf{1}_n & 0 \\ -[D_y F(x, y)]^{-1} \cdot D_x F(x, y) & [D_y F(x, y)]^{-1} \end{pmatrix}. \end{aligned}$$

Hence $Df(\phi(x, y)) = -[D_y F(x, y)]^{-1} \cdot D_x F(x, y)$. □

THEOREM 12.8: (2) $\implies$ (1)

Let $f \in C^1(\mathbb{R}^k, \mathbb{R}^n)$ with $0 < k < n$ such that $Df(0)$ has rank k . Then there is $r > 0$ small and $g \in C^1(B_r(f(0)), \mathbb{R}^{n-k})$ such that $Dg(f(0))$ has rank $n - k$ and $f(\mathbb{R}^k) \cap B_r(f(0)) = \{g = 0\} \cap B_r(f(0))$.

Proof. Pick some vectors $v_1, \dots, v_{n-k}$ in $\mathbb{R}^n$, and $s > 0$ small in such a way that

$$\det(Df(z) \mid v_1 \mid \dots \mid v_{n-k}) \neq 0, \text{ for all } |z| < s.$$

Consider the map $\phi \in C^1(B_s \times \mathbb{R}^{n-k}, \mathbb{R}^n)$ given by $\phi(z, w) := f(z) + \sum_{j=1}^{n-k} w_j v_j$, whose differential at the origin is invertible being

$$D\phi(0) = (Df(z) \mid v_1 \mid \dots \mid v_{n-k}).$$

By the Inverse function Theorem there are functions $\zeta \in C^1(B_r(f(0), \mathbb{R}^k)$ and $\omega \in C^1(B_r(f(0), \mathbb{R}^{n-k})$ such that

$$x = f(\zeta(x)) + \sum_{j=1}^{n-k} \omega_j(x) v_j, \text{ for all } x \in B_r(f(0)),$$

and such a representation of x is unique in that ball. The choice $g(x) := \omega(x)$ thus provides the sought function. □

12.2.2 Tangent space to a manifold

DEFINITION 12.9:

Let $X \subset \mathbb{R}^n$ be a k -dimensional manifold. The **tangent space** of X at $p \in M$ is the vector subspace of $\mathbb{R}^n$ defined as

$$T_p M := \{\gamma'(0) \mid \gamma : (-1, 1) \rightarrow M \text{ differentiable with } \gamma(0) = p\} \subset \mathbb{R}^n.$$

12.10. — In plain terms, the set $T_p M$ is the set of velocity vectors of short paths through p in M . If we think of elements of $T_p M$ as vectors with base point p , then $T_p M$ corresponds to a k -dimensional affine subspace of $\mathbb{R}^n$ that touches M at the point p . We note that the choice of $(-1, 1)$ as the domain of differentiable paths is arbitrary; we could just as well have chosen another open neighborhood of $0 \in \mathbb{R}$, such as an open interval $(-\delta, \delta)$ for a ideally small $\delta > 0$.

The following Proposition shows how to practically compute $T_p X$ and that it is indeed a vector subspace of $\mathbb{R}^n$ (which is not transparent from the given definition).

PROPOSITION 12.11:

According to how X is given around p , we equivalently have

- (1) $T_p X = \ker Df(p)$ if $X = \{f = 0\}$ in $B_r(p)$ for some $f \in C^1(B_r(p), \mathbb{R}^{n-k})$;
- (2) $T_p X = \text{ran } Df(0)$ if $X = f(B_s)$ in $B_r(p)$ and $f(0) = p$, for some $f \in C^1(B_s, \mathbb{R}^n)$ with $B_s \subset \mathbb{R}^k$;
- (3) $T_p X$ is the tangent plane to f at p , if X is the graph of $f: \mathbb{R}^k \rightarrow \mathbb{R}^{n-k}$ in $B_r(p)$.

In particular, $T_p X$ is a vector subspace of $\mathbb{R}^n$.

Proof. ...

□

DEFINITION 12.12:

Let $X \subset \mathbb{R}^n$ be a k -dimensional manifold. The **normal space** of X at $p \in M$ is the $(n - k)$ dimensional vector subspace of $\mathbb{R}^n$ which is orthogonal to $T_p X$.

12.2.3 Examples and Non-Examples

Chapter 13

NEW: Multidimensional Integration

13.1 The n -volume

DEFINITION 13.1: DYADIC CUBES AND THEIR MEASURE

We call any subset of the form $Q = 2^{-k}(a + [0, 1)^n)$, for some $a \in \mathbb{Z}^n$ and $k \in \mathbb{Z}$ a **dyadic cube**. Given such Q we define its $\mu(Q) = 2^{-kn}$. For a finite disjoint union of dyadic cubes $E = \cup_i Q_i$, we define $\mu(E) = \sum_i \mu(Q_i)$.

PROPOSITION 13.2: PROPERTIES OF μ

The measure μ , defined over finite disjoint unions of dyadic cubes, satisfies the following properties:

1. *Translation invariance:* $\mu(E + \tau) = \mu(E)$ for all τ such that $2^k \tau \in \mathbb{Z}^n$ for some $k \in \mathbb{N}$.
2. *Additivity:* $\mu(E \cup F) = \mu(E) + \mu(F) - \mu(E \cap F)$
3. *Normalization:* $\mu([0, 1)^n) = 1$

Proof. ... the key advantage of using dyadic cubes is that it allows to refine any union or intersection of dyadic cubes with sufficiently small cubes, all of the same size. At the same time we can approximate any real box. $\square$

13.3. — Actually, it is not difficult to prove that the properties (1)-(3) below completely characterize μ .

DEFINITION 13.4: JORDAN MEASURABLE SETS AND THEIR MEASURE

Given a set $E \subset \mathbb{R}^n$ let us define

$$\mu_{\text{in}}(E) = \sup \left\{ \sum_i \text{vol}_n(Q_i) \mid Q_i \subset E \text{ pairwise disjoint} \right\}$$

and

$$\mu_{\text{out}}(E) = \inf \{ \text{vol}_n(Q_i) \mid E \subset \cup_i Q_i \}.$$

Here, the supremum and the infimum are taken among finite families of rational cubes.

If $\mu_{\text{in}}(E) = \mu_{\text{out}}(E)$ then we say that E is **Jordan measurable**

We then denote $\text{vol}_n(E) = \mu_{\text{in}}(E) = \mu_{\text{out}}(E)$

PROPOSITION 13.5: FIRST PROPERTIES OF vol_n

The measure vol_n , defined over all Jordan measurable sets $E \subset \mathbb{R}^n$:

1. *Translation invariance: if E is Jordan measurable and $\tau \in \mathbb{R}^n$ then $E + \tau$ is also Jordan measurable and $\text{vol}_n(E + \tau) = \text{vol}_n(E)$ for all $\tau \in \mathbb{R}^n$*
2. *Additivity: If E, F are Jordan measurable then their intersection and union also are, and $\text{vol}_n(E \cup F) = \text{vol}_n(E) + \text{vol}_n(F) - \text{vol}_n(E \cap F)$.*
3. *Normalization: $\text{vol}_n([0, 1]^n) = 1$*

PROPOSITION 13.6: AFFINE TRANSFORMATIONS

Given some invertible linear map $L : \mathbb{R}^n \rightarrow \mathbb{R}^n$. If a set $E \subset \mathbb{R}^n$ is Jordan measurable then so is its image under L , which we denote LE . Moreover there exist a positive factor $\lambda(L)$ such that

$$\text{vol}_n(LE) = \lambda(L) \text{vol}_n(E) \quad \text{for all } E \text{ Jordan measurable.}$$

LEMMA 13.7: THE FACTOR FOR SPECIAL STRETCHINGS

Given $\lambda_1, \dots, \lambda_n$ given positive real numbers, consider the 'special stretching' (affine transformation) $Sx = (\lambda_1 x_1, \dots, \lambda_n x_n)$. Then $\lambda(S) = \prod_{i=1}^n \lambda_i$

LEMMA 13.8: BALLS ARE JORDAN MEASURABLE

Balls $B_r(x) \subset \mathbb{R}^n$ are Jordan measurable and $\text{vol}_n(B_r(x)) = r^n \text{vol}_n(B_1)$

Proof. Show that the number of cubes in the family $\{[0, 2^{-k})^n + \varepsilon T \mid T \in \mathbb{Z}^n\}$ that intersect the boundary $|x| = 1$ of unit ball is bounded by $C(n)(2^{-k})^{1-n} \dots$ □

PROPOSITION 13.9: THE FACTOR IS THE DETERMINANT

For every invertible linear map $L : \mathbb{R}^n \rightarrow \mathbb{R}^n$ we have $\lambda(L) = |\det L|$

Proof. Using the polar decomposition of a matrices (see...) we have that $L = R_2SR_1$, where S is an special stretching and R_i are orthogonal matrices (i.e., rotations; characterized by $RR^T = \text{Id}$). Since rotations leave the unit ball invariant we have $\text{vol}_n(B_1) = \text{vol}_n(R_iB_1) = \lambda(R_i)\text{vol}_n(B_1)$ and hence $\lambda(R_i) = 1$. On the other hand λ is clearly multiplicative, that is $\lambda(L) = \lambda(R_2SR_1) = \lambda(R_2)\lambda(S)\lambda(R_1) = \lambda S = |\det S|$.

Since orthogonal matrices have determinant equal to one, and the determinant is also multiplicative, we are done. $\square$

COROLLARY 13.10: BOXES ARE JORDAN MEASURABLE

Any box $E = [a_1, b_1) \times [a_2, b_2) \times \cdots \times [a_n, b_n)$ where $b_i > a_i$ is Jordan measurable and $\text{vol}_n(E) = (b_1 - a_1)(b_2 - a_2) \cdots (b_n - a_n)$.

COROLLARY 13.11:

vol_n is also invariant under rotations.

13.2 Measurable Sets and the Integral

13.2.1 Measurable sets and Fubini's Theorem

DEFINITION 13.12: JORDAN MEASURABLE SET

A set $E \subset \mathbb{R}^n$ is measurable if the two numbers

$$\sup\left\{\sum_i \text{vol}(P_i) : \{P_i\} \text{ is a finite disjoint collection of parallelograms contained in } E\right\},$$

$$\inf\left\{\sum_i \text{vol}(P_i) : \{P_i\} \text{ is a finite collection of parallelograms containing } E\right\}.$$

coincide. Their common value is $\text{vol}(E)$.

PROPOSITION 13.13: WELL-POSEDNESS

If P is a parallelogram the symbol $\text{vol}(P)$ is well-defined.

Proof. Is it clear that if $P \cup P' \subset Q$, then $\text{vol}(P) + \text{vol}(P') \leq \text{vol}(Q)$? □

PROPOSITION 13.14: ONE CAN USE STRAIGHT CUBES

If E is Jordan measurable then one can use cubes with sides parallel to the axis to find its volume.

Proof. By induction on the dimension one can show that an ε tiling of a cube is off by $C_n \varepsilon$, so by sandwiching cube tilings suffices. □

13.2.2 The integral of a continuous function

13.2.3 The change of variables formula

13.3 Computation of Multiple integrals

Chapter 14

OLD: Multidimensional Integration

In this chapter, we will extend the Riemann integral over an interval $[a, b] \subset \mathbb{R}$ to a multidimensional Riemann integral over sufficiently nice open subsets of $\mathbb{R}^n$. Again, it suffices to consider real-valued functions, as the generalization to complex-valued or vector-valued functions is done component-wise.

14.1 The Riemann Integral for Boxes

In this section, we fix a non-negative integer n and aim to replace intervals as integration domains with subsets in $\mathbb{R}^n$. We begin with the case of a box.

14.1.1 Definition and Initial Properties

14.1. — A **box** is a subset Q of $\mathbb{R}^n$ that is a product of intervals, i.e.,

$$Q = I_1 \times \dots \times I_n$$

for intervals $I_1, \dots, I_n \subset \mathbb{R}$. If the lengths of the intervals $I_1, \dots, I_n$ are all equal, we also call Q a **cube**. For $n = 2$, we refer to them as **rectangles**.

DEFINITION 14.2. — For bounded non-empty intervals $I_1, \dots, I_n$, the **volume** of the box $Q = I_1 \times \dots \times I_n$ in $\mathbb{R}^n$ is defined as

$$\text{vol}(Q) = \prod_{k=1}^n (b_k - a_k)$$

with $a_k = \inf I_k$ and $b_k = \sup I_k$.

14.3. — Let $Q = I_1 \times \dots \times I_n$ be a bounded closed box with $I_k = [a_k, b_k]$. A **partition** of Q means specifying a partition for each interval I_k , in the sense of Definition ???. If such a partition is given, i.e.,

$$a_k = x_{k,0} < x_{k,1} < x_{k,2} < \dots < x_{k,l(k)} = b_k \tag{14.1}$$

for each k , then, for $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n) \in \mathbb{N}^n$ with $1 \leq \alpha_k \leq l(k)$, an **address** for this partition is defined. For each such address, we write

$$Q_\alpha = \prod_{k=1}^n [x_{k,\alpha_k-1}, x_{k,\alpha_k}]$$

for the corresponding closed sub-box of Q . In other words, $Q_\alpha \subseteq Q \subseteq \mathbb{R}^n$ is the subset of all $(t_1, \dots, t_n) \in \mathbb{R}^n$ for which $x_{k,\alpha_k-1} \leq t_k \leq x_{k,\alpha_k}$ holds for all $k = 1, 2, \dots, n$. The **addition formula**

$$\text{vol}(Q) = \sum_{\alpha} \text{vol}(Q_\alpha) \quad (14.2)$$

is shown using complete induction, where the sum extends over all addresses for the given partition (14.1). A **refinement** of the partition (14.1) is a partition

$$a_k = y_{k,0} \leq y_{k,1} \leq y_{k,2} \leq \dots \leq y_{k,m(k)} = b_k \quad k = 1, 2, \dots, n$$

such that, for each fixed k , the partition $a_k = y_{k,0} \leq \dots \leq y_{k,m(k)} = b_k$ of I_k is a refinement of $a_k = x_{k,0} \leq \dots \leq x_{k,l(k)} = b_k$, as discussed in ???. Two arbitrary partitions of Q always have a common refinement.

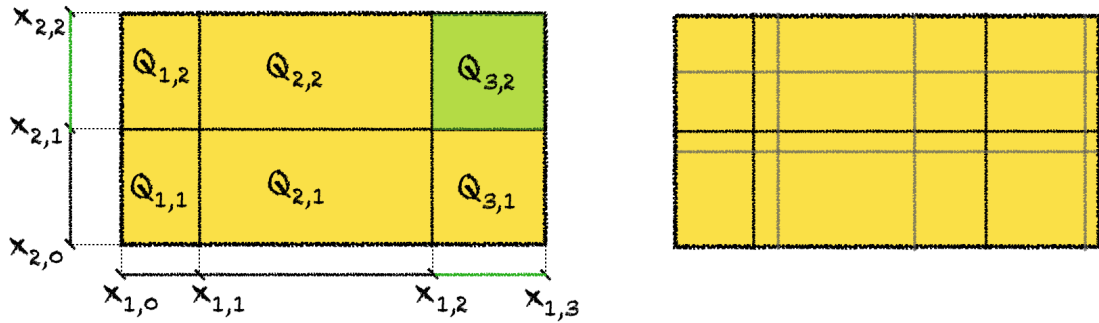


Figure 14.1: On the left, a partition of a box $[a_1, b_1] \times [a_2, b_2] \subseteq \mathbb{R}^2$ is shown. The horizontally represented interval $[a_1, b_1]$ is divided into three parts, and the vertically represented interval $[a_2, b_2]$ is divided into two parts. The address $\alpha = (3, 2)$ corresponds to the sub-box $Q_\alpha = [x_{1,2}, x_{1,3}] \times [x_{2,1}, x_{2,2}]$. On the right, a refinement of this partition is shown.

14.4. — Let $Q \subseteq \mathbb{R}^n$ be a box. A **step function** on Q is a bounded function $f : Q \rightarrow \mathbb{R}$ such that there exists a partition of Q , for each address α , the function f is constant on the open sub-box Q_α° . In this case, we also say that f is a step function with respect to this partition of Q . If c_α is the constant value of f on the open sub-box Q_α° , we write

$$\int_Q f(x) dx = \sum_{\alpha} c_\alpha \text{vol}(Q_\alpha) \quad (14.3)$$

where the sum extends over all addresses for the given partition. We call this number the **integral** of f over Q . In the case $n = 1$, this brings back the definitions ??? and ??? of step functions and their integral.

EXERCISE 14.5. — Let $Q \subseteq \mathbb{R}^n$ be a box. Show that the set of all step functions $\mathcal{TF}(Q)$ on Q forms a vector space with respect to pointwise addition and multiplication. Show that (14.3) is independent of the choice of partition and thus defines a monotone and linear mapping

$$\int_Q : \mathcal{TF}(Q) \rightarrow \mathbb{R}$$

14.6. — Let $Q \subseteq \mathbb{R}^n$ be a box, $\mathcal{TF}$ denote the vector space of step functions on Q , and $f : Q \rightarrow \mathbb{R}$ be a function. We define the sets of **lower sums** $\mathcal{U}(f)$ and **upper sums** $\mathcal{O}(f)$ of f as

$$\mathcal{U}(f) = \left\{ \int_Q u dx \mid u \in \mathcal{TF} \text{ and } u \leq f \right\} \quad \mathcal{O}(f) = \left\{ \int_Q o dx \mid o \in \mathcal{TF} \text{ and } f \leq o \right\}$$

If f is bounded, these sets are non-empty. Due to the monotonicity of the integral for step functions, the inequality

$$\sup \mathcal{U}(f) \leq \inf \mathcal{O}(f)$$

holds if f is bounded.

DEFINITION 14.7. — Let $Q \subseteq \mathbb{R}^n$ be a box, and $f : Q \rightarrow \mathbb{R}$ be a bounded function. We call $\sup \mathcal{U}(f)$ the **lower**, and $\inf \mathcal{O}(f)$ the **upper integral** of f . The function f is called **Riemann integrable** if $\sup \mathcal{U}(f) = \inf \mathcal{O}(f)$. In this case, the common value is referred to as the **Riemann integral** of f , and is written as

$$\int_Q f(x) dx = \sup \mathcal{U}(f) = \inf \mathcal{O}(f)$$

14.8. — Step functions are Riemann integrable, and the Riemann integral of a step function is precisely given by (14.3). The following alternative notations for the Riemann integral are common:

$$\int_Q f(x_1, x_2, \dots, x_n) dx_1 dx_2 \cdots dx_n \quad \int_Q f(x) d\text{vol}$$

The left-hand notation is useful when f is given as a function of variables $x, y, z, \dots$ whose combination has no specific name. The right-hand notation indicates that the integral is defined with respect to the standard volume for boxes in $\mathbb{R}^n$ - alternatives are conceivable and the subject of the so-called **measure theory**. The following proposition is a generalization of Proposition ??.

PROPOSITION 14.9. — Let $f : Q \rightarrow \mathbb{R}$ be bounded. The function f is Riemann integrable if and only if, for every $\varepsilon > 0$, there are step functions u and o on Q such that

$$u \leq f \leq o \quad \text{and} \quad \int_Q (o - u) dx < \varepsilon$$

Proof. The proof is the same as for Proposition ??. □

PROPOSITION 14.10. — Let $Q \subseteq \mathbb{R}^n$ be a box, and $\mathcal{R}(Q)$ denote the set of all Riemann integrable functions on Q . Then $\mathcal{R}(Q)$ is an $\mathbb{R}$ -vector space with respect to pointwise addition and multiplication, and integration

$$\int_Q : \mathcal{R}(Q) \rightarrow \mathbb{R}$$

is an $\mathbb{R}$ -linear map. Integration is also monotonic and satisfies the triangle inequality: It holds

$$f \leq g \implies \int_Q f(x)dx \leq \int_Q g(x)dx \quad \text{and} \quad \left| \int_Q f(x)dx \right| \leq \int_Q |f(x)|dx$$

for all $f, g \in \mathcal{R}(Q)$. In particular, $|f|$ is Riemann integrable.

Proof. The proof is analogous to the proof of the corresponding statements for the Riemann integral on an interval; see Theorem ?? for linearity, Proposition ?? for monotonicity, and Theorem ?? for the triangle inequality. $\square$

EXERCISE 14.11. — Let $Q \subseteq \mathbb{R}^2$ be the box $[-1, 1]^2$. Calculate the two-dimensional integral

$$\int_Q (2 - x^2 - y^2) dx dy$$

according to the definitions of this section. Then calculate

$$F(y) = \int_{-1}^1 f(x, y) dx \quad \text{and} \quad \int_{-1}^1 F(y) dy$$

and compare the results.

EXERCISE 14.12. — Let $Q \subseteq \mathbb{R}^n$ be a closed box and $f : Q \rightarrow \mathbb{R}$ a bounded function. Assume an arbitrary partition of Q is given. Show that f is Riemann integrable if and only if for each sub-box Q_α of the given partition, the restricted function $f|_{Q_\alpha} : Q_\alpha \rightarrow \mathbb{R}$ is Riemann integrable. Show that in this case

$$\int_Q f dx = \sum_{\alpha} \int_{Q_\alpha} f|_{Q_\alpha} dx$$

holds.

14.1.2 Null Sets

DEFINITION 14.13. — A subset $N \subseteq \mathbb{R}^n$ is called a **Lebesgue null set** or simply a **null set** if, for every $\varepsilon > 0$, there exists a countable family of open boxes $(Q_k)_{k \in \mathbb{N}}$ in $\mathbb{R}^n$ such that

$$N \subset \bigcup_{k=0}^{\infty} Q_k \quad \text{and} \quad \sum_{k=0}^{\infty} \text{vol}(Q_k) < \varepsilon \quad (14.4)$$

holds.

14.14. — We interpret the condition in the definition of null sets such that 0 is the only reasonable value for the volume of a null set N . In fact, (14.4) can be read as stating that N is contained in a set $\bigcup_{\ell=1}^{\infty} Q_{\ell}$ whose volume is small. Here, the volume of the countable union $\bigcup_{\ell=1}^{\infty} Q_{\ell}$ is not defined, but $\sum_{\ell=1}^{\infty} \text{vol}(Q_{\ell})$ can be regarded as an upper bound for it. We say that a statement A about elements $x \in \mathbb{R}^n$ is true **almost everywhere** if

$$\{x \in \mathbb{R}^n \mid \neg A(x)\}$$

is a null set. For example, if $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a function, we say f is **almost everywhere continuous** if the set of $x \in \mathbb{R}^n$ at which f is discontinuous is a null set.

LEMMA 14.15. — *A subset of a null set is a null set. A countable union of null sets is again a null set.*

Proof. The first statement follows directly from (14.4). Let $(N_j)_{j \in \mathbb{N}}$ be a countable family of null sets in $\mathbb{R}^n$, and let N be their union. Let $\varepsilon > 0$. Then, by the definition of null sets, for each $j \in \mathbb{N}$, there exists a countable family of open boxes $(Q_{jk})_{k \in \mathbb{N}}$ such that

$$N_j \subset \bigcup_{k=0}^{\infty} Q_{jk} \quad \text{and} \quad \sum_{k=0}^{\infty} \text{vol}(Q_{jk}) < \frac{\varepsilon}{2^{j+1}}.$$

This implies

$$N = \bigcup_{j=0}^{\infty} N_j \subset \bigcup_{j=0}^{\infty} \bigcup_{k=0}^{\infty} Q_{jk} \quad \text{and} \quad \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \text{vol}(Q_{jk}) < \sum_{j=0}^{\infty} \frac{\varepsilon}{2^{j+1}} = \varepsilon.$$

Since $\mathbb{N} \times \mathbb{N}$ is countable, and $\varepsilon > 0$ was arbitrary, this shows that N is a null set. □

EXAMPLE 14.16. — Every countable subset in $\mathbb{R}^n$ is a null set. In particular, for example, $\mathbb{Q}^n \subseteq \mathbb{R}^n$ is a null set, since $\mathbb{Q}^n$ is countable. Every box with an empty interior is a null set. Every linear subspace $V \subsetneq \mathbb{R}^n$ is a null set in $\mathbb{R}^n$. In Proposition 14.18, we will show that an open subset $U \subseteq \mathbb{R}^n$ is a null set if and only if U is empty.

EXERCISE 14.17. — Let $N \subseteq \mathbb{R}^n$ be a null set, and $\alpha : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear map. Show that $\alpha(N)$ is a null set.

PROPOSITION 14.18. — *A set $X \subseteq \mathbb{R}^n$ with a non-empty interior is not a null set.*

Proof. If the interior of X is not empty, then X contains a closed box with a non-empty interior. Let $Q = [a_1, b_1] \times \cdots \times [a_n, b_n]$ be such a box, $a_1 < b_1, \dots, a_n < b_n$. It suffices to show that Q is not a null set. If Q were a null set, there would exist open boxes $O_1, O_2, \dots$

with

$$Q \subset \bigcup_{k=1}^{\infty} O_k \quad \text{and} \quad \sum_{k=1}^{\infty} \text{vol}(O_k) < \frac{\text{vol}(Q)}{2}.$$

By the Heine-Borel Theorem ?? and Theorem ??, Q is compact, so there exists an $m \in \mathbb{N}$ such that

$$Q \subset \bigcup_{k=1}^m O_k \quad \text{and} \quad \sum_{k=1}^m \text{vol}(O_k) < \frac{\text{vol}(Q)}{2}. \quad (14.5)$$

We define the boxes $Q_k = \overline{O_k} \cap Q$ and write $Q_k = [a_{k,1}, b_{k,1}] \times \dots \times [a_{k,n}, b_{k,n}]$ for all $k \in \{1, \dots, m\}$. In particular, $Q = \bigcup_{k=1}^m Q_k$.

For each $j \in \{1, \dots, n\}$, we can define a partition of $[a_j, b_j]$ by arranging the points $\{a_{1,j}, b_{1,j}, \dots, a_{m,j}, b_{m,j}\}$. This gives us a partition of Q such that each closed sub-box $Q_k \subseteq Q$ is a finite union of boxes $\overline{Q}_\alpha$, where α are addresses in this partition of Q . It follows that

$$\text{vol}(Q) = \sum_{\alpha} \text{vol}(Q_\alpha) \quad \text{and} \quad \text{vol}(Q_k) = \sum_{\alpha | Q_\alpha \subseteq Q_k} \text{vol}(Q_\alpha)$$

which, after summing over k , results in the inequality

$$\text{vol}(Q) \leq \sum_{k=1}^n \text{vol}(Q_k)$$

which contradicts (14.5). □

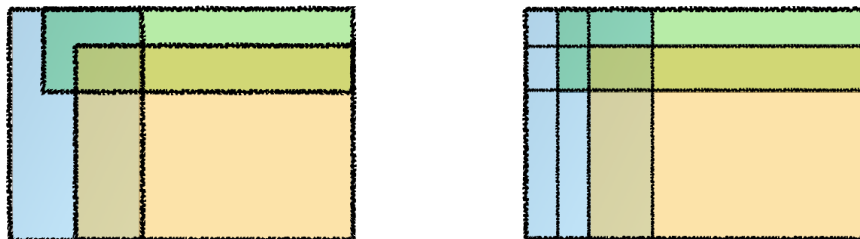


Figure 14.2: After reducing to a finite cover of the cube Q , it is much easier to show that the total volume of the covering cubes is at least as large as the volume of Q . In fact, for the latter, a partition is fabricated from $Q_1, \dots, Q_m$ as illustrated, and then the addition formula (??) is applied.

EXERCISE 14.19. — Show that a subset $N \subseteq \mathbb{R}^n$ is a null set if and only if, for every $\varepsilon > 0$, there exists a family $(Q_k)_{k \in \mathbb{N}}$ of *closed* cubes, with

$$N \subset \bigcup_{\ell=1}^{\infty} Q_\ell \quad \text{and} \quad \sum_{\ell=1}^{\infty} \text{vol}(Q_\ell) < \varepsilon.$$

PROPOSITION 14.20. — Let $Q \subset \mathbb{R}^{n-1}$ be a closed cube, and $f : Q \rightarrow \mathbb{R}$ be Riemann integrable. Then, the graph $\{(x, f(x)) \mid x \in Q\} \subseteq \mathbb{R}^n$ of f is a null set.

Proof. Let $\varepsilon > 0$. Since f is Riemann integrable, for every $\varepsilon > 0$, there exist step functions u and o on Q with $u \leq f \leq o$ and $\int_Q (o - u) dx < \varepsilon$. We choose a partition of Q such that for each address α , both functions u and o are constant on Q_α° . Let c_α be the constant value of u , and d_α be the constant value of o on Q_α° . For each address α , we define the cube $P_\alpha = Q_\alpha \times [c_\alpha, d_\alpha]$ and obtain

$$\text{graph}(f) \subseteq \bigcup_{\alpha} P_\alpha \cup \bigcup_{\alpha} (\partial Q_\alpha \times \mathbb{R})$$

where the second union corresponds to the grid of the partition of Q . It is a finite union of axis-parallel hyperplanes in $\mathbb{R}^n$, which we can cover with countably many cubes with volume 0. It holds

$$\sum_{\alpha} \text{vol}(P_\alpha) = \sum_{\alpha} (d_\alpha - c_\alpha) \text{vol}(Q_\alpha) = \int_Q (o - u) dx < \varepsilon,$$

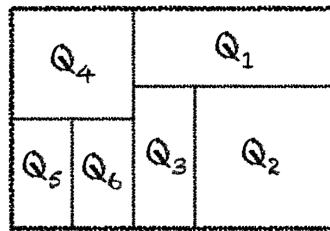
showing that $\text{graph}(f)$ can be covered by countably many cubes with a sum of volumes less than ε . Since $\varepsilon > 0$ was arbitrary, the proposition follows. $\square$

EXERCISE 14.21. — Let $U \subseteq \mathbb{R}^n$ be open, and let $\Phi : U \rightarrow \mathbb{R}^n$ be a Lipschitz map. Show that the image of a Lebesgue null set under Φ is again a Lebesgue null set.

EXERCISE 14.22. — Let $n \geq 1$, $k \in \{1, \dots, n-1\}$, $Q \subseteq \mathbb{R}^{n-k}$ be a closed cube, and f be a continuous function on Q with values in $\mathbb{R}^k$. Show that $\text{graph}(f) \subseteq \mathbb{R}^n$ is a null set.

EXERCISE 14.23. — Use the previous exercise to show that every k -dimensional submanifold of $\mathbb{R}^n$ for $k < n$ is a null set in $\mathbb{R}^n$.

EXERCISE 14.24. — Given is a rectangle $Q \subseteq \mathbb{R}^2$, which has a finite cover with rectangles $Q_1, \dots, Q_n \subset Q$ that intersect only along their edges.



Assume that each of the rectangles $Q_1, \dots, Q_n$ has at least one edge of integer length. Show that then Q also has at least one edge of integer length.

14.1.3 The Lebesgue Criterion

THEOREM 14.25 (Lebesgue Criterion). — *A real-valued, bounded function on a closed cube Q is Riemann integrable if and only if it is continuous almost everywhere, i.e., if the set*

$$N = \{x \in Q \mid f \text{ is discontinuous at } x\}$$

is a null set.

COROLLARY 14.26. — *Let Q be a closed cube with non-empty interior. Then, every continuous function $f : Q \rightarrow \mathbb{R}$ is Riemann integrable.*

Proof. This follows directly from the Lebesgue Criterion. Since Q is compact, all continuous functions on Q are bounded. $\square$

14.27. — In Definition ??, the oscillation $\omega(f, x)$ of $f : Q \rightarrow \mathbb{R}$ was introduced, describing the fluctuation of f around a point $x \in Q$. For $\delta > 0$, we define

$$\omega(f, x, \delta) = \sup f(Q \cap B_\infty(x, \delta)) - \inf f(Q \cap B_\infty(x, \delta)),$$

where $B_\infty(x, \delta)$ denotes the ball with respect to the supremum norm. Such a ball is an open, axis-parallel cube with center x and side length 2δ . The oscillation $\omega(f, x)$ is defined as the limit, independent of the shape of the balls, of $\omega(f, x, \delta)$ as $\delta \rightarrow 0$.

LEMMA 14.28. — *Let $f : X \rightarrow \mathbb{R}$ be a bounded function on a metric space X . For every $\eta \geq 0$, the subset $N_\eta = \{x \in X \mid \omega(f, x) \geq \eta\} \subseteq X$ is closed.*

Proof. Let $\eta \geq 0$, and let $(x_k)_{k=0}^\infty$ be a convergent sequence in X with $\omega(f, x_k) \geq \eta$ for all $k \in \mathbb{N}$ and with limit $x \in X$. Take $\delta > 0$ arbitrarily. Then, there exists a k such that $x_k \in B(x, \delta)$. Since $B(x, \delta)$ is open, there is a $\delta_k > 0$ with $B(x_k, \delta_k) \subseteq B(x, \delta)$. From

$$\sup f(B(x, \delta)) \geq \sup f(B(x_k, \delta_k)), \quad \inf f(B(x, \delta)) \leq \inf f(B(x_k, \delta_k)),$$

it follows $\omega(f, x, \delta) \geq \omega(f, x_k, \delta_k)$. With $\omega(f, x_k, \delta_k) \geq \omega(f, x_k) \geq \eta$, we obtain $\omega(f, x, \delta) \geq \eta$, and since $\delta > 0$ was arbitrary, the lemma follows. $\square$

EXERCISE 14.29. — Let $f : X \rightarrow \mathbb{R}$ be a bounded function on a metric space X . Is the set $\{x \in X \mid f \text{ is discontinuous at } x\}$ closed?

Proof of Theorem 14.25. First, we assume that the bounded real-valued function f on Q is Riemann-integrable. Let $\eta > 0$ and $\varepsilon > 0$ be arbitrary. Then, according to Proposition 14.9, there exist step functions u and o on Q with $u \leq f \leq o$ and $\int_Q (o - u) dx < \varepsilon \eta$. We choose a partition of Q so that for each address α , the function u is constant on Q_α° with value c_α , and o is constant on Q_α° with value d_α . It holds

$$\sum_{\alpha} (d_\alpha - c_\alpha) \text{vol}(Q_\alpha) < \varepsilon \eta.$$

We define the set of addresses $A(\eta) = \{\alpha \mid (d_\alpha - c_\alpha) \geq \eta\}$ and obtain

$$\sum_{\alpha \in A(\eta)} \text{vol}(Q_\alpha) \leq \sum_{\alpha \in A(\eta)} \eta^{-1}(d_\alpha - c_\alpha) \text{vol}(Q_\alpha) < \varepsilon.$$

Now we consider the closed set $N_\eta = \{x \in Q \mid \omega(f, x) \geq \eta\}$ from Lemma 14.28. For each $\alpha \notin A(\eta)$ and $x \in Q_\alpha^\circ$, there exists a $\delta > 0$ such that $B(x, \delta)$ is contained in Q_α° . Therefore,

$$\omega(f, x) \leq \omega(f, x, \delta) \leq \sup f(Q_\alpha^\circ) - \inf f(Q_\alpha^\circ) \leq d_\alpha - c_\alpha < \eta$$

for all $\alpha \notin A(\eta)$ and $x \in Q_\alpha^\circ$. Every element $x \in N_\eta$ is either an element of ∂Q_α for an address α , or an element of Q_α° for an address $\alpha \in A(\eta)$. From

$$N_\eta \subset \bigcup_{\alpha} \partial Q_\alpha \cup \bigcup_{\alpha \in A(\eta)} Q_\alpha^\circ \quad \text{and} \quad \sum_{\alpha \in A(\eta)} \text{vol}(Q_\alpha) < \varepsilon$$

follows that N_η is a null set. From Lemma ??, it follows that

$$N = \{x \in Q \mid f \text{ is discontinuous at } x\} = \{x \in Q \mid \omega(f, x) > 0\} = \bigcup_{k=0}^{\infty} N_{2^{-k}}$$

is a null set. This establishes one direction of the Lebesgue Criterion.

To prove the reverse direction, assume that $f : Q \rightarrow \mathbb{R}$ is bounded, and $N = \{x \in Q \mid f \text{ is discontinuous at } x\}$ is a null set. Write $M = \sup |f(Q)|$ and let $\varepsilon > 0$. The set $N_\varepsilon = \{x \in Q \mid \omega(f, x) \geq \varepsilon\} \subseteq N$ is, according to Lemma ??, a null set, closed according to Lemma 14.28, and compact according to the Heine-Borel Theorem ?. Therefore, there exist finitely many open cubes $O_1, \dots, O_m$ such that

$$N_\varepsilon \subseteq \bigcup_{\ell=1}^m O_\ell \quad \text{and} \quad \sum_{\ell=1}^m \text{vol}(O_\ell) < \varepsilon. \quad (14.6)$$

The subset $K = Q \setminus \bigcup_{\ell=1}^m O_\ell$ is compact according to the Heine-Borel Theorem. By construction, $K \subseteq Q \setminus N_\varepsilon$, and thus, $\omega(f, x) < \varepsilon$ for all $x \in K$. According to Proposition ??, there exists a $\delta > 0$ such that for all $x \in K$, the estimate

$$\omega(f|_K, x, \delta) < 2\varepsilon \quad (14.7)$$

holds. We now choose a partition of Q so that the mesh size is smaller than δ , and such that each of the cubes $\overline{O_k} \cap Q$ is a union of closed cubes Q_α . We define step functions u and o with $u \leq f \leq o$ by

$$u(x) = \begin{cases} \inf f(Q_\alpha^\circ) & \text{if } x \in Q_\alpha^\circ \\ f(x) & \text{if } x \notin Q_\alpha^\circ \quad \forall \alpha \end{cases} \quad o(x) = \begin{cases} \sup f(Q_\alpha^\circ) & \text{if } x \in Q_\alpha^\circ \\ f(x) & \text{if } x \notin Q_\alpha^\circ \quad \forall \alpha \end{cases}$$

and want to show that $\int_Q (o - u) dx$ is small. To do this, we separate the sum corresponding to (??) into two parts. In the first part, we sum over addresses α such that Q_α is part of a

cube $\overline{O_\ell}$, and in the second part, we sum over the remaining addresses α . On the first sum, we then apply (14.6), and on the second sum, we apply (14.7). This results in

$$\int_Q (o - u) dx \leq \sum_{\ell=1}^m \sum_{\alpha | Q_\alpha \subseteq O_\ell} 2M \operatorname{vol}(Q_\alpha) + \sum_{\alpha | Q_\alpha \subseteq K} 2\varepsilon \operatorname{vol}(Q_\alpha) \leq 2M\varepsilon + 2\varepsilon \operatorname{vol}(Q) = 2(M + \operatorname{vol}(Q))\varepsilon,$$

and since $\varepsilon > 0$ was arbitrary, this implies the Riemann integrability of f . $\square$

14.1.4 Riemann Integrability and Continuity

Continuous functions on rectangles are integrable, as stated in Corollary 14.26, and are in many ways simpler to understand than general Riemann-integrable functions. In this section, we show that Riemann-integrable functions can be *sandwiched* between continuous functions. This will later allow us to reduce some statements about Riemann-integrable functions to statements about continuous functions.

PROPOSITION 14.30. — *Let $Q \subseteq \mathbb{R}^n$ be a closed rectangle. A bounded function $f : Q \rightarrow \mathbb{R}$ is Riemann-integrable if and only if, for every $\varepsilon > 0$, there exist continuous functions $f_-, f_+ : Q \rightarrow \mathbb{R}$ satisfying*

$$f_- \leq f \leq f_+ \quad \text{and} \quad \int_Q (f_+ - f_-) dx < \varepsilon. \quad (14.8)$$

Proof. Suppose there exist continuous functions f_- and f_+ on Q satisfying (14.8) for $\varepsilon > 0$. Since f_- and f_+ are Riemann-integrable according to Corollary 14.26, there exist step functions u and o with $u \leq f_-$ and $\int_Q (f_- - u) dx < \varepsilon$, as well as $f_+ \leq o$ and $\int_Q (o - f_+) dx < \varepsilon$. It follows that $u \leq f \leq o$ and

$$\int_Q (o - u) dx \leq \int_Q (o - f_+ + f_- - u) dx \leq 2\varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, this implies the Riemann integrability of f from Proposition 14.9.

For the reverse direction, we proceed step by step, making progressively weaker assumptions about the functions f .

Case 1: $f = \mathbb{1}_{Q'}$ for a closed rectangle $Q' \subseteq Q$. For $\delta > 0$, consider the functions f_+ and f_- on Q , given by

$$\begin{aligned} f_+(x) &= 1 - \min\{1, \delta^{-1} \inf\{\|y - x\| \mid y \in Q'\}\}, \\ f_-(x) &= \min\{1, \delta^{-1} \inf\{\|y - x\| \mid y \in Q \setminus Q'\}\}. \end{aligned}$$

The functions f_+ and f_- are both continuous, and it holds $f_- \leq \mathbb{1}_{Q'} \leq f_+$. Indeed, the function f_+ is constant with value 1 on Q' and constant with value 0 outside a δ -neighborhood of Q' . The function f_- is constant with value 0 on the complement of Q' and constant with value 0 outside a δ -neighborhood of $Q \setminus Q'$. In particular, $f_-(x) = \mathbb{1}_{Q'}(x) = f_+(x)$ for all x

outside a δ -neighborhood of $\partial Q'$, which implies the estimate for the integral in (14.8) as long as δ is chosen small enough.

Case 2: f is a step function. We choose a partition of Q that suits f . By duplicating each partition point, we can refine the partition so that for each subrectangle Q_α , each side face is again a subrectangle in the partition. Let N be the number of rectangles in this partition, and let $M \geq 0$ with $-M \leq f(x) \leq M$ for all $x \in Q$. According to the previous case, for each α there exist continuous functions $f_{\alpha,-} \leq \mathbb{1}_{Q_\alpha} \leq f_{\alpha,+}$ with

$$\int_Q (f_{\alpha,+} - f_{\alpha,-}) dx \leq \frac{\varepsilon}{2MN}$$

If the interior of Q_α is empty, then $\int_Q \mathbb{1}_{Q_\alpha} dx = 0$, and thus

$$-\frac{\varepsilon}{2MN} < \int_Q f_{\alpha,-} dx \quad \text{and} \quad \int_Q f_{\alpha,+} dx < \frac{\varepsilon}{2MN}$$

according to the reverse triangle inequality. We consider the continuous functions

$$f_- = \sum_{\alpha|c_\alpha \geq 0} c_\alpha f_{\alpha,-} + \sum_{\alpha|c_\alpha < 0} c_\alpha f_{\alpha,+} - \sum_{\beta} M f_{\beta,-}, \quad f_+ = \sum_{\alpha|c_\alpha \geq 0} c_\alpha f_{\alpha,+} + \sum_{\alpha|c_\alpha < 0} c_\alpha f_{\alpha,-} + \sum_{\beta} M f_{\beta,-}$$

where in each sum, the first two sums run over all addresses α with $Q_\alpha^\circ \neq \emptyset$ and c_α denotes the constant value of f on Q_α° , and the second sum runs over all addresses β with $Q_\beta^\circ = \emptyset$. It holds $f_- \leq f \leq f_+$ by construction, and

$$\begin{aligned} \int_Q (f_+ - f_-) dx &= \sum_{\alpha} |c_\alpha| \int_Q (f_{\alpha,+} - f_{\alpha,-}) dx + 2M \sum_{\beta} \int_Q (f_{\beta,+} + f_{\beta,-}) dx < \\ &\sum_{\alpha} \frac{|c_\alpha| \varepsilon}{2MN} + 2M \sum_{\beta} \frac{\varepsilon}{2MN} \leq \varepsilon \end{aligned}$$

which was to be shown.

Case 3: f is Riemann-integrable. Let $\varepsilon > 0$ and let $u \leq f \leq o$ be step functions with $\int_Q (o - u) dx < \varepsilon$. We have already shown that there exist continuous functions $u_- \leq u \leq u_+$ and $o_- \leq o \leq o_+$ with $\int_Q (u_+ - u_-) dx \leq \varepsilon$ and $\int_Q (o_+ - o_-) dx \leq \varepsilon$. It follows that $u_- \leq f \leq o_+$ and

$$\int_Q (o_+ - u_-) dx \leq \varepsilon + \int_Q (o_+ - o + u - u_-) dx \leq \varepsilon + \int_Q (o_+ - o_-) dx + \int_Q (u_+ - u_-) dx \leq 3\varepsilon$$

which proves the statement, as $\varepsilon > 0$ was arbitrary. $\square$

14.2 The Riemann Integral over Jordan-Measurable Sets

We now turn away from integration over axis-parallel rectangles and aim to consider more general sets, such as the circular disk in $\mathbb{R}^2$.

14.2.1 Jordan Measurability

DEFINITION 14.31. — A subset B of $\mathbb{R}^n$ is called **Jordan-measurable** if there exists a closed rectangle Q in $\mathbb{R}^n$ with $Q \supseteq B$, such that the characteristic function $\mathbb{1}_B$ on Q is Riemann-integrable. The **volume** or **Jordan measure** of B is defined in this case by

$$\text{vol}(B) = \int_Q \mathbb{1}_B dx$$

COROLLARY 14.32 (Lebesgue's Criterion). — *A subset $B \subset \mathbb{R}^n$ is Jordan-measurable if and only if B is bounded and the boundary ∂B is a null set. If $B_1, B_2 \subset \mathbb{R}^n$ are Jordan-measurable, then $B_1 \cup B_2$, $B_1 \cap B_2$, and $B_1 \setminus B_2$ are also Jordan-measurable.*

Proof. The first statement follows from Lebesgue's Criterion and the equality

$$\partial B = \{x \in \mathbb{R}^n \mid \mathbb{1}_B \text{ is not continuous at } x\} \quad (14.9)$$

which can be verified directly. Now, let $B_1, B_2 \subset Q$ be Jordan-measurable. It holds

$$\mathbb{1}_{B_1 \cap B_2} = \mathbb{1}_{B_1} \cdot \mathbb{1}_{B_2}, \quad \mathbb{1}_{B_1 \cup B_2} = \mathbb{1}_{B_1} + \mathbb{1}_{B_2} - \mathbb{1}_{B_1 \cap B_2}, \quad \mathbb{1}_{B_1 \setminus B_2} = \mathbb{1}_{B_1} - \mathbb{1}_{B_1 \cap B_2}$$

and, according to Equation (14.9), the boundaries $\partial(B_1 \cup B_2)$, $\partial(B_1 \cap B_2)$, and $\partial(B_1 \setminus B_2)$ are contained in the union $\partial B_1 \cup \partial B_2$ and thus null sets. Again, by Lebesgue's Criterion, $B_1 \cup B_2$, $B_1 \cap B_2$, and $B_1 \setminus B_2$ are Jordan-measurable. $\square$

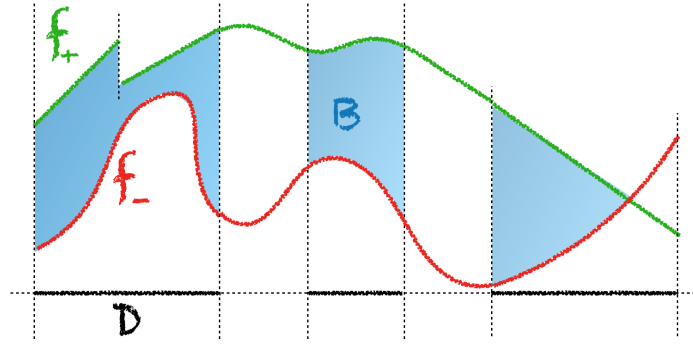
PROPOSITION 14.33. — *Let $Q \subset \mathbb{R}^{n-1}$ be a closed rectangle, $f_-, f_+ : Q \rightarrow \mathbb{R}$ Riemann integrable, and let $D \subseteq Q$ be Jordan-measurable. Then, the set*

$$B = \{(x, y) \in \mathbb{R}^n \mid x \in D, f_-(x) \leq y \leq f_+(x)\}$$

is Jordan-measurable.

Proof. The functions f_- and f_+ are bounded, so there exists $M \geq 0$ such that $-M < f_-(x) < M$ and $-M < f_+(x) < M$ for all $x \in Q$. Let $N_- \subseteq D$ and $N_+ \subseteq D$ be the set of discontinuity points of f_- and f_+ , respectively, and write $N = \partial D \cup N_- \cup N_+$. The set B is bounded since it is contained in $Q \times [-M, M]$. The boundary of B is contained in the union

$$N \times [-M, M] \cup \text{graph}(f_-) \cup \text{graph}(f_+)$$



By Proposition 14.20, $\text{graph}(f_-)$ and $\text{graph}(f_+)$ are null sets. According to Lebesgue's Criterion, N as the union of three null sets is also a null set. Therefore, $N \times [-M, M]$ is a null set. In fact, for a countable cover by open rectangles $\{Q_k | k \in \mathbb{N}\}$ of A , the family $\{Q_k \times [-M, M] | k \in \mathbb{N}\}$ is a countable cover by open rectangles of $N \times [-M, M]$, and from

$$\sum_{k=0}^{\infty} \text{vol}(Q_k) < \frac{\varepsilon}{2M} \implies \sum_{k=0}^{\infty} \text{vol}(Q_k \times [-M, M]) < \varepsilon$$

the claim follows. This implies that ∂B is a null set, so B is Jordan-measurable. $\square$

EXERCISE 14.34. — Show that for a null set $N \subseteq \mathbb{R}^{n-1}$, $N \times \mathbb{R}$ is also a null set in $\mathbb{R}^n$. Show that for a Jordan-measurable set $D \subseteq \mathbb{R}^{n-1}$ and any $a < b$, $D \times [a, b]$ is a Jordan-measurable subset of $\mathbb{R}^n$.

DEFINITION 14.35. — Let $B \subseteq \mathbb{R}^n$ be a Jordan-measurable subset, and let f be a real-valued function on B . Then, f is called **Riemann-integrable** if there exists a closed rectangle $Q \subseteq \mathbb{R}^n$ with $B \subseteq Q$, such that the function given by

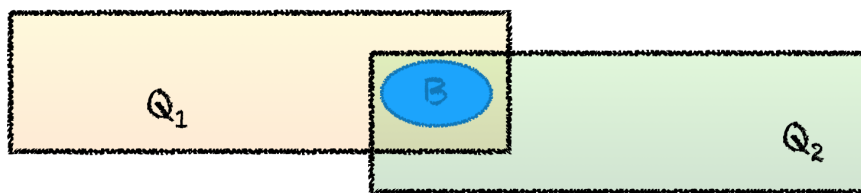
$$f_1(x) = \begin{cases} f(x) & \text{if } x \in B \\ 0 & \text{if } x \in Q \setminus B \end{cases}$$

is Riemann-integrable on Q . In this case, we write

$$\int_B f dx = \int_Q f_1 dx$$

and call this number the **Riemann integral** of f over B .

14.36. — Let $B \subseteq \mathbb{R}^n$ be a Jordan-measurable subset, and let $f : Q \rightarrow \mathbb{R}$ be a Riemann-integrable function. The integral of f over B , as defined in ??, depends a priori on the choice of a closed box Q containing the set B . The independence of the choice of the box is shown as follows: Suppose $Q_1, Q_2 \subset \mathbb{R}^n$ are two closed boxes with $B \subset Q_1$ and $B \subset Q_2$. Then, $Q = Q_1 \cap Q_2$ is also a box with $B \subseteq Q$.



The integral of the function f_1 extended by the constant value 0 over Q_1 is equal to the corresponding integral over Q , as becomes evident when considering a partition of Q_1 for which $Q \subseteq Q_1$ is a sub-box, as shown in Exercise ??.

EXERCISE 14.37. — Let $a \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$. Show that for every Jordan-measurable subset $B \subseteq \mathbb{R}^n$, the subset $a + \lambda B = \{a + \lambda b \mid b \in B\}$ is Jordan-measurable with volume

$$\text{vol}(a + \lambda B) = |\lambda^n| \text{vol}(B).$$

COROLLARY 14.38 (Lebesgue's Criterion). — Let $B \subseteq \mathbb{R}^n$ be Jordan-measurable, and let $f : B \rightarrow \mathbb{R}$ be bounded. Then, f is Riemann-integrable if and only if f is almost everywhere continuous on B , meaning the set of points of discontinuity of f is a null set. In particular, every bounded continuous function on a Jordan-measurable set is Riemann-integrable.

Proof. Let Q be a closed box with $B \subseteq Q$. Since, by assumption, B is Jordan-measurable, $\mathbb{K}_B$ is Riemann-integrable on Q , and the set ∂B of points of discontinuity of $\mathbb{K}_B$ is a null set according to Lebesgue's Criterion in Theorem ??. Define $f_1 : Q \rightarrow \mathbb{R}$ by

$$f_1(x) = \begin{cases} f(x) & \text{if } x \in B \\ 0 & \text{if } x \in Q \setminus B \end{cases}$$

then

$$\{x \in Q \mid f_1 \text{ is discontinuous at } x\} \subseteq \partial B \cup \{x \in B \mid f \text{ is discontinuous at } x\}.$$

The corollary follows from Lebesgue's Criterion in Theorem ??. □

COROLLARY 14.39. — Let $B \subseteq \mathbb{R}^n$ be a Jordan-measurable subset. Linearity of the Riemann integral, monotonicity, and the triangle inequality as in Proposition ?? hold analogously for the Riemann integral of Riemann-integrable functions on B .

Proof. All statements follow directly by applying the propositions in Section 14.1.1 to f_1 as in Definition ??. □

PROPOSITION 14.40. — Let $B_1 \subseteq \mathbb{R}^n$ and $B_2 \subseteq \mathbb{R}^n$ be Jordan-measurable, and let $f : B_1 \cup B_2 \rightarrow \mathbb{R}$ be a Riemann-integrable function. Then, $f|_{B_1}$ is Riemann-integrable, and it

holds

$$\int_{B_1 \cup B_2} f dx = \int_{B_1} f dx + \int_{B_2} f dx - \int_{B_1 \cap B_2} f dx.$$

Proof. Integrability of the restriction of f to B_1 , B_2 , or $B_1 \cap B_2$ follows from Lebesgue's Criterion. We have $\mathcal{K}_{B_1 \cup B_2} = \mathcal{K}_{B_1} + \mathcal{K}_{B_2} - \mathcal{K}_{B_1 \cap B_2}$. The proposition follows by multiplying with f and using the linearity of the Riemann integral in Proposition ??.

□

EXERCISE 14.41. — A subset $J \subset \mathbb{R}^n$ is called a **Jordan null set** if, for every $\epsilon > 0$, there exists a finite family $Q_1, \dots, Q_m \subset \mathbb{R}^n$ of open boxes such that

$$J \subseteq \bigcup_{k=1}^m Q_k, \quad \sum_{k=1}^m \text{vol}(Q_k) < \epsilon$$

Show the equivalence of the following statements for a subset $J \subseteq \mathbb{R}^n$:

1. J is a Jordan null set.
2. J is Jordan-measurable, and $\text{vol}(J) = 0$.
3. $\overline{J}$ is a bounded Lebesgue null set.

EXERCISE 14.42. — Let J be a Jordan null set, and let $f : J \rightarrow \mathbb{R}$ be a bounded function. Show that f is Riemann-integrable, and $\int_J f dx = 0$.

EXERCISE 14.43. — Let Q be a closed box, and let $f : Q \rightarrow \mathbb{R}$ be a Riemann-integrable function with $f \geq 0$ and $\int_Q f dx = 0$. Show, in analogy to Exercise ??, that $f = 0$ almost everywhere. Formulate and prove the analogous statement for functions on Jordan-measurable subsets.

14.44. — Let Q be a closed box with non-empty interior, and $\mathcal{R}(Q)$ be the vector space of Riemann-integrable functions on Q . Then, the expression

$$\|\cdot\|_1 : \mathcal{R}(Q) \rightarrow \mathbb{R} \quad \|f\|_1 = \int_Q |f| dx$$

defines a so-called **seminorm** on $\mathcal{R}(Q)$. In fact, $\|\cdot\|_1$ satisfies all properties of a norm from Definition ??, except definiteness. Instead, for $f \in \mathcal{R}(Q)$ with $\|f\|_1 = 0$ according to Exercise 14.43, f is almost everywhere equal to zero. The set

$$\mathcal{N}(Q) = \{f \in \mathcal{R}(Q) \mid \|f\|_1 = 0\}$$

is a linear subspace of $\mathcal{R}(Q)$, and the seminorm $\|\cdot\|_1$ can be interpreted as a norm on the quotient space $\mathcal{R}(Q)/\mathcal{N}(Q)$. The resulting normed vector space is not complete and thus not as useful for advanced analysis. However, the completion of this or similarly constructed spaces plays a crucial role in **measure theory**, extensively studied.

14.2.2 The Fubini's Theorem

So far, we have not seen a general method for the explicit calculation of multidimensional integrals. We address this in this section with the discussion of Fubini's theorem (1879-1943). This theorem states that an integral $\int_Q f dx$ over a box can be calculated as an iterated parameter integral, as hinted at in Exercise ???. We use the notation introduced in ??? for lower and upper sums $\mathcal{U}(f)$, $\mathcal{O}(f)$ for bounded functions f on a box Q .

THEOREM 14.45 (Fubini). — *Let $P \subset \mathbb{R}^n$ and $Q \subset \mathbb{R}^m$ be closed boxes, and $f : P \times Q \rightarrow \mathbb{R}$ be a Riemann-integrable function. For $x \in P$, define $f_x : Q \rightarrow \mathbb{R}$ as the function $f_x(y) = f(x, y)$, and define*

$$F_-(x) = \sup \mathcal{U}(f_x) \quad \text{and} \quad F_+(x) = \inf \mathcal{O}(f_x)$$

There exists a null set $N \subseteq P$ such that for all $x \notin N$, the function f_x is Riemann integrable, and thus

$$F_-(x) = F_+(x) = \int_Q f_x(y) dy = \int_Q f(x, y) dy$$

holds. The functions F_- and F_+ on P are both Riemann-integrable, and it holds

$$\int_{P \times Q} f(x, y) d(x, y) = \int_P F_-(x) dx = \int_P F_+(x) dx.$$

Proof. A partition of the box $P \times Q$ is given by a partition of P and a partition of Q . If $P_\alpha \subseteq P$ and $Q_\beta \subseteq Q$ are sub-boxes with respect to partitions of P and Q , then $P_\alpha \times Q_\beta$ is a sub-box with respect to the corresponding partition of $P \times Q$, formally with address (α, β) . We will use this correspondence implicitly several times in the proof. Furthermore, according to the definition of the volume of boxes, we have

$$\text{vol}(P_\alpha \times Q_\beta) = \text{vol}(P_\alpha) \cdot \text{vol}(Q_\beta) \quad (14.10)$$

where we do not distinguish in the notation between volumes of boxes in $\mathbb{R}^{n+m}$, $\mathbb{R}^n$, or $\mathbb{R}^m$.

Let $h : P \times Q \rightarrow \mathbb{R}$ be a step function with constant value $c_{\alpha, \beta}$ on $P_\alpha^\circ \times Q_\beta^\circ$ for a corresponding partition. For each address α and each fixed $x \in P_\alpha^\circ$, $h_x : y \mapsto h(x, y)$ is then a step function on Q with respect to the given partition of Q , and

$$\int_Q h_x(y) dy = \sum_{\beta} c_{\alpha, \beta} \text{vol}(Q_\beta)$$

is independent of $x \in P_\alpha^\circ$. It follows that the functions defined by

$$H_-(x) = \sup \mathcal{U}(h_x) \quad \text{and} \quad H_+(x) = \inf \mathcal{O}(h_x) \quad (14.11)$$

are both step functions on P with respect to the given partition of P . It holds

$$\int_{P \times Q} h(x, y) d(x, y) = \int_P H_-(x) dx = \int_P H_+(x) dx \quad (14.12)$$

due to (14.10).

With these preparations, we now consider a function $F : P \rightarrow \mathbb{R}$ with $F_- \leq F \leq F_+$, choose $\epsilon > 0$ and step functions u and o on $P \times Q$ with $u \leq f \leq o$ and $\int_{P \times Q} (o - u) d(x, y) < \epsilon$. Defining step functions U_- and O_+ according to (14.11), it follows

$$U_- \leq F \leq O_+ \quad \text{and} \quad \int_P (O_+ - U_-) dx = \int_{P \times Q} (o - u) d(x, y) < \epsilon$$

from elementary properties of supremum and infimum as well as (14.12). This shows that F is Riemann-integrable, and also that

$$\int_P F(x) dx = \int_{P \times Q} f(x, y) d(x, y)$$

holds. Choosing $F = F_-$ or $F = F_+$, the second statement of Fubini's theorem is obtained.

It remains to show that the function $f_x : Q \rightarrow \mathbb{R}$ is Riemann-integrable for almost all $x \in X$, meaning that $F_+(x) = F_-(x)$ holds for almost all $x \in P$. We know that $g = F_+ - F_-$ is a non-negative, Riemann-integrable function on P with integral $\int_P g dx = 0$. According to the Lebesgue Criterion, the set of discontinuity points $N \subseteq P$ of g is a null set. If g is continuous at $x_0 \in P$, then $g(x_0) = 0$ must hold; otherwise, there would be a $\delta > 0$ such that $g(x) \geq \frac{1}{2}g(x_0)$ for all $x \in B(x_0, \delta)$, leading to the contradiction

$$0 = \int_P g dx \geq \frac{1}{2}g(x_0)\text{vol}(B(x_0, \delta)) > 0$$

It follows that $g(x) > 0 \implies x \in N$, and therefore

$$f_x \text{ not integrable} \iff g(x) > 0 \implies x \in N$$

concluding the proof of the theorem. $\square$

COROLLARY 14.46. — *Let $Q = [a_1, b_1] \times \cdots \times [a_n, b_n]$ be a box, and let $f : Q \rightarrow \mathbb{R}$ be Riemann-integrable. Then,*

$$\int_Q f dx = \int_{a_1}^{b_1} \cdots \int_{a_n}^{b_n} f(x_1, \dots, x_n) dx_n \dots dx_1$$

if all parameter integrals exist. Otherwise, the parameter integrals can be replaced by suprema of lower sums or infima of upper sums.

Proof. This follows by repeatedly applying Fubini's theorem. $\square$

COROLLARY 14.47. — *Let $D \subseteq \mathbb{R}^{n-1}$ be a Jordan-measurable set, let $\varphi_-, \varphi_+ : D \rightarrow \mathbb{R}$ be Riemann-integrable with $\varphi_- \leq \varphi_+$, and let $B \subseteq \mathbb{R}^n$ be the Jordan-measurable subset*

$$B = \{(x, y) \in D \times \mathbb{R} \mid \varphi_-(x) \leq y \leq \varphi_+(x)\}.$$

For any Riemann-integrable function f on B , it holds

$$\int_B f(x, y) d(x, y) = \int_D \left(\int_{\varphi_-(x)}^{\varphi_+(x)} f(x, y) dy \right) dx,$$

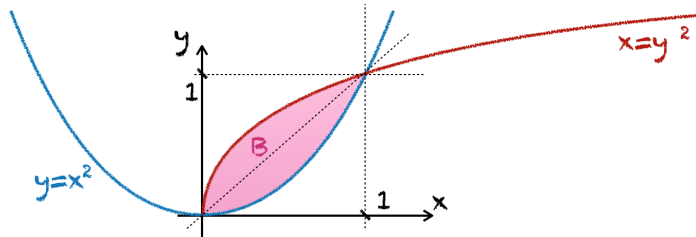
where the same complications as in Theorem 14.45 may arise.

Proof. According to Proposition 14.33, B is indeed Jordan-measurable. Applying Fubini's theorem to a box $P \subseteq \mathbb{R}^{n-1}$ with $D \subseteq Q$ and the interval $Q = [\inf \varphi_-(D), \sup \varphi_+(D)]$, and denoting $f_! : P \times Q \rightarrow \mathbb{R}$ as the zero-extension of f , we get

$$\int_B f(x, y) d(x, y) = \int_{P \times Q} f_! d(x, y) = \int_P \int_Q f_! dy dx = \int_D \int_{\varphi_-(x)}^{\varphi_+(x)} f(x, y) dy dx$$

as claimed. $\square$

EXAMPLE 14.48. — Consider the subset $B \subseteq \mathbb{R}^2$ enclosed between the parabolas with equations $y = x^2$ and $x = y^2$. We view this set as a uniformly thick, homogeneous plate and want to calculate its center of mass.



From Section ??, we already know that the region defined by $0 \leq y \leq x^2$ and $0 \leq x \leq 1$ has area $\frac{1}{3}$. Due to symmetry, the area of B is also $\frac{1}{3}$. Also, due to symmetry, the center of mass of B lies on the line with equation $x = y$. The x -coordinate x_S , and thus the y of the center of mass S , is given by definition through $x_S = \frac{1}{\text{vol}(B)} \int_B x dy dx$ and is calculated with Corollary 14.47 as

$$x_S = \frac{1}{\text{vol}(B)} \int_B x dy dx = 3 \int_0^1 \int_{x^2}^{\sqrt{x}} x dy dx = 3 \int_0^1 x(\sqrt{x} - x^2) dx = \frac{9}{20}.$$

EXERCISE 14.49. — Calculate the integral of the function $f(x, y, z) = xyz$ over the set $B \subseteq \mathbb{R}^3$ for

$$B = \{(x, y, z) \in \mathbb{R}^3 \mid 0 \leq x \leq y \leq z \leq 1\} \quad \text{and} \quad B = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 \leq 1\}.$$

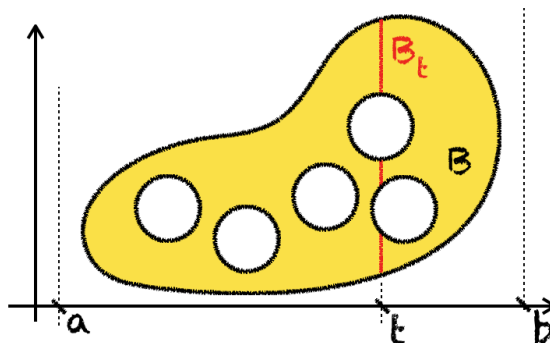
Calculate the volume of the set $A \subseteq \mathbb{R}^2$ enclosed between the curves $x^2 + y^2 = 8$ and $4y = x^2 + 4$.

COROLLARY 14.50 (Cavalieri's Principle). — Let $B \subseteq [a, b] \times \mathbb{R}^{n-1}$ be a bounded and Jordan-measurable set. Then,

$$\text{vol}(B) = \int_a^b \text{vol}(B_t) dt$$

where for $t \in [a, b]$, the subset $B_t \subseteq \mathbb{R}^{n-1}$ is given by $B_t = \{y \in \mathbb{R}^{n-1} \mid (t, y) \in B\}$ and is Jordan-measurable for almost all $t \in [a, b]$.

Proof. Follows directly from Theorem 14.45 with $f = \mathbb{K}_B$. □



EXERCISE 14.51. — In Section ??, the volume of rotational bodies was defined. Here, we verify that the introduced volume of Jordan-measurable subsets in the last section is compatible with it. Let $f : [a, b] \rightarrow \mathbb{R}_{\geq 0}$ be continuous, and let

$$K = \{(x, y, z) \in \mathbb{R}^3 \mid a \leq x \leq b, 0 \leq \sqrt{y^2 + z^2} \leq f(x)\}$$

be the rotational body given by f . Show that K is Jordan-measurable and that the volume of K_f in the sense of Definition 14.31 is given by $\pi \int_a^b f(x)^2 dx$.

EXERCISE 14.52. — Calculate $\int_0^1 \int_x^1 \exp(y^2) dy dx$.

14.53. — Theorems 14.45 and Corollary 14.50 are, among other things, necessary even for very elementary volume calculations. For example, one might wonder whether the volume of a pyramid in $\mathbb{R}^3$ can be geometrically calculated by decomposing the pyramid into finitely many operations of cutting and gluing into boxes. This method works excellently for polygons in $\mathbb{R}^2$. However, already for pyramids in $\mathbb{R}^3$, it can be proven, using the so-called **Dehn Invariant**, that this method generally does not work - thus, for the volume calculation of polyhedra in $\mathbb{R}^3$, one has to resort to integration methods of analysis.

14.3 Multidimensional Substitution Rule

14.3.1 The Substitution Rule and First Examples

In this section, we introduce the multidimensional substitution rule for integrable functions with compact support and illustrate it with some examples. We use the concept of a diffeomorphism from Definition 12.5. In this context, diffeomorphisms can be understood as smooth coordinate transformations. The proof is discussed in the following two sections 14.3.2 and 14.3.3.

14.54. — Let $n \geq 1$ and $U \subseteq \mathbb{R}^n$ be open. The **support** of a function $f : U \rightarrow \mathbb{R}$ is the set

$$\text{supp}(f) = \overline{\{x \in U \mid f(x) \neq 0\}}.$$

We say that f has **compact support** if $\text{supp}(f)$ is a compact subset of U . If $U \subseteq \mathbb{R}^n$ is open and Jordan-measurable, and $f : U \rightarrow \mathbb{R}$ is integrable, then the extension by zero of f to $\mathbb{R}^n$, usually denoted as $f_! : \mathbb{R}^n \rightarrow \mathbb{R}$, has compact support in $\mathbb{R}^n$. Since U is bounded, $\overline{U}$ and therefore $\text{supp}(f_!)$ are also bounded. Moreover, as $\text{supp}(f_!)$ is closed by definition, the assertion follows from the Heine-Borel theorem. However, having a compact support for $f : U \rightarrow \mathbb{R}$ is not necessary. Conversely, if $f : U \rightarrow \mathbb{R}$ is a function with compact support, we define the integral

$$\int_U f dx$$

as the integral of f over a box containing $\text{supp}(f)$, if it exists, and in that case, we call f Riemann-integrable. Here, U does not necessarily need to be bounded or Jordan-measurable.

THEOREM 14.55. — *Let $X, Y \subseteq \mathbb{R}^n$ be open subsets, $\Phi : X \rightarrow Y$ be a C^1 diffeomorphism, and $f : Y \rightarrow \mathbb{R}$ be a Riemann-integrable function with compact support. Then, the function $\Phi^* f : X \rightarrow \mathbb{R}$ defined by $(\Phi^* f)(x) = (f \circ \Phi(x)) |\det(D\Phi(x))|$ is Riemann-integrable, has compact support, and satisfies*

$$\int_Y f(y) dy = \int_X (f \circ \Phi(x)) |\det(D\Phi(x))| dx.$$

14.56. — The reason why the hypothesis about the support of f is necessary is as follows: The function $\arctan : \mathbb{R} \rightarrow (-\frac{\pi}{2}, \frac{\pi}{2})$ is a diffeomorphism. According to the substitution rule, here in dimension 1, for the constant function $f : (-\frac{\pi}{2}, \frac{\pi}{2}) \rightarrow \mathbb{R}$ with value 1, it should hold

$$\pi = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 1 dy = \int_{\mathbb{R}} |\arctan'(x)| dx = \int_{-\infty}^{\infty} \frac{1}{x^2 + 1} dx$$

This is not entirely incorrect, but in this case, the integral on the right must be understood as an improper Riemann integral, which we have not introduced in the multidimensional case

yet. According to Definition ??, the function $x \mapsto (x^2 + 1)^{-1}$ on $\mathbb{R}$ is not Riemann-integrable, as $\mathbb{R}$ as a subset of $\mathbb{R}$ is not Jordan-measurable.

PROPOSITION 14.57. — Let X and Y be open subsets of $\mathbb{R}^n$, and $\Phi : X \rightarrow Y$ be a homeomorphism. Let $f : Y \rightarrow \mathbb{R}$ be a function, and denote $g = f \circ \Phi$.

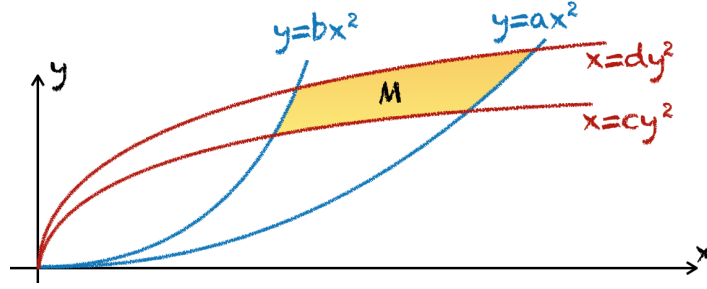
1. If f is Riemann-integrable, then g is also Riemann-integrable.
2. It holds $\text{supp}(g) = \Phi^{-1}(\text{supp}(f))$.

Proof. □

EXAMPLE 14.58. — Let $0 < a < b$ and $0 < c < d$ be fixed parameters used to define the curvilinear bounded rectangle $M \subseteq X = \mathbb{R}_{>0}^2$, which is bounded by the four curves

$$y = ax^2, \quad y = bx^2, \quad x = cy^2, \quad x = dy^2.$$

This is given by $M = \{(x, y) \in X \mid \frac{1}{b} \leq x^2 y^{-1} \leq \frac{1}{a}, \frac{1}{d} \leq x^{-1} y^2 \leq \frac{1}{c}\}$.



We want to use the substitution rule to calculate the area $\text{vol}(M)$ of M . To do this, we introduce the variables $u = x^2 y^{-1}$ and $v = x^{-1} y^2$. Through

$$\Psi \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x^2 y^{-1} \\ x^{-1} y^2 \end{pmatrix} \quad \text{and} \quad \Phi \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} u^{\frac{2}{3}} v^{\frac{1}{3}} \\ u^{\frac{1}{3}} v^{\frac{2}{3}} \end{pmatrix},$$

we obtain mutually inverse diffeomorphisms $\Psi : X \rightarrow X$ and $\Phi : X \rightarrow X$, and $\mathbb{1}_M \circ \Phi = \mathbb{1}_Q$ for the rectangle $Q = [\frac{1}{b}, \frac{1}{a}] \times [\frac{1}{d}, \frac{1}{c}]$. The Jacobian determinant of Φ is

$$\det(D\Phi(u, v)) = \det \begin{pmatrix} \frac{2}{3} u^{-\frac{1}{3}} v^{\frac{1}{3}} & \frac{1}{3} u^{\frac{2}{3}} v^{-\frac{2}{3}} \\ \frac{1}{3} u^{-\frac{2}{3}} v^{\frac{2}{3}} & \frac{2}{3} u^{\frac{1}{3}} v^{-\frac{1}{3}} \end{pmatrix} = \frac{4}{9} - \frac{1}{9} = \frac{1}{3}.$$

for all $(u, v) \in X$. Thus, we obtain

$$\text{vol}(M) = \int_X \mathbb{1}_M(x, y) dx dy = \int_X \mathbb{1}_Q(u, v) \cdot \frac{1}{3} du dv = \frac{1}{3} \left(\frac{1}{a} - \frac{1}{b} \right) \left(\frac{1}{c} - \frac{1}{d} \right).$$

EXAMPLE 14.59. — Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a Riemann-integrable function. We can apply Theorem 14.55 as follows:

$$\int_{\mathbb{R}^2} f(x, y) dx dy = \int_U f(x, y) dx dy = \int_0^\infty \int_0^{2\pi} f(r \cos \varphi, r \sin \varphi) r d\varphi dr.$$

where $U = \{(x, y) \in \mathbb{R}^2 \mid y \neq 0 \text{ or } x < 0\}$. In the first equation, we omit the integration over the Jordan null set $\{(x, y) \mid y = 0, x \geq 0\}$ using Proposition 14.40, and in the second equation, we apply Theorem 14.55 to the polar coordinate transformation discussed in Section ??

$$\Phi : (0, R) \times (0, 2\pi) \rightarrow U \quad \Phi \begin{pmatrix} r \\ \varphi \end{pmatrix} = \begin{pmatrix} r \cos \varphi \\ r \sin \varphi \end{pmatrix}$$

The factor r in the right integral is the Jacobian determinant of the polar coordinate transformation Φ .

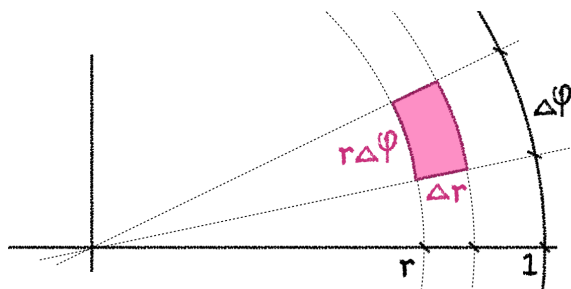


Figure 14.3: The factor r as the Jacobian determinant has a geometric meaning. A box with side lengths Δr and $\Delta \varphi$ corresponds to an almost rectangular section of a circular ring with “side lengths” Δr and approximately $r \Delta \varphi$, which is often informally denoted as dr and $r d\varphi$ due to the desired size of these differences.

EXAMPLE 14.60. — Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be a Riemann-integrable function. Then, with similar arguments as in Example 14.59, we have

$$\int_{\mathbb{R}^3} f(x, y, z) dx dy dz = \int_0^\infty \int_0^\pi \int_0^{2\pi} f(r \sin \theta \cos \varphi, r \sin \theta \sin \varphi, r \cos \theta) r^2 \sin \theta d\varphi d\theta dr,$$

where the Jacobian determinant $r^2 \sin \theta$ can again be geometrically interpreted.

EXERCISE 14.61. — Compute the area for given constants $0 < a < b$ and $0 < c < d$ of the set $\{(x, y) \in \mathbb{R}^2 \mid a \leq y \exp(-x) \leq b, c \leq y \exp(x) \leq d\}$.

EXERCISE 14.62. — Compute the following integral.

$$\int_V x^2 y z dx dy dz \quad V = \{(x, y, z) \in \mathbb{R}_{>0}^3 \mid x^2 + y^2 \leq 1, \frac{1}{\sqrt{3}} \leq \frac{y}{x} \leq \sqrt{3}, z \leq 1\}$$

14.3.2 Linear Substitution

First, we consider the important special case of a linear coordinate transformation, which will be significant in the proof of the general case.

LEMMA 14.63. — *Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be an invertible linear transformation given by an upper triangular matrix, and let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a continuous function with compact support. Then,*

$$\int_{\mathbb{R}^n} f(x) dx = |\det(T)| \int_{\mathbb{R}^n} f(T(x)) dx$$

Proof. Since T is invertible, the function $f \circ T : \mathbb{R}^n \rightarrow \mathbb{R}$ has compact support, namely $\text{supp}(f \circ T) = T^{-1}(\text{supp } f)$, and is Riemann-integrable. We express the integral $\int f(T(x)) dx$ using Fubini's theorem as a multiple parameter integral

$$\iiint \cdots \int f(t_{11}x_1 + t_{12}x_2 + \cdots + t_{1n}x_n, t_{22}x_2 + t_{23}x_3 + \cdots + t_{2n}x_n, \dots, t_{nn}x_n) dx_1 dx_2 \cdots dx_n$$

where t_{ij} are the coefficients of the matrix associated with T . Each of these parameter integrals extends from $-\infty$ to ∞ or alternatively from $-R$ to R for a sufficiently chosen $R > 0$, as f has compact support. For the integral with respect to the variable x_k , starting with $k = 1$, then for $k = 2$, and so on until $k = n$, we perform the one-dimensional linear substitution $y_k = |t_{kk}|x_k + t_{k,k+1}x_{k+1} + \cdots + t_{k,n}x_n$. In Leibniz notation, $dy_k = |t_{kk}|dx_k$, or $dx_k = |t_{kk}|^{-1}dy_k$. The multiple integral becomes

$$|t_{11}t_{22} \cdots t_{nn}|^{-1} \iiint \cdots \int f(y_1, y_2, \dots, y_n) dy_n \cdots dy_2 dy_1$$

which, again using Fubini's theorem, is precisely the integral $|\det T|^{-1} \int f(y) dy$. $\square$

LEMMA 14.64. — *Let $L : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be an invertible linear transformation, and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a Riemann-integrable function. Then, $f \circ L$ is Riemann-integrable, and it holds*

$$\int_{\mathbb{R}^n} f(x) dx = |\det(L)| \int_{\mathbb{R}^n} f(L(x)) dx$$

Proof. Let's assume initially that the function f is continuous. From linear algebra, we know that L can be written as the product of invertible matrices $L = PST$, where P is a permutation matrix, S is a lower triangular matrix, and T is an upper triangular matrix. The statement of Lemma 14.63 holds equally for lower triangular matrices with the same proof, and also for permutation matrices. It follows by applying Lemma 14.63 three times:

$$\int_{\mathbb{R}^n} f(PST(x)) dx = |\det(P)| |\det(S)| |\det(T)| \int_{\mathbb{R}^n} f(x) dx = |\det(L)| \int_{\mathbb{R}^n} f(x) dx$$

which proves the statement for continuous functions. Now, let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be any Riemann-integrable function with support in an open box $Q \subseteq \mathbb{R}^n$. According to Proposition ??, for

every $\epsilon > 0$, there exist continuous functions f_- and f_+ on $\mathbb{R}^n$ with support in Q such that

$$f_- \leq f \leq f_+ \quad \text{and} \quad \int_{\mathbb{R}^n} (f_+ - f_-) < \epsilon.$$

From this, using the already treated case of continuous functions, we get

$$f_- \circ L \leq f \circ L \leq f_+ \circ L \quad \text{and} \quad \int_{\mathbb{R}^n} (f_+ \circ L - f_- \circ L) < |\det L| \epsilon$$

which, by Proposition ??, implies that $f \circ L$ is Riemann integrable. We have

$$\begin{aligned} -\epsilon &< \int_{\mathbb{R}^n} f_-(x) dx - |\det L| \int_{\mathbb{R}^n} f_+(L(x)) dx \leq \int_{\mathbb{R}^n} f(x) dx - |\det L| \int_{\mathbb{R}^n} f(L(x)) dx \\ &\leq \int_{\mathbb{R}^n} f_+(x) dx - |\det L| \int_{\mathbb{R}^n} f_-(L(x)) dx < \epsilon \end{aligned}$$

which proves the lemma since $\epsilon > 0$ was arbitrary. $\square$

COROLLARY 14.65. — *Let $L : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation, and $B \subseteq \mathbb{R}^n$ be a Jordan-measurable subset. Then, $L(B)$ is Jordan-measurable, and $\text{vol}(L(B)) = |\det L| \text{vol}(B)$.*

Proof. If L is not invertible, then $L(B)$ is bounded and contained in a proper linear subspace of $\mathbb{R}^n$, and thus, a Jordan null set. If L is invertible, the volume $\text{vol}(L(B))$ is given by Lemma 14.64 as

$$\int_{\mathbb{R}^n} \mathbf{1}_{L(B)}(x) dx = |\det L| \int_{\mathbb{R}^n} \mathbf{1}_{L(B)}(L(x)) dx = |\det L| \int_{\mathbb{R}^n} \mathbf{1}_B(x) dx = |\det L| \text{vol}(B)$$

which completes the proof. $\square$

EXERCISE 14.66. — Compute, for a symmetric positive definite matrix $A \in \text{Mat}_n(\mathbb{R})$, the volume of the **ellipsoid** $\{x \in \mathbb{R}^n \mid \langle Ax, x \rangle \leq 1\}$, assuming knowledge of the volume of the unit ball.

COROLLARY 14.67. — *Let $L \in \text{Mat}_n(\mathbb{R})$ with columns $v_1, \dots, v_n \in \mathbb{R}^n$. Then, the parallelepiped*

$$P = L([0, 1]^n) = \left\{ \sum_{i=1}^n s_i v_i \mid 0 \leq s_i \leq 1 \right\}$$

is Jordan-measurable, and $\text{vol}(P) = |\det(L)| = \sqrt{\text{gram}(v_1, \dots, v_n)}$.

Proof. This follows from Corollary 14.65, applied to the Jordan-measurable set $[0, 1]^n$ with volume 1. Here, $\text{gram}(v_1, \dots, v_n)$ denotes the **Gram determinant**, which is the determinant of the matrix A with coefficients $a_{ij} = \langle v_i, v_j \rangle$. It holds $A = L^t L$, and therefore $|\det L|^2 = \det A$. $\square$

14.3.3 Proof of the Substitution Rule

In this section, we prove the substitution rule for functions with compact support, Theorem 14.55. We will repeatedly make use of the following fact.

14.68. — Let $X \subset \mathbb{R}^n$ be an open subset, and let $K_0 \subset X$ be a compact subset. Then, there exists $\delta_0 > 0$, such that the compact set

$$K_1 = K_0 + \overline{B_\infty(0, 2\delta_0)} = \{x + y \mid x \in K_0, \|y\|_\infty \leq 2\delta_0\}$$

is contained in X . The compactness of K_1 follows from the Heine-Borel theorem. The containment of K_1 in X implies that any closed box $Q \subseteq \mathbb{R}^n$ with maximum edge length less than δ_0 and $Q \cap K_0 \neq \emptyset$ is contained in K_1 . To see this, we can consider the continuous function, as per Exercise 9.54:

$$h : K_0 \rightarrow \mathbb{R}, \quad h(x) = \sup\{\delta > 0 \mid B_\infty(x, 2\delta) \subseteq X\}$$

Since X is open, we have $h(x) > 0$ for all $x \in K_0$, and by Theorem ??, this function attains its minimum on K_0 . Then, $\delta_0 = \frac{1}{3} \min\{h(x) \mid x \in K_0\}$ fulfills the required condition.

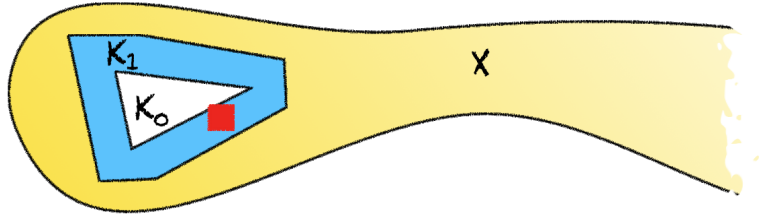


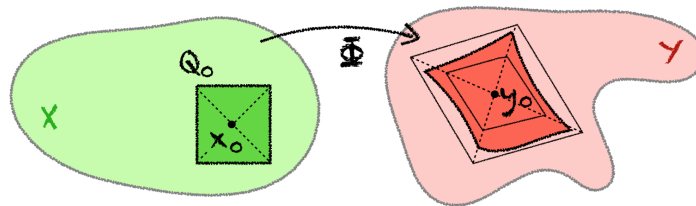
Figure 14.4: The set K_1 corresponds to a slightly inflated version of K_0 within X , defined in such a way that sufficiently small boxes intersecting K_0 are fully contained in K_1 .

LEMMA 14.69. — Let $X \subseteq \mathbb{R}^n$ and $Y \subseteq \mathbb{R}^n$ be open, and let $\Phi : X \rightarrow Y$ be a C^1 diffeomorphism. Consider an axis-parallel closed cube $Q_0 \subseteq X$ with side length $2r > 0$ and center $x_0 \in X$. Set $y_0 = \Phi(x_0)$, $L = D\Phi(x_0)$, and

$$\sigma = \max\{\|D\Phi(x) - L\|_{\text{op}} \mid x \in Q_0\}. \quad (14.13)$$

For any real number s satisfying $\sigma\|L^{-1}\|_{\text{op}}\sqrt{n} \leq s < 1$, we have

$$y_0 + (1 - s)L(Q_0 - x_0) \subseteq \Phi(Q_0) \subseteq y_0 + (1 + s)L(Q_0 - x_0). \quad (14.14)$$



Proof. We can replace Φ with $\Phi(x - x_0) - y_0$, without loss of generality, assuming $x_0 = 0$ and $y_0 = \Phi(x_0) = 0$. In this case, the inclusions (14.14) become

$$(1 - s)Q_0 \subseteq L^{-1}\Phi(Q_0) \subseteq (1 + s)Q_0,$$

which, in turn, is equivalent to the inequalities

$$\|x\|_\infty \leq r(1 - s) \implies x \in L^{-1}\Phi(Q_0) \quad (14.15)$$

$$\|x\|_\infty \leq r \implies \|L^{-1}\Phi(x)\|_\infty \leq (1 + s)r \quad (14.16)$$

For all $x_1, x_2 \in Q_0$, we have

$$\Phi(x_2) - \Phi(x_1) - L(x_2) + L(x_1) = \int_0^1 (D\Phi(x_1 + t(x_2 - x_1)) - L)(x_2 - x_1) dt$$

from which, using the vector-valued integral triangle inequality in (??), the estimation

$$\|(\Phi(x_2) - L(x_2)) - (\Phi(x_1) - L(x_1))\| \leq \sigma\|x_2 - x_1\| \leq \sigma\sqrt{n}\|x_2 - x_1\|_\infty \quad (14.17)$$

follows. For $x_2 = x \in Q_0$ and $x_1 = 0$, we obtain $\|\Phi(x) - L(x)\| \leq \sigma\sqrt{n}r$, and thus

$$\|L^{-1}\Phi(x) - x\|_\infty \leq \|L^{-1}\Phi(x) - x\| \leq \|L^{-1}\|_{\text{op}}\sigma\sqrt{n}r \leq sr \quad (14.18)$$

Therefore, the claimed inequality in (14.16) holds. For the first inclusion, we apply the Banach Fixed Point Theorem 9.56 to the map

$$T_x : Q_0 \rightarrow Q_0, \quad T_x(y) = x - (L^{-1}\Phi(y) - y)$$

For $y \in Q_0$ and $x \in (1 - s)Q_0$, we have, by (14.18)

$$\|T_x(y)\|_\infty \leq \|x\|_\infty + \|L^{-1}\Phi(y) - y\|_\infty \leq (1 - s)r + sr = r.$$

Thus, T_x indeed maps every point in Q_0 back to Q_0 . For $y_1, y_2 \in Q_0$, (14.17) implies

$$\|T_x(y_1) - T_x(y_2)\| = \|(\Phi(y_1) - y_1) - (\Phi(y_2) - y_2)\| \leq \|L^{-1}\|_{\text{op}}\sigma\|y_1 - y_2\|.$$

This shows $\|T_x(y_1) - T_x(y_2)\| \leq s\|y_1 - y_2\|$. Since $s < 1$ by assumption, we can apply the Banach Fixed Point Theorem, showing that for every $x \in (1 - s)Q_0$, there exists a unique $y \in Q_0$ such that $T_x(y) = y$ or equivalently $L^{-1}\Phi(y) = x$. This proves (14.16). $\square$

LEMMA 14.70. — *Let $X, Y \subset \mathbb{R}^n$ be open, let $\Phi : X \rightarrow Y$ be a C^1 diffeomorphism, and let $K_0 \subseteq X$ be a compact subset. Then, for any $\epsilon \in (0, 1)$, there exists $\delta > 0$ with the following property: For any cube Q_0 with center x_0 , side length smaller than δ , and $Q_0 \cap K_0 \neq \emptyset$, we have*

$$\frac{\text{vol}(\Phi(Q_0))}{1 + \epsilon} \leq |\det D\Phi(x_0)| \text{vol}(Q_0) \leq \frac{\text{vol}(\Phi(Q_0))}{1 - \epsilon}$$

Proof. We will construct parallelotopes P^+ and P^- such that $P^- \subseteq \Phi(Q_0) \subseteq P^+$ and fulfill the inequalities

$$\frac{\text{vol}(P^+)}{1 + \epsilon} < |\det D\Phi(x_0)| \text{vol}(Q_0) < \frac{\text{vol}(P^-)}{1 - \epsilon}$$

Let $s \in (0, 1)$ be small enough such that

$$(1 - s)^n > 1 - \epsilon, \quad (1 + s)^n < 1 + \epsilon,$$

and let $\delta_0 > 0$ be small enough such that the compact set $K_1 = K_0 + \overline{B_\infty(0, 2\delta_0)}$ is contained in X . According to Theorem ??, the continuous function $x \mapsto \|D\Phi(x)^{-1}\|_{\text{op}}$ is bounded on K_1 , so we have

$$\|D\Phi(x)^{-1}\|_{\text{op}} < M \quad \text{for all } x \in K_1$$

for some suitable $M > 0$. According to Proposition ??, $D\Phi$ is uniformly continuous on K_1 , so there exists $\delta \in (0, \delta_0)$ such that for all $x_0, x \in K_1$,

$$\|x_0 - x\|_\infty < \delta \implies M \cdot \|D\Phi(x) - D\Phi(x_0)\|_{\text{op}} \cdot \sqrt{n} < s$$

Hold. Now, let $x_0 \in K_0$ and Q_0 be a closed cube with center x_0 and side length smaller than δ . Write $y_0 = \Phi(x_0)$ and $L = D\Phi(x_0)$. According to Lemma 14.69, we have

$$y_0 + (1 - s)L(Q_0 - x_0) \subseteq \Phi(Q_0) \subseteq y_0 + (1 + s)L(Q_0 - x_0)$$

We set $P^+ = y_0 + (1 + s)L(Q_0 - x_0)$ and $P^- = y_0 + (1 - s)L(Q_0 - x_0)$. Therefore, $P^- \subseteq \Phi(\overline{Q_0}) \subseteq P^+$. According to Corollary 14.65,

$$\text{vol}(P^-) = (1 - s)^n |\det L| \text{vol}(Q_0) \quad \text{and} \quad \text{vol}(P^+) = (1 + s)^n |\det L| \text{vol}(Q_0)$$

which implies the inequalities in the lemma given the choice of s . □

Proof of Theorem ??. Let $f : Y \rightarrow \mathbb{R}$ be a Riemann integrable function. Writing $f = f_+ - f_-$ for non-negative functions f_+ and f_- as in ?? and using linearity of the integral, we can assume without loss of generality that $f \geq 0$. Write $g = f \circ \Phi$. Then, according to Proposition ??, $g : X \rightarrow \mathbb{R}$ is also Riemann-integrable, and the support $K_0 = \text{supp}(g) \subseteq X$ is compact. Choose $\delta_0 > 0$ small enough so that the compact set

$$K_1 := K_0 + \overline{B_\infty(0, \delta_0)}$$

is contained in X , and choose a closed box $Q \subseteq \mathbb{R}^n$ containing K_1 . Write g_1 for the extension by 0 of g to Q .

Let $\epsilon > 0$, and let $\delta \in (0, \delta_0)$ be small enough such that δ satisfies the statement for $\epsilon > 0$ in Lemma 14.70 for $K_0 \subseteq X$. Since the function $x \mapsto \det D\Phi(x)$ is continuous on X and thus

uniformly continuous on K_1 , we can additionally choose δ to be small enough that

$$\|x_0 - x\| \leq \delta \implies |\det D\Phi(x_0) - \det D\Phi(x)| < \epsilon$$

holds for all $x_0, x \in K_1$. Choose non-negative step functions u and o on Q with respect to a partition of Q with mesh smaller than δ , such that

$$u \leq g \leq o \quad \text{and} \quad \int_Q (o - u) dx < \epsilon$$

Write $c_\alpha \geq 0$ for the constant value of u on Q_α° and $d_\alpha \geq 0$ for the constant value of o on Q_α° . Let A be the set of addresses α in this partition for which $Q_\alpha \cap K_0 \neq \emptyset$. For $\alpha \in A$, we have $Q_\alpha \subseteq K_1$, and

$$\text{supp}(g) \subseteq \bigcup_{\alpha \in A} Q_\alpha \subseteq X \quad \text{and} \quad \text{supp}(f) \subseteq \bigcup_{\alpha \in A} \Phi(Q_\alpha) \subseteq Y.$$

For $\alpha \in A$, both ∂Q_α and $\Phi(\partial Q_\alpha)$ are Jordan measurable with volume zero. We can thus calculate the integral of $g|\det D\Phi|$ over X as

$$\int_X g(x) |\det D\Phi(x)| dx = \sum_{\alpha \in A} \int_{Q_\alpha^\circ} g(x) |\det D\Phi(x)| dx$$

and the integral of f over Y as

$$\int_Y f(y) dy = \sum_{\alpha \in A} \int_{\Phi(Q_\alpha^\circ)} f(y) dy$$

We now estimate the individual summands in these representations. For brevity, write $v_\alpha = \text{vol}(Q_\alpha)$ and $v = \text{vol}(Q)$, and choose $M > 0$ such that

$$|\det D\Phi(x_\alpha)| \leq M, \quad |c_\alpha| \leq M \quad \text{and} \quad |d_\alpha| \leq M$$

for all $\alpha \in A$, where x_α denotes the center of Q_α .

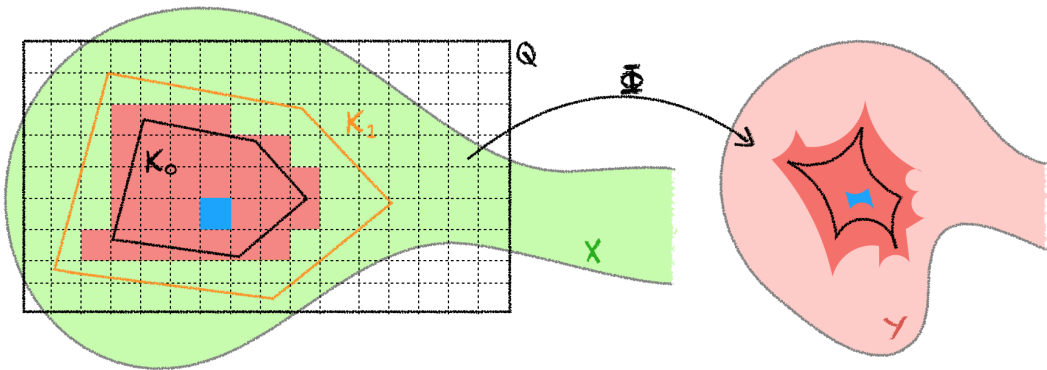


Figure 14.5: Geometric setup in the proof of the substitution rule. The boxes Q_α with $\alpha \in A$ are highlighted.

The given partition of Q having mesh smaller than δ means that for all $\alpha \in A$ and $x \in Q_\alpha$, the estimate $\|x_\alpha - x\| \leq \delta$ holds. By choosing δ accordingly, this implies $|\det(D\Phi(x)) - \det(D\Phi(x_\alpha))| \leq \epsilon$ for all $\alpha \in A$ and $x \in Q_\alpha$. This leads to the upper estimate

$$\int_{Q_\alpha} g(x) |\det(D\Phi(x))| dx \leq v_\alpha d_\alpha (|\det(D\Phi(x_\alpha))| + \epsilon) \leq v_\alpha d_\alpha |\det(D\Phi(x_\alpha))| + \epsilon v_\alpha M.$$

Similarly, we can derive a lower estimate, using the step function $u \leq g$ with constant value c_α on Q_α° . Summing over $\alpha \in A$ gives

$$-\epsilon Mv + \sum_{\alpha \in A} v_\alpha c_\alpha |\det(D\Phi(x_\alpha))| \leq \int_X g(x) |\det(D\Phi(x))| dx \leq \epsilon Mv + \sum_{\alpha \in A} v_\alpha d_\alpha |\det(D\Phi(x_\alpha))| \quad (14.19)$$

By choosing δ and using Lemma 14.70, we have

$$\text{vol}(\Phi(Q_\alpha^\circ)) d_\alpha \leq (1 + \epsilon) \det(\Phi(x_\alpha)) v_\alpha d_\alpha \leq \epsilon Mv_\alpha + \det(D\Phi(x_\alpha)) v_\alpha d_\alpha$$

for all $\alpha \in A$. The function $o \circ \Phi$ is constant with value d_α on $\Phi(Q_\alpha^\circ)$ and an upper bound for $f = g \circ \Phi^{-1}$, which implies

$$\int_{\Phi(Q_\alpha^\circ)} f dy \leq \int_{\Phi(Q_\alpha^\circ)} (o \circ \Phi^{-1}) dy \leq \epsilon Mv_\alpha + \det(D\Phi(x_\alpha)) v_\alpha d_\alpha$$

Summing over $\alpha \in A$ gives

$$\int_Y f dy \leq \sum_{\alpha \in A} \epsilon Mv_\alpha + \det(D\Phi(x_\alpha)) v_\alpha d_\alpha \leq \epsilon Mv + \sum_{\alpha \in A} \det(D\Phi(x_\alpha)) v_\alpha d_\alpha$$

Similarly, using the step function $u \leq g$ with constant value c_α on Q_α° , we derive a lower bound for the integral of f , which finally leads to the estimates

$$-\epsilon Mv + \sum_{\alpha \in A} \det(D\Phi(x_\alpha)) v_\alpha c_\alpha \leq \int_Y f dy \leq \epsilon Mv + \sum_{\alpha \in A} \det(D\Phi(x_\alpha)) v_\alpha d_\alpha \quad (14.20)$$

□

14.4 Improper Multiple Integrals and applications

For nonnegative continuous $f : \mathbb{R}^n \rightarrow \mathbb{R}$ define

$$\int_{\mathbb{R}^m} f(x) dx = \lim_{R \rightarrow \infty} \int_{B_R} f(x) dx$$

Notice that change of variable formula holds and Fubini theorem hold.

EXAMPLE 14.71. — The **Gaussian curve** $f : \mathbb{R} \rightarrow \mathbb{R}$, also known as the **probability density function**, is defined by $f(x) = \exp(-x^2)$. The normalized cumulative distribution function

$$F(x) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^x \exp(-t^2) dt$$

is called the **distribution function** of the normal distribution and is an indispensable function in statistics. We compute the converging integral

$$I = \int_{-\infty}^{\infty} \exp(-x^2) dx$$

by using polar coordinates in $\mathbb{R}^2$ to calculate I^2 . Indeed, we have

$$\begin{aligned} I^2 &= \left(\int_{\mathbb{R}} e^{-x^2} dx \right) \left(\int_{\mathbb{R}} e^{-y^2} dy \right) = \int_{\mathbb{R}^2} e^{-x^2-y^2} dx dy = \int_0^{\infty} \int_0^{2\pi} e^{-r^2} r dr d\varphi \\ &= \pi \int_0^{\infty} e^{-r^2} 2r dr = \pi \end{aligned}$$

which implies $I = \sqrt{\pi}$. We have applied Fubini's Theorem to the entire $\mathbb{R}^2$, which can be justified by considering suitable exhaustions, such as $([-m, m]^2)_{m=0}^{\infty}$ for $\mathbb{R}^2$.

EXERCISE 14.72. — Let $A \in \text{Mat}_{n,n}(\mathbb{R})$ be symmetric and positive definite. Show

$$\int_{\mathbb{R}^n} \exp(-\langle Ax, x \rangle) dx = \frac{\pi^{n/2}}{\sqrt{\det A}}$$

EXAMPLE 14.73. — As an application of the theory in this chapter, we want to prove the equation known as the **Basel problem**:

$$\zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

The approach we take here is from T. Apostol [Apo1983]. To do this, we will evaluate the integral

$$\int_0^1 \int_0^1 \frac{1}{1-xy} dx dy \tag{14.21}$$

in two different ways.

First, using the geometric series, we observe that

$$\begin{aligned} \int_0^1 \int_0^1 \frac{1}{1-xy} dx dy &= \int_0^1 \int_0^1 \sum_{k=0}^{\infty} (xy)^k dx dy = \sum_{k=0}^{\infty} \int_0^1 \int_0^1 x^k y^k dx dy \\ &= \sum_{k=0}^{\infty} \int_0^1 \frac{1}{k+1} y^k dy = \sum_{k=0}^{\infty} \frac{1}{(k+1)^2} = \zeta(2) \end{aligned} \quad (14.22)$$

It is left to the reader to show that the function $f : (x, y) \mapsto \frac{1}{1-xy}$ on $[0, 1]^2$ is improperly Riemann-integrable and that the manipulations in (14.22) are meaningful.

Now, we want to calculate the integral (14.21) using a linear substitution. For this, we consider new integration variables u, v given by the rotation

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}.$$

The new integration domain in the u, v -coordinates is the square with corners at $(0, 0)$, $(1, 1)$, $(2, 0)$, and $(1, -1)$, as can be easily verified by substitution. It holds that

$$\frac{1}{1-xy} = \frac{1}{1 - \frac{1}{4}(u-v)(u+v)} = \frac{4}{4 - u^2 + v^2}$$

and with substitution and Fubini, we have

$$\begin{aligned} \int_0^1 \int_0^1 \frac{1}{1-xy} dx dy &= \int_0^1 \int_{-u}^u \frac{4}{4 - u^2 + v^2} \frac{1}{2} dv du + \int_1^2 \int_{-(2-u)}^{2-u} \frac{4}{4 - u^2 + v^2} \frac{1}{2} dv du \\ &= \underbrace{4 \int_0^1 \int_0^u \frac{1}{4 - u^2 + v^2} dv du}_{=I_1} + \underbrace{4 \int_1^2 \int_0^{2-u} \frac{1}{4 - u^2 + v^2} dv du}_{=I_2} \end{aligned}$$

A direct calculation (see (??)) shows

$$I_1 = 4 \int_0^1 \frac{1}{\sqrt{4-u^2}} \arctan\left(\frac{u}{\sqrt{4-u^2}}\right) du = 4 \int_0^{\frac{\pi}{6}} \theta d\theta = 2\left(\frac{\pi}{6}\right)^2$$

where $u = 2 \sin(\theta)$ is substituted. Similarly, we calculate

$$I_2 = 4 \int_1^2 \frac{1}{\sqrt{4-u^2}} \arctan\left(\frac{2-u}{\sqrt{4-u^2}}\right) du = 8 \int_0^{\frac{\pi}{6}} \phi d\phi = 4\left(\frac{\pi}{6}\right)^2,$$

where $u = 2 \cos(2\phi)$ is substituted. In summary, we obtain

$$\zeta(2) = \int_0^1 \int_0^1 \frac{1}{1-xy} dx dy = I_1 + I_2 = 2\left(\frac{\pi}{6}\right)^2 + 4\left(\frac{\pi}{6}\right)^2 = \frac{\pi^2}{6}$$

as claimed.

EXERCISE 14.74. — Carry out the above calculations in detail, justify all the formal steps.

EXERCISE 14.75. — Calculate the value of the alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$.

EXERCISE 14.76. — Let $n \in \mathbb{N}$. The set $\Delta_n = \{(x_1, \dots, x_n) \in [0, 1]^n \mid x_1 + \dots + x_n \leq 1\}$ is called the n -dimensional **standard simplex**. Calculate the volume $\text{vol}(\Delta_n)$ and the integral

$$\int_{\Delta_n} e^{x_1 + \dots + x_n} dx_1 \cdots dx_n.$$

EXERCISE 14.77. — For $n \in \mathbb{N}$, let ω_n denote the volume of the n -dimensional unit ball $B(0, 1) \subseteq \mathbb{R}^n$. Show that

$$\omega_n = \frac{\pi^{n/2}}{\Gamma(n/2 + 1)}.$$

Calculate ω_{100} with computer assistance to 30 correct decimal places. For which n is ω_n maximal?

Hint: Show or use without proof that the one-dimensional integral

$$I_n = \int_0^\pi \sin^n(x) dx$$

satisfies the equation $I_n I_{n-1} = \frac{1}{n} I_1 I_0 = \frac{2\pi}{n}$ and use this to derive a recursion formula for ω_n .

EXERCISE 14.78. — Let $(B_m)_{m=0}^\infty$ be an exhaustion of a subset $B \subseteq \mathbb{R}^n$. Suppose $f, g : B \rightarrow \mathbb{R}$ such that $|f| \leq g$ and the functions $f|_{B_m}$ and $g|_{B_m}$ are Riemann-integrable for all $m \in \mathbb{N}$. Assume g is improperly Riemann-integrable. Show that f and $|f|$ are also improperly Riemann-integrable on B , and

$$\left| \int_B f dx \right| \leq \int_B |f| dx \leq \int_B g dx$$

holds.

EXERCISE 14.79. — For $x, y > 0$, the **beta function** is defined by

$$B(x, y) = \int_0^1 t^{x-1} (1-t)^{y-1} dt$$

Show that

$$B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$$

holds.

14.5 Parameter Integrals

14.5.1 Interchanging Differentiation and Integration

Let $a < b$ be real numbers, $U \subseteq \mathbb{R}^n$ be open, and $f : U \times [a, b] \rightarrow \mathbb{R}$ be a function. An integral of the form

$$F(x) = \int_a^b f(x, t) dt$$

is referred to as a **parameter integral**. Typically, we assume that the function f in $n + 1$ variables is at least continuous so that, in particular, for each fixed $x \in U$, the map $t \mapsto f(x, t)$ is continuous and thus Riemann integrable.

DO THE GENERAL THEOREM!

THEOREM 14.80: INTEGRALS DEPENDING ON A PARAMETER

Let $U \subset \mathbb{R}^n$ be an open subset, $a < b$ be real numbers, and $f : U \times [a, b] \rightarrow \mathbb{R}$ be continuous. Then, the parameter integral

$$F(x) = \int_a^b f(x, t) dt$$

defines a continuous function $F : U \rightarrow \mathbb{R}$. If the partial derivatives $\partial_k f$ for $k = 1, \dots, n$ exist and are continuous on $U \times [a, b]$, then F is continuously differentiable, and it holds that

$$\partial_k F(x) = \int_a^b \partial_k f(x, t) dt$$

for all $x \in U$ and $k \in \{1, \dots, n\}$.

Proof. We check the continuity of F at $x_0 \in U$. Let $\epsilon > 0$. Choose $r > 0$ such that $\overline{B(x_0, r)}$ is contained in U . According to the Heine-Borel Theorem, let $K := \overline{B(x_0, r)} \times [a, b]$ be compact, and $f|_K$ is uniformly continuous by Heine-Cantor Theorem. Thus, there exists $\delta \in (0, r)$ such that for all $x \in B(x_0, \delta)$ and $t \in [a, b]$, the inequality $|f(x, t) - f(x_0, t)| < \epsilon(b - a)^{-1}$ holds. This implies

$$|F(x) - F(x_0)| \leq \int_a^b |f(x, t) - f(x_0, t)| dt < \epsilon$$

for all $x \in B(x_0, \delta)$, proving the continuity of F at x_0 .

Assuming the partial derivative $\partial_k f$ exists and is continuous. For $s \in (-r, r) \setminus \{0\}$ and $t \in [a, b]$, according to the Mean Value Theorem 10.22, there exists $\xi \in (0, 1)$ such that

$$\frac{f(x_0 + se_k, t) - f(x_0, t)}{s} = \partial_k f(x_0 + \xi se_k, t).$$

Let $\epsilon > 0$. Due to the uniform continuity of $\partial_k f$ on K , there exists $\delta \in (0, r)$ such that for $x \in B(x_0, \delta)$ and all $t \in [a, b]$, the estimate

$$|\partial_k f(x, t) - \partial_k f(x_0, t)| < \epsilon(b - a)^{-1}$$

holds. Combining this, for $s \in (-\delta, \delta) \setminus \{0\}$, we get

$$\begin{aligned} \left| \frac{F(x_0 + se_k) - F(x_0)}{s} - \int_a^b \partial_k f(x_0, t) dt \right| &= \left| \int_a^b \left(\frac{f(x_0 + se_k, t) - f(x_0, t)}{s} - \partial_k f(x_0, t) \right) dt \right| \\ &= \left| \int_a^b (\partial_k f(x_0 + \xi se_k, t) - \partial_k f(x_0, t)) dt \right| \leq \epsilon(b-a). \end{aligned}$$

As $\epsilon > 0$ was arbitrary, we conclude

$$\partial_k F(x_0) = \lim_{s \rightarrow 0} \frac{F(x_0 + se_k) - F(x_0)}{s} = \int_a^b \partial_k f(x_0, t) dt.$$

By the first part of the theorem, $\partial_k F$ is continuous, and since k was arbitrary, continuous differentiability of F follows from Theorem 10.10. $\square$

EXAMPLE 14.81. — Theorem 14.80 allows us to analyze functions given by integrals. An example of such a function is obtained by calculating the circumference of an ellipse with axis lengths $2a$ and $2b$. Let $a \geq b > 0$ be real. The ellipse with parameters a, b is defined by the solution set of the equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

A parametrization of this ellipse is given by $\gamma(t) = (a \cos(t), b \sin(t))$ for $t \in [0, 2\pi]$. The circumference of the ellipse is then, as explained in Section ??:

$$\begin{aligned} L(\gamma) &= \int_0^{2\pi} \sqrt{a^2 \sin^2(t) + b^2 \cos^2(t)} dt = a \int_0^{2\pi} \sqrt{\sin^2(t) + \left(\frac{b}{a}\right)^2 \cos^2(t)} dt \\ &= a \int_0^{2\pi} \sqrt{1 - \varepsilon^2 \cos^2(t)} dt = 4a \int_0^{\pi/2} \sqrt{1 - \varepsilon^2 \sin^2(t)} dt, \end{aligned}$$

where $\varepsilon = \sqrt{1 - \frac{b^2}{a^2}}$ denotes the **eccentricity** of the ellipse, which measures the deviation of the ellipse from a circle. The parameter integral

$$F(x) = \int_0^{\pi/2} \sqrt{1 - x^2 \sin^2(t)} dt$$

defines, for $x \in (0, 1)$, the so-called “complete elliptic integral of the second kind”. According to Theorem 14.80 and induction, the function $F : (0, 1) \rightarrow \mathbb{R}$ is smooth.

COROLLARY 14.82:

Let $U \subset \mathbb{R}^n$ be open, $a < b$ be real numbers, and $f : U \times (a, b) \rightarrow \mathbb{R}$ be continuous with continuous partial derivatives $\partial_k f$ for $k \in \{1, \dots, n\}$. Let $\alpha, \beta : U \rightarrow (a, b)$ be continuously differentiable. Then, the parameter integral with varying limits

$$F(x) = \int_{\alpha(x)}^{\beta(x)} f(x, t) dt$$

is continuously differentiable on U , and it holds that

$$\partial_k F(x) = f(x, \beta(x)) \partial_k \beta(x) - f(x, \alpha(x)) \partial_k \alpha(x) + \int_{\alpha(x)}^{\beta(x)} \partial_k f(x, t) dt$$

for all $x \in U$.

Proof. We combine Theorem 14.80, the Fundamental Theorem of Calculus from Analysis I, and the chain rule (10.16). For this, we define the auxiliary function

$$\varphi : U \times (a, b)^2 \rightarrow \mathbb{R}, \quad (x, \alpha, \beta) \mapsto \int_{\alpha}^{\beta} f(x, t) dt$$

First, we show that φ is continuous. Let $(x_n, \alpha_n, \beta_n) \in U \times (a, b)^2$ be a sequence converging to $(x, \alpha, \beta) \in U \times (a, b)^2$. Choose $\epsilon > 0$ and $c, d \in (a, b)$ such that

$$\overline{B(x, \epsilon)} \subset U \quad \text{and} \quad c \leq \alpha_n, \beta_n \leq d$$

for all $n \geq 0$. Thus, $K = \overline{B(x, \epsilon)} \times [c, d]$ is a compact subset of $U \times (a, b)$ and

$$M = \max\{|f(x', t')| \mid (x', t') \in K\}$$

exists. For all sufficiently large n , $x_n \in B(x, \epsilon)$ holds, and consequently

$$\begin{aligned} & |\varphi(x_n, \alpha_n, \beta_n) - \varphi(x, \alpha, \beta)| \\ & \leq \left| \int_{\alpha_n}^{\beta_n} f(x_n, t) dt - \int_{\alpha}^{\beta} f(x_n, t) dt \right| + \left| \int_{\alpha}^{\beta} f(x_n, t) dt - \int_{\alpha}^{\beta} f(x, t) dt \right| \\ & \leq M|\alpha_n - \alpha| + M|\beta_n - \beta| + \left| \int_{\alpha}^{\beta} f(x_n, t) dt - \int_{\alpha}^{\beta} f(x, t) dt \right|, \end{aligned}$$

using the triangle inequality for integrals over the subintervals between α_n and α , as well as β_n and β , and the bound M for the function values of f . For $n \rightarrow \infty$, it follows from Theorem 14.80 that this expression tends to 0, hence φ is continuous.

According to Theorem 14.80, the partial derivatives $\partial_k \varphi$ for $k = 1, 2, \dots, n$ exist and are given by

$$\partial_k \varphi(x, \alpha, \beta) = \int_{\alpha}^{\beta} \partial_k f(x, t) dt.$$

As argued above, $\partial_k \varphi$ is also continuous. By The fundamental Theorem of Integral Calculus, the partial derivatives of φ with respect to α and β exist and are given by

$$\partial_\alpha \varphi(x, \alpha, \beta) = -f(x, \alpha) \quad \text{and} \quad \partial_\beta \varphi(x, \alpha, \beta) = f(x, \beta)$$

In particular, $\partial_\alpha \varphi$ and $\partial_\beta \varphi$ are continuous. By Theorem 10.10, φ is continuously differentiable.

The function F is given by $F(x) = \varphi(x, \alpha(x), \beta(x))$, and can thus be seen as the composition of φ with the function $\psi : U \rightarrow U \times (a, b)^2$

$$\psi(x) = (x, \alpha(x), \beta(x))$$

According to the assumption, ψ is also continuously differentiable, with a total derivative given by

$$D\psi(x)(v) = (v, D\alpha(x)(v), D\beta(x)(v))$$

We can apply the chain rule and obtain that F is continuously differentiable, with a total derivative

$$DF(x)(v) = D\varphi(\psi(x))(D\psi(x)(v)) = D\varphi(\psi(x))(v, D\alpha(x)(v), D\beta(x)(v)).$$

For $v = e_k$ and thus $\partial_k F(x) = DF(x)(e_k)$, it follows, in particular,

$$\begin{aligned} \partial_k F(x) &= \partial_k \varphi(x, \alpha(x), \beta(x)) + \partial_\alpha \varphi(x, \alpha(x), \beta(x)) \partial_k \alpha(x) + \partial_\beta \varphi(x, \alpha(x), \beta(x)) \partial_k \beta(x) \\ &= \int_{\alpha(x)}^{\beta(x)} \partial_k f(x, t) dt - f(x, \alpha(x)) \partial_k \alpha(x) + f(x, \beta(x)) \partial_k \beta(x) \end{aligned}$$

which was to be shown. □

14.5.2 The Bessel Differential Equation

As an application of the general theory of parameter integrals, we want to use it here to solve a differential equation. For a parameter $n \geq 0$, usually a natural number, the differential equation

$$x^2 u''(x) + x u'(x) + (x^2 - n^2) u(x) = 0 \tag{14.23}$$

for $u \in C^2((0, \infty), \mathbb{R})$ is called the **Bessel Differential Equation**. It is linear, homogeneous of the second order. From the Picard-Lindelöf existence and uniqueness theorem, which we will prove towards the end of the semester, it follows that (14.23), together with any two initial values $u(1) = a$ and $u'(1) = b$ for $a, b \in \mathbb{R}$, has a uniquely determined solution on $(0, \infty)$. In particular, the vector space of solutions to (14.23) is two-dimensional. We aim to provide two linearly independent solutions. For this purpose, we assume $n \in \mathbb{N}$.

14.83. — The function defined by the parameter integral

$$J_n(x) = \frac{1}{\pi} \int_0^\pi \cos(x \sin(t) - nt) dt \quad (14.24)$$

is called the **Bessel function of the first kind**, and it solves the differential equation (14.23), as we can verify using Theorem 14.80. Indeed, we have $\partial_x(\cos(x \sin(t) - nt)) = -\sin(x \sin(t) - nt) \sin(t)$, and therefore,

$$J'_n(x) = \frac{1}{\pi} \int_0^\pi (-\sin(x \sin(t) - nt) \sin(t)) dt$$

and similarly

$$J''_n(x) = \frac{1}{\pi} \int_0^\pi (-\cos(x \sin(t) - nt) \sin^2(t)) dt.$$

For the expression $x^2 J''_n(x) + (x^2 - n^2) J_n(x)$, we obtain

$$\begin{aligned} & \frac{1}{\pi} \int_0^\pi (-x^2 \cos(x \sin(t) - nt) \sin^2(t) + (x^2 - n^2) \cos(x \sin(t) - nt)) dt \\ &= \frac{1}{\pi} \int_0^\pi \cos(x \sin(t) - nt) (x^2 \cos^2(t) - n^2) dt \\ &= \frac{1}{\pi} \int_0^\pi \cos(x \sin(t) - nt) \underbrace{(x \cos(t) - n)(x \cos(t) + n)}_{\partial_t(x \sin(t) - nt)} dt \\ &= \frac{1}{\pi} \sin(x \sin(t) - nt) (x \cos(t) + n) \Big|_0^\pi \\ &\quad + \frac{1}{\pi} \int_0^\pi \sin(x \sin(t) - nt) x \sin(t) dt \\ &= -x J'_n(x) \end{aligned}$$

using integration by parts and the assumption $n \in \mathbb{N}$. Therefore, (14.24) satisfies the differential equation (14.23).

14.84. — The **Bessel function of the second kind** is defined by the improper integral

$$Y_n(x) = \frac{1}{\pi} \int_0^\pi \sin(x \sin(t) - nt) x \sin(t) dt - \frac{1}{\pi} \int_0^\infty (\exp(t) + (-1)^n \exp(-nt)) \exp(-x \sinh(t)) dt$$

for $x \in (0, \infty)$. It can be shown that Y_n also satisfies the differential equation (14.23). We have

$$\lim_{x \rightarrow 0} J_n(x) = \frac{1}{\pi} \int_0^\pi \cos(nt) dt$$

according to Theorem 14.80, and

$$\lim_{x \rightarrow 0} Y_n(x) = -\infty. \quad (14.25)$$

This shows that J_n and Y_n are linearly independent.

EXERCISE 14.85. — Let $n \in \mathbb{N}$.

- (a) Show that the Bessel function Y_n of the second kind is well-defined and prove the asymptotics in (14.25).
- (b) Assume a suitable generalization of differentiation under the integral for the improper integrals and use it to prove that Y_n is a solution to the Bessel differential equation (14.23).
- (c) For the proof of the appropriate generalization of differentiation under the integral, consider the real-valued function

$$f(x, t) = (\exp(t) + (-1)^n \exp(-nt)) \exp(-x \sinh(t))$$

on $(0, \infty) \times [1, \infty)$ and

$$F(x, s) = \begin{cases} 0 & \text{if } s = 0 \\ \frac{f(x, s^{-1})}{s^2} & \text{if } s > 0 \end{cases}$$

on $(0, \infty) \times [0, 1]$. Show that

$$\int_1^\infty f(x, t) dt = \int_0^1 F(x, s) ds$$

holds and that F satisfies all the conditions of Theorem 14.80.

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